



ViewPoint™

XML Distiller

User Guide

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About this Document

The XML Distiller User Manual explains how to use XML Distiller and associated tools to process enterprise documents (managed in the ViewPoint system) and periodicals (managed in the ActivePaper system) into XML files. This manual explains in stepped procedures how to perform the processing and how to use *Job Manager* for fully-automated processing of multiple jobs. This manual also instructs how to improve profiles and perform other programming tasks.

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Style Conventions

Convention	Used for
Verdana	Regular text.
Arial Bold	Names of menus, commands, buttons, steps, and other elements of the user interface.
<i>Arial Italics</i>	Special terms, the first time they appear.
Monospace	Text entered in a text file, values of parameters, and usage examples.
	Notes , which offer an additional explanation or a hint on how to overcome a common problem.
	Warnings , which indicate potentially damaging user operations and explain how to avoid them.

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1 Introduction

Olive Software's comprehensive ePublishing infrastructure transforms any document into intelligent XML-based content, bridging the gap between the print, archive, and online worlds. Olive's systems can take any legacy hardcopy, paper, microfilm, microfiche, or unstructured digital documents and transform them to rich and intelligent, platform independent XML content.

Olive's solutions process various document types, from newspapers and magazines (*Periodicals*) to books, catalogs, technical documentation, etc. (*Enterprise Documents*). Each of these document groups, periodicals and enterprise documents, is processed and managed by a separate Olive product. *ActivePaper*™ processes Periodicals and *ViewPoint*™ processes Enterprise Documents.

Both Viewpoint and ActivePaper solutions use an interactive process that analyzes the documents and create the XML. XML Distiller handles this process, which is the subject of this manual.

1.1 About XML Distiller

XML Distiller automatically transforms a variety of electronic and hardcopy documents to a standard XML format. The Distiller captures, analyzes, identifies, extracts the data form the original documents, and then it tags and converts them. It can accept a variety of complex and unstructured content sources, including PDF, TIFF and MS Office files, while preserving the original documents' look as well as content.

XML Distiller not only creates an accurate XML representation of the source document, it also discovers its hidden intelligence, including hierarchical structure, semantics, metadata and links. It recognizes a variety of document types, including complex *information intensive* documents such as books, academic papers, journals, research reports, procedures, guide books, manuals, catalogs and more. It also processes data-intensive documents such as claims, invoices and statements.

XML Distiller (serving ActivePaper and ViewPoint) is installed on a Windows desktop environment as a plug-in to Adobe® Acrobat®. The user interface is the Olive menu that is added to the Acrobat menu bar, and an associated tool bar.

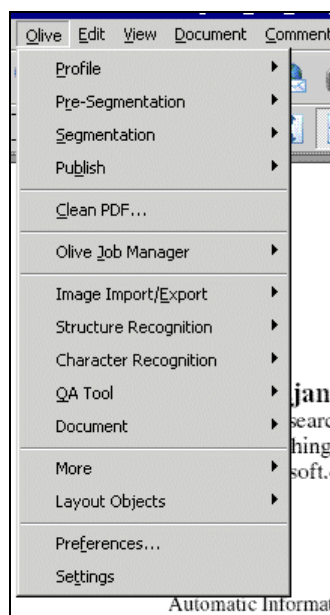


Figure 1 – Olive Menu for XML Distiller

The Distillation Process

Converting scanned and electronic documents to XML brings many challenges, of which the main challenge is transforming a "physical" package of text and graphics (a document) into a "logical" representation. Such a process normally requires a human-like "understanding" of the document's style, structure and semantics.

Another problem is the number and variety of digital documents and formats. These variations create a non-standard and unexpected environment when converting scanned, PDF, or Microsoft Office documents to XML.

Olive XML Distiller is based on the Olive Recognition Framework, an open and extensible recognition infrastructure. This framework supports three document recognition approaches:

- **Style-based recognition** works according to style based rules, including intrinsic knowledge to recognize embedded objects and resolve ambiguities
- **TOC recognition** looks for a Table of Contents and uses it to recognize document structure
- **Olive Recognition Language (ORL)** provides DOM model and language (JavaScript) to recognize structure and extract metadata.

Introduction

About XML Distiller

XML Distiller automatically "transforms" document content to XML using a synthesis of image processing, style and text analysis, pattern recognition and metadata extraction techniques. The resulting XML files also contain advanced information that is unavailable to the human eye in the source document format, such as metadata, document structure and internal hierarchy. Olive's unique artificial intelligence and heuristic algorithms generate a multi-dimensional document with "understanding" - knowledge that reflects a document's formatting, semantics and logic - resulting in an XML structure that maps all aspects of the source document.

The preserved physical representation, including layout style and building blocks (Contextual Components), enables on-demand reconstruction of the original document or generating a completely new customized view. The semantic knowledge of the document's structure and metadata enables efficient retrieval and management of relevant information components, such as delivering the most relevant chapters of a long document.

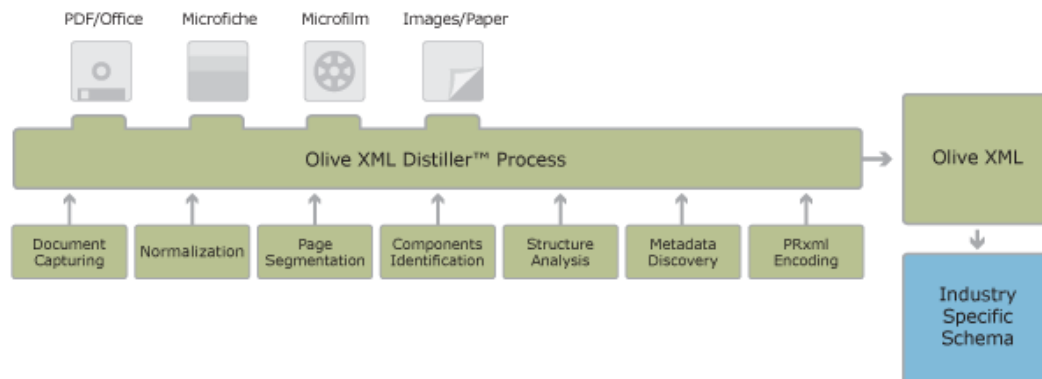


Figure 2 – XML Distiller Process

1.2 About ActivePaper

ActivePaper™ is Olive's e-publishing platform for periodicals such as newspapers, magazines, and periodic journals. It provides a comprehensive solution for online publishing, archiving, content personalization, advertising, multi-channel publishing and content e-commerce.

The ActivePaper™ e-publishing platform taps into existing print production lines, converting existing print data for Internet and Intranet publishing – integrating multiple publishing objectives into one workflow. It transforms legacy digital content, microfilm, paper-based content and other resources into a highly reusable, non-database, XML-based Warehouse (Repository).

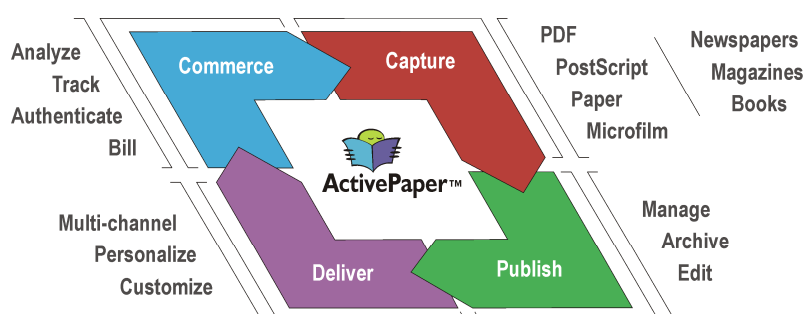


Figure 3 – ActivePaper™ Platform Functionality

ActivePaper™ is a multi-application e-publishing platform. Publishers may wish to purchase applications developed by Olive Software, or develop custom applications using Olive's ActivePaper™ API (Application Programming Interface).

The currently available ActivePaper™ applications are:

- ActivePaper Daily™ – Electronic version of a daily newspaper
- ActivePaper Archive™ – Online newspaper archive
- ActiveTearsheet™ – Ad tracking, archiving, invoice matching, and tearsheet delivery
- ActiveMagazine™ – Electronic version of a magazine
- ActiveMagazine Archive™ – Online magazine archive

1.2.1 Multi-Channel Publishing

ActivePaper's rich XML concept and its advanced XSL enables instant formatting and publishing to multiple on-line channels, including HTML, XML and ASCII. The Channel Output filters the required information based on *Element Tags*. For example, output to mobile devices may include only the Headlines. New custom channels for new electronic devices or even interfaces to database applications and content management systems can be easily created.

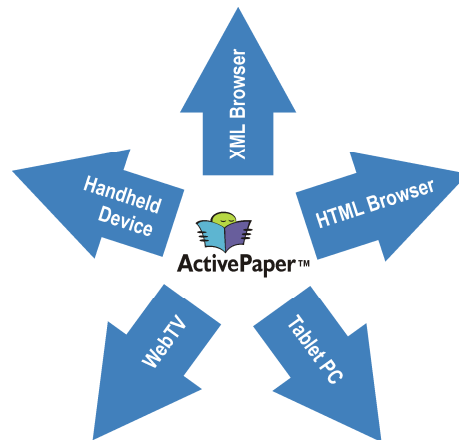


Figure 4 – Multi-channel Platform

1.2.2 ActivePaper's XML Warehouse

The XML Warehouse (Repository) holds the output of the tagging process in a file system format. The Warehouse is structured as a tree of files and folders that reside in a file server that is accessible locally or over a network.

- The Warehouse's **Root** directory contains a separate folder for each publication and two special folders for internal usage - *commonFiles* and *commonStyles*.
- Each **Publication** directory bears a unique publication name (e.g. "Houston Chronicle").
- Each Publication directory holds **Year** directories, named with a four-digit number.
- Each Year directory holds **Month** directories, named with a two-digit number (01 to 12).
- Each Month directory holds **Day** directories, named a two-digit number (01 to 31), containing the following:
 - A subdirectory for each page
 - A Table of Contents XML file named *TOC.xml*
 - A Font Style Gallery file named *StyleGallery.txt*
 - A copy of the source PDF file, given the name of the original file
 - A subdirectory named Res (optional), which may contain:
 - a subdirectory **PDF** with one-page PDF-files and ASCII file named *OlvData.txt* containing segmentation
 - a multi-page TIFF file named *Document.tif* containing images of all pages of the document (for scanned documents only)
 - a subdirectory named **PageImg** containing per-page images (for scanned documents only).

- Each **Page** directory contains the following:
 - IMG directory, containing all the image files for the page
 - Entity XML files
 - Page XML file
- Each **IMG** directory contains
 - Primitive images
 - Scaled page images
 - Page thumbnails
 - All images are provided in GIF, JPEG or PNG format.

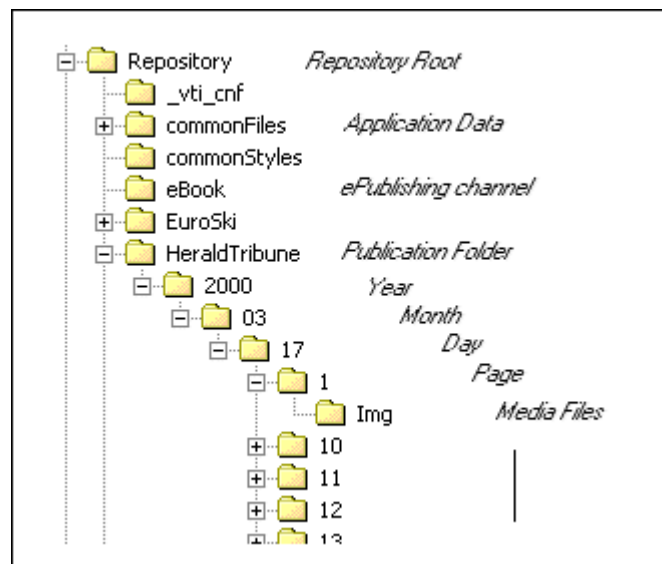


Figure 5 – XML Warehouse (Repository) Structure

1.3 About ViewPoint™

ViewPoint™ is Olive's e-publishing platform for enterprise documents such as technical documentation, books, and legal documents. The documents can be "born digital," such as Microsoft Office documents and Adobe Acrobat PDF files, or non digital scanned documents, TIFF files from scanned paper, microfilm, or microfiche. A wide variety of documents, such as technical procedures, illustrated manuals, manufacturing and parts catalogs, and reports, can be made available for online access and search. ViewPoint provides powerful knowledge mining, search, and retrieval tools, enhancing information access. It enables pinpoint controlled access to specific data - without the need to download large files.

1.3.1 Search and Retrieval

ViewPoint's powerful XML-based search engine can search through the text of an entire document warehouse, including metadata, to immediately retrieve components from multiple documents. It is possible, for example, to simultaneously search a specific term in a group of separate documents, or in a specific section in those documents, such as the summary or abstract. The search engine provides an efficient display of search results based on the proprietary "*Relational Relevance Ranking*" algorithm. The displayed search results enables immediate retrieval of the relevant components from the specific documents that contain the search term, relative to neighboring components, allowing the user to quickly select and retrieve the most relevant information component. All this is done without downloading the entire document. Using unique Bitmap Indexing technology, ViewPoint enables searching scanned documents while ensuring 100% readability. This technology highlights the searched term directly on the "image," saving the need to deal with corrupted or badly scanned text.

1.3.2 ViewPoint's Architecture

ViewPoint is comprised of three main modules – XML Distiller, Olive XML Warehouse™ and the Olive ViewPoint Server™. The XML Distiller captures unstructured, multiple-format document resources and transforms them into a unified, structured XML schema (PrXML). The content is stored in the Olive XML warehouse file system, where all the content is accessible, fully indexed and ready to be searched. Olive XML application server — the ViewPoint Server™, provides dynamic access, viewing and search services.

In addition to these modules, which are server based, ViewPoint also includes two client applications, ViewPoint Client and XML Distiller Editor. ViewPoint Client enables users to process documents over an enterprise network. It sends documents to an XML Distiller that operates on a server, processing the documents of an entire enterprise. XML Distiller Editor is a plug-in to Adobe Acrobat that turns the user's computer into a stand-alone document publishing station.

The published documents can be viewed on any web browser as part of an external web application or using Olive's Sample Application. The viewing components can be programmed using a special API when developing external application. Whereas in Olive's Sample Application, the viewing components have been built into the document viewer and it is ready to be used.

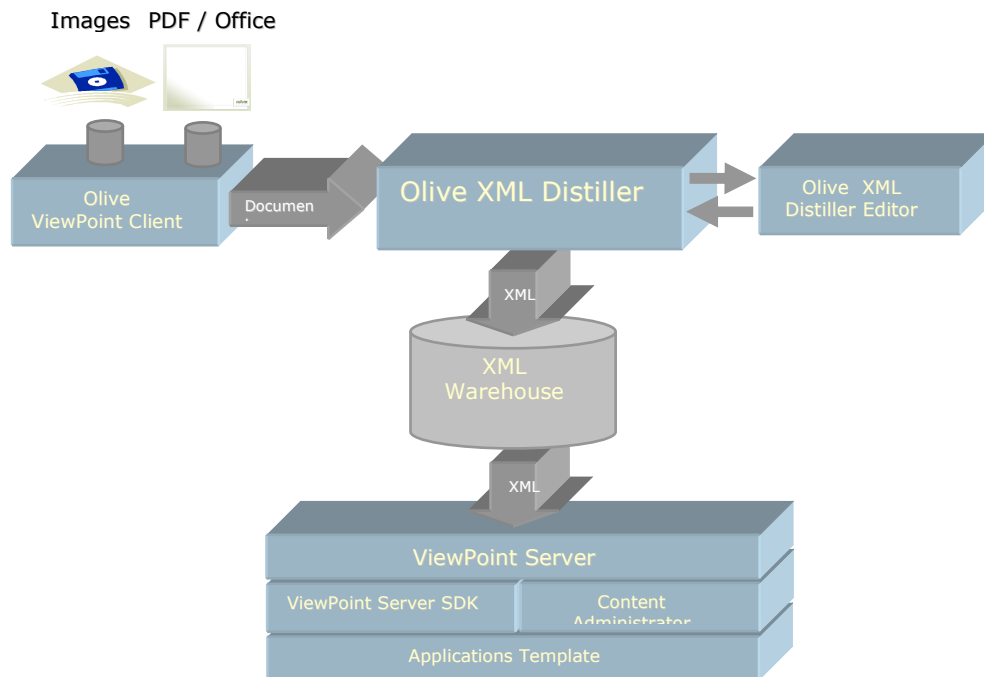


Figure 6 – ViewPoint's Solution Architecture

1.3.3 ViewPoint's XML Warehouse

ViewPoint's XML Warehouse (Repository) structure is based on ActivePaper's XML repository (see *ActivePaper's XML Warehouse* on page 16), in which each document receives a unique identifier that reflects its location in the repository's tree. ViewPoint's Warehouse tree is based on an XXXX/XX/XX hierarchy, like ActivePaper, although the identifier is not associated with a meaningful date.

- The Warehouse's **Root** directory contains a separate folder for each publication and two special folders for internal usage. These are *commonFiles* and *commonStyles*.
- The **Publication** directory can represent the publication name, document type, company name, or any other publication type.
- Three directory levels represent subdivisions within the Publication directory. The number structure can range from 1600/01/01 to 9999/12/31, like the date structure in ActivePaper, but with no correlation between the number and an actual date.
- The bottom-most directory (comparable to the **Day** directory in ActivePaper) contains the content of the document itself, including the following files and subdirectories:
 - A Table of Contents XML file for the entire document, named *TOC.xml*
 - A Font Style Gallery file named *StyleGallery.txt*
 - A copy of the source PDF file, given the name of the original file (optional)

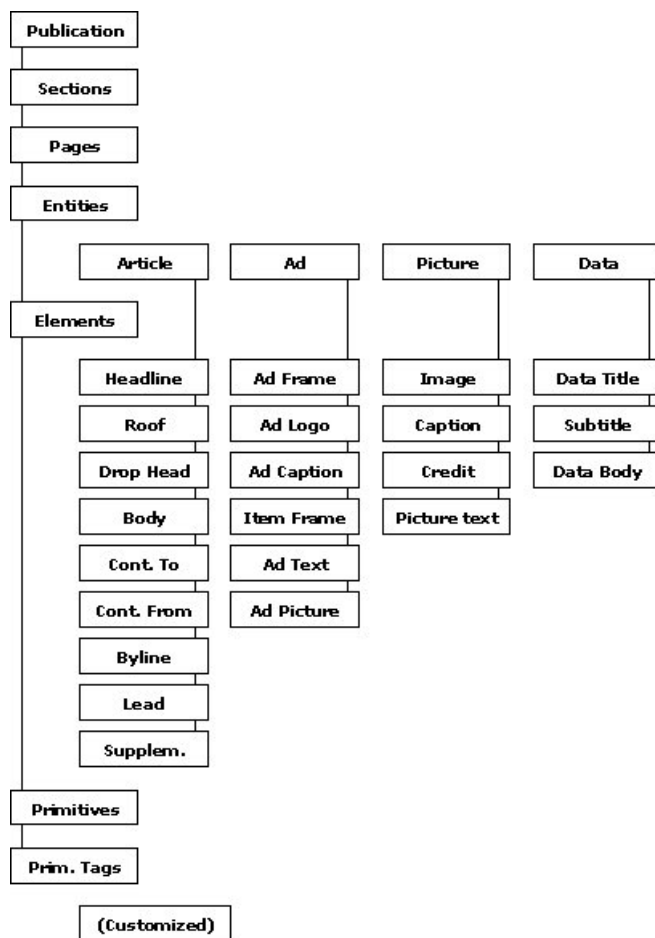
- A subdirectory named **Res** (optional) which can contain:
 - a subdirectory **PDF** with one-page PDF-files and ASCII file named *OlvData.txt* containing segmentation
 - a multi-page TIFF file named *Document.tif* containing images of all pages of the document (for scanned documents only)
 - a subdirectory named **PageImg** containing per-page images (for scanned documents only)
- A **Page** directory with the contents of each individual page
- Each **Page** directory contains the following:
 - IMG directory
 - Entity XML files
 - Page XML file.
- Each **IMG** directory contains
 - Primitive images
 - Scaled page images
 - Page thumbnails.

All images are provided in GIF, JPEG or PNG format.

1.4 Tagging Characteristics

1.4.1 Periodical Tagging

The tagging process divides (or segments) the periodical into its elements. ActivePaper identifies the basic elements and labels, or **tags**, them accordingly. It is therefore important to know these elements and understand how they are related in order to evaluate the tagging results. ActivePaper views a periodical's elements in a hierarchical perspective, as follows:



- **Publication:** is the largest unit in the hierarchy. A publication can be a newspaper or any other type of publication that contains similar page objects.
- **Section:** is the largest division of a publication, such as the newspaper sections "International", "Culture", "News", and "Sport".
- **Pages:** Each section is made up of one or more pages. A page is a single page within an Adobe Acrobat document. A page has two editable properties: Section Name, defining the section to which the page belongs, and Page ID, defining what the page is called within the newspaper (for example, "Page VII").
- **Entities:** Each page is made up of one or more entities. There are several types of entities, as follows:
 - **Article:** Text that includes a story including its related picture, headline, byline, and other elements.

- **Ad:** An advertisement. Ads are not automatically recognized unless an *Ads Manifest* is loaded. The Manifest provides information on the exact locations of the ads and related metadata (See *Creating Ad Frames from Manifest* on page 53). You can segment a newspaper without an Ads Manifest by marking the ads in *Predefined Areas* and tagging them manually (see *Predefining Areas* on page 54).
- **Picture:** A picture that may be associated with an Article entity or stand on its own.
- **Data:** Tabular textual information such as weather tables, stocks, and the like. Data is not automatically recognized. To tag data, you can mark *Predefined Areas* and segment them manually (see *Predefining Areas* on page 54).

Each entity has three editable properties: **Entity Type**, defining the type of entity, **Title**, defining the entity's title, and **Publish**, defining when and how entity is to be output to XML. Picture entities can be associated with an article entity, for pictures that relate to an article and don't stand on their own. Pictures that are associated with an article become nested entities under the article entity.

- **Elements:** Entities are made up of one or more elements. An element is a part of an entity that performs a certain function, as described in the element type. An element has one editable property: **Entity**. This property defines the logical position of the element.
 - **Article Elements** can be made up of the following element types:
 - **Headline:** The title of the article.
 - **Roof:** Text appearing above the headline.
 - **Drop Head:** Text appearing below the headline.
 - **Body:** The body text of the article.
 - **Continuation to:** Text that indicates that the article is continued on another page.
 - **Continuation from:** Text that indicates that the article is continued from another page. If an article is continued on another page (or on several other pages), each continuation is considered a separate entity.
 - **Byline:** Text identifying who wrote and/or contributed to the article.
 - **Lead:** Text appearing within the article, generally in a different style, highlighting important points.
 - **Supplementary:** A "general" element type - Any text which appears in the article and cannot be classified as one of the other element types can be classified as supplementary.
 - **Ad Elements:** Ad entities are defined for display ads and can be made up of the following element types:
 - **Ad Frame:** Complete area of a display ad.

- **Ad Logo:** Image element, as described above (such as the advertiser's logo).
- **Ad Caption:** Text, as described above.
- **Item Frame:** Section of a multiple item display ad that defines one of its items.
- **Ad Text:** Ad Text, may define data of any ad item.
- **Ad Picture:** Image element, as described above. May define any important graphic section on the ad.
- **Picture Elements:** Picture entities can be made up of the following four element types:
 - **Image:** An image, such as a photo or drawing.
 - **Caption:** Text, is usually located below or alongside the image.
 - **Credit:** Name of the organization that holds the copyright for the picture.
 - **Picture Text:** Text, as described above. Usually located on the image.
- **Primitives** are the primary units defined when manually editing structural definitions. Element that consist of more than one primitive, the logical order of primitives within the element (called the reading order) is defined by numbers (e.g., Article1, TextBlock1, TextBlock2, Article 2, TextBlock3, TextBlock4). The reading order of the primitives should match the logical order of the text or images contained within the article. An element with only one primitive has a reading order where its primitive is defined by the number 1.

A primitive has two editable properties: **Element** and **Entity**. These properties define the logical position of the primitive.

1.4.2 Segmentation of Enterprise Documents

The segmentation of enterprise documents is based on the same principal as periodicals but less complex. Unlike periodicals, enterprise documents are usually not compound documents. The document includes a single article that runs across the entire document. The images in the document are therefore related to the single article. On the other hand, these documents typically include elements that are not commonly found in periodicals, such as a table of contents (TOC), headers/footers, and a deeper hierarchy. ViewPoint views a document's elements in a hierarchical perspective as follows:

- **Publication:** Company name or document type, or any other primary category.
- **Entities:** There are several entity types:
 - **Article:** Body of text. In enterprise documents, articles usually run across the entire document.
 - **Picture:** A picture that may be associated with the Article entity or stand on its own.

- **Elements:** Entities are made up of one or more elements, which perform certain functions, as described in the element type. An element has one editable property: Entity. This property defines the logical position of the element.
 - **Article Elements** are defined as needed, such as unit, sub-chapter, section, and paragraph.
 - **Picture Elements** can be made up of the following element types:
 - **Image:** An image, such as a photo or drawing.
 - **Caption:** Text, is usually located below or alongside the image.
 - **Credit:** Name of the organization that holds the copyright for the picture.
 - **Picture Text** is usually located within the image, as described above.

1.5 XML Distiller Workflows

XML Distiller follows certain workflows processing digital and scanned periodicals as well as digital and scanned enterprise documents. The workflow diagram displays how the user interacts with the system and not the internal processes that occur within the XML Distiller software. Each object generally represents a menu function.

1.5.1 Digital Periodicals Workflow

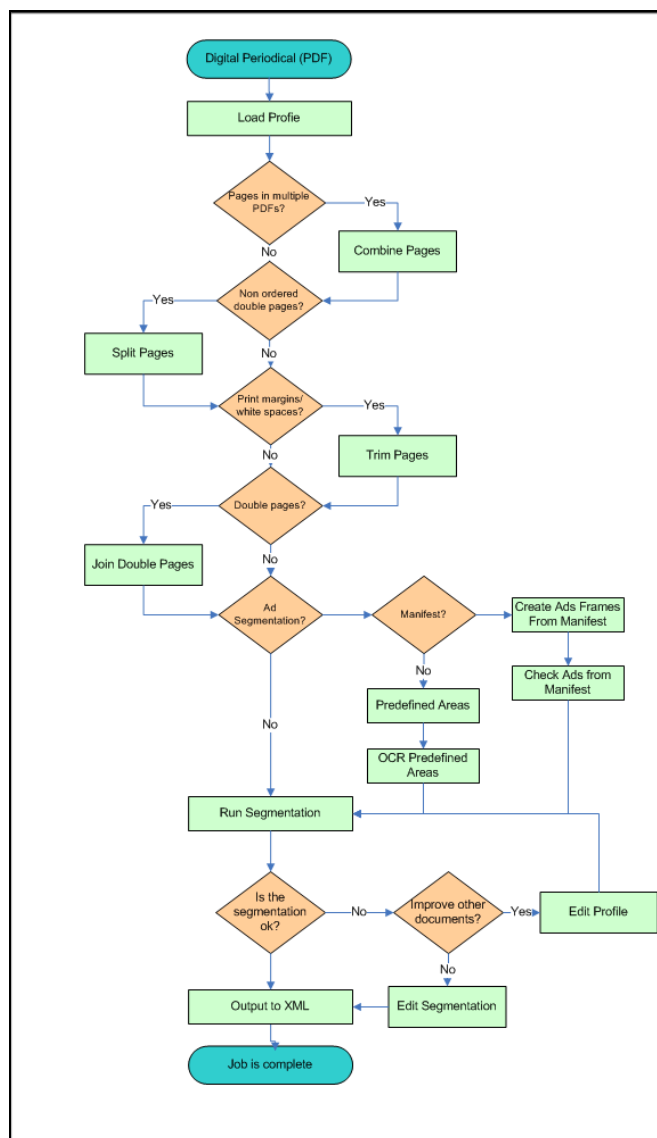


Figure 7 – Digital Periodical's Workflow

1. The process begins by loading the appropriate profile using the **Load Profile** function (page 35). The default profiles are created by Olive Professional Service and already includes the characteristics of your publication.
2. If the issue is scattered across multiple PDF files, first combine them into a single PDF file using the **Combine Pages** function (page 42).
3. If the issue includes pages that are not in the reading order but are in print order, such as a spread with the first and then the last page of the issue, split these pages using the **Split Pages** function (page 43).
4. If the issue has white spaces or print markings (bleed), trim them out from the pages using the **Trim Pages** function (page 45).

5. Pages in the issue that are intended to be “spreads” should be joined using the **Join Double Pages** function (page 49).
6. If the publication includes ads that you would like to segment, mark these ads manually using the **Predefine Areas** function (page 54), then run OCR to extract the texts using the **OCR Predefined Areas** function (page 57), since ads usually do not include embedded text in the PDF. If you have a manifest, a file containing the exact location of each ad and its metadata information, you can create Predefined Areas for the ads using **Create Ads Frames From Manifest** (page 53) and then verify the segmentation using **Check Ads From Manifest** (page 53).
7. You can now run segmentation using the **Run Segmentation** function (page 62).
8. If the segmentation is accurate, proceed to the output stage using **Output to XML** (page 107).
9. If the segmentation is not accurate, correct it using **Edit Segmentation** (page 70).
10. Segmentation editing improves the segmentation of the issue itself, but will not improve the segmentation for other issues. Improve the profile itself using **Edit Profile** (page 185) if similar issues or the same problems occur in other issues.
11. Run segmentation again upon editing the profile, and output to XML (steps 7 and 8 above) if it is accurate.

1.5.2 Scanned Periodicals Workflow (Microfilm)

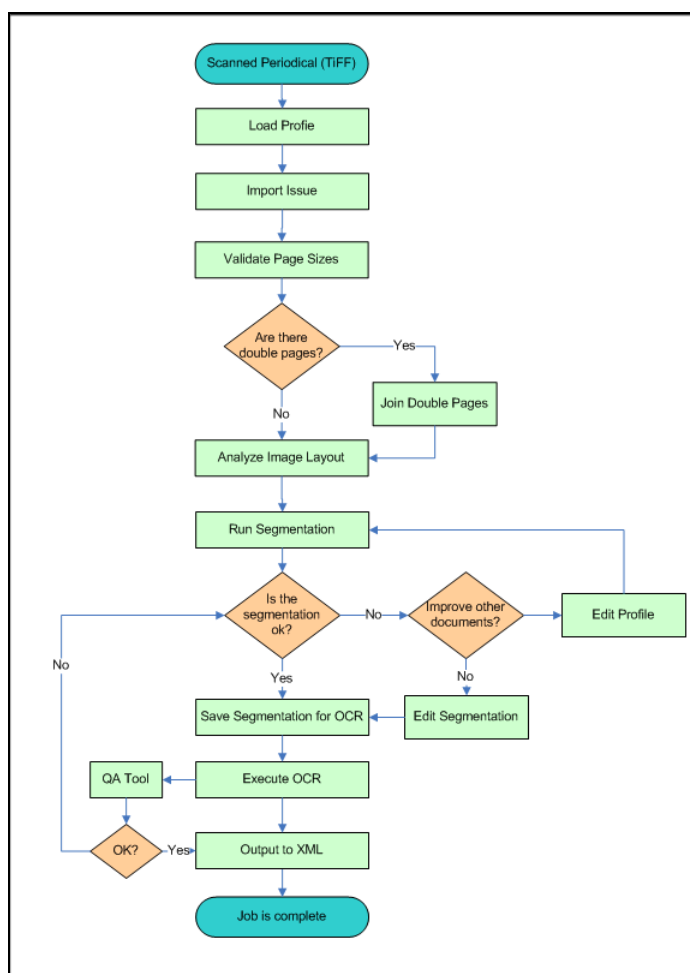


Figure 8 – Scanned Periodicals Workflow

1. Load the appropriate profile using **Load Profile** (page 35). This profile was created by Olive Professional Service and already includes the publication's characteristics.
2. Import the page images (TIFF files) using **Import Issue** (page 36).
3. If the pages sizes may vary, run **Validate Page Sizes** (page 40) to solve these problems.
4. If there are pages in the issue that are intended to be "spreads", join them using **Join Double Pages** (page 49).
5. Use **Analyze Image Layout** (page 41) to differentiate between lines, text areas (single lines or more complex structures) and image areas, and provide detailed measurements of each recognized component as input to segmentation, such as font, font size, and the ratio between uppercase and lowercase characters in a line of text.
6. Run segmentation using **Run Segmentation** (page 61). The segmentation assists the OCR in recognizing the correct reading order.

7. If the segmentation is accurate, proceed to save the segmentation results for the OCR engine using **Save Segmentation for OCR** (page 103).
8. If the segmentation is not accurate, correct it by editing the segmentation using **Edit Segmentation** (page 70).
9. Editing segmentation improves the segmentation of the issue itself, but will not improve the segmentation for other issues. If similar issues or the same problems are likely to occur in other issues, improve the profile itself using **Edit Profile** (page 185).
10. Upon editing the profile, run segmentation again, and proceed to saving the segmentation for the OCR if it is accurate.
11. Run the OCR using the **Execute OCR** function (page 104).
12. Use the **QA Tool** (page 112) to receive statistical information about the quality of the output based on its *readability* and *searchability* levels. It does not fix problems, but provides information on the quality of recent segmentation procedures.
13. If the quality results are not satisfactory, improve the various OCR and segmentation parameters and repeat steps 6-12.
14. If the result is satisfactory, proceed to output to XML using **Output to XML** (page 107).

1.5.3 Digital Enterprise Documents Workflow

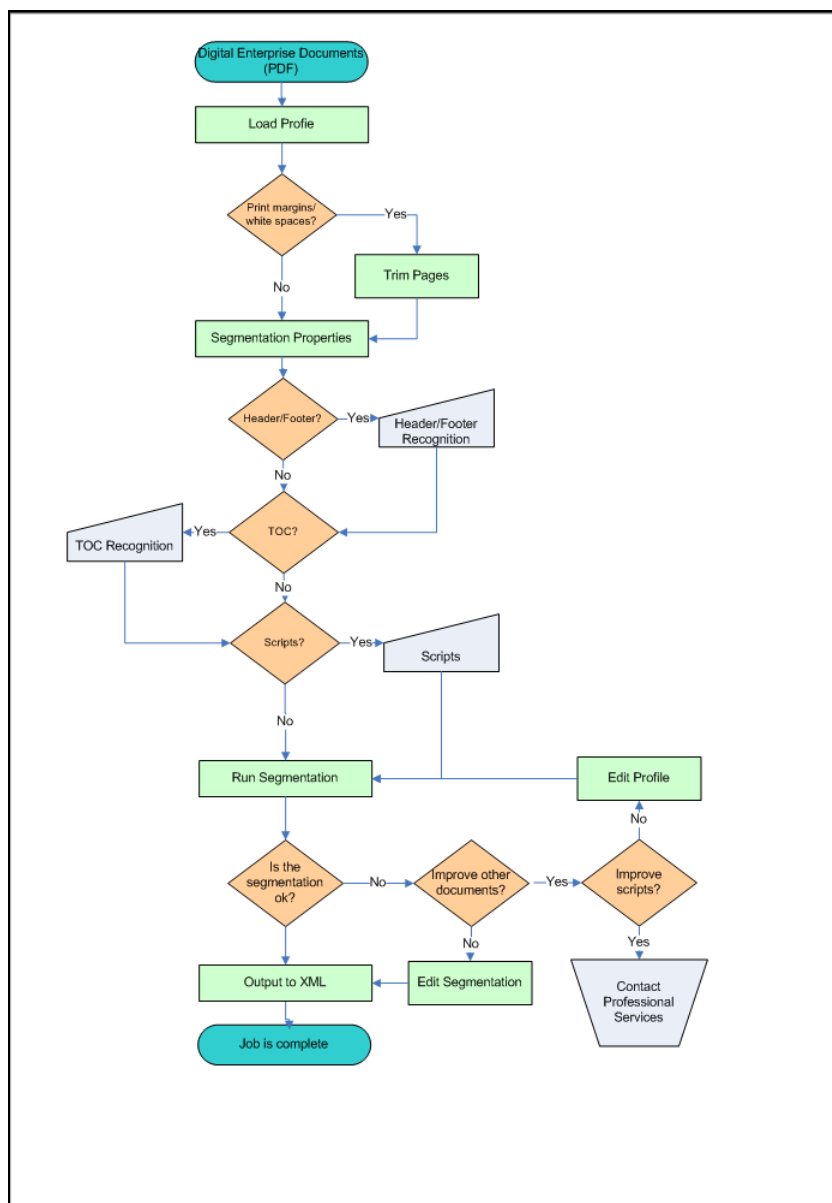


Figure 9 – Digital Enterprise Documents Workflow

1. Load the appropriate profile using **Load Profile** (page 136). This profile was created by Olive Professional Services and already includes the publication's characteristics.
2. Use **Trim Pages** (page 45) to trim the pages of a document that has white spaces or print markings (bleed).
3. Configure the segmentation properties using **Segmentation Properties** (page 146):

- Check the **Header/Footer** checkbox if you have headers/footers in your document. XML Distiller will identify the headers and footer and not include them in the segmentation.
 - Check the **TOC** checkbox if the document has a table of contents. XML Distiller identifies the TOC, segments each entry, and links it to the chapter titles.
 - Check the **Scripts** checkboxes if the segmentation process is to use scripts, such as extraction photo credits. The script is activated during the segmentation.
4. Run segmentation using **Run Segmentation** (page 148).
 5. If the segmentation is accurate, proceed to output using **Output to XML** (page 151).
 6. If the segmentation is not accurate, edit the segmentation using **Edit Segmentation** (pages 70 and 149).
 7. Editing segmentation improves the segmentation of the issue itself, but will not improve the segmentation for other issues. If similar issues or the same problems are likely to occur in other issues, improve the profile itself using **Edit Profile** (page 185).
 8. Contact Olive's Professional Services department to improve scripts or TOC recognition.
 9. Run the segmentation again upon editing the profile, and output to XML (steps 7 and 8 above) if it is accurate.

1.5.4 Scanned Enterprise Documents Workflow

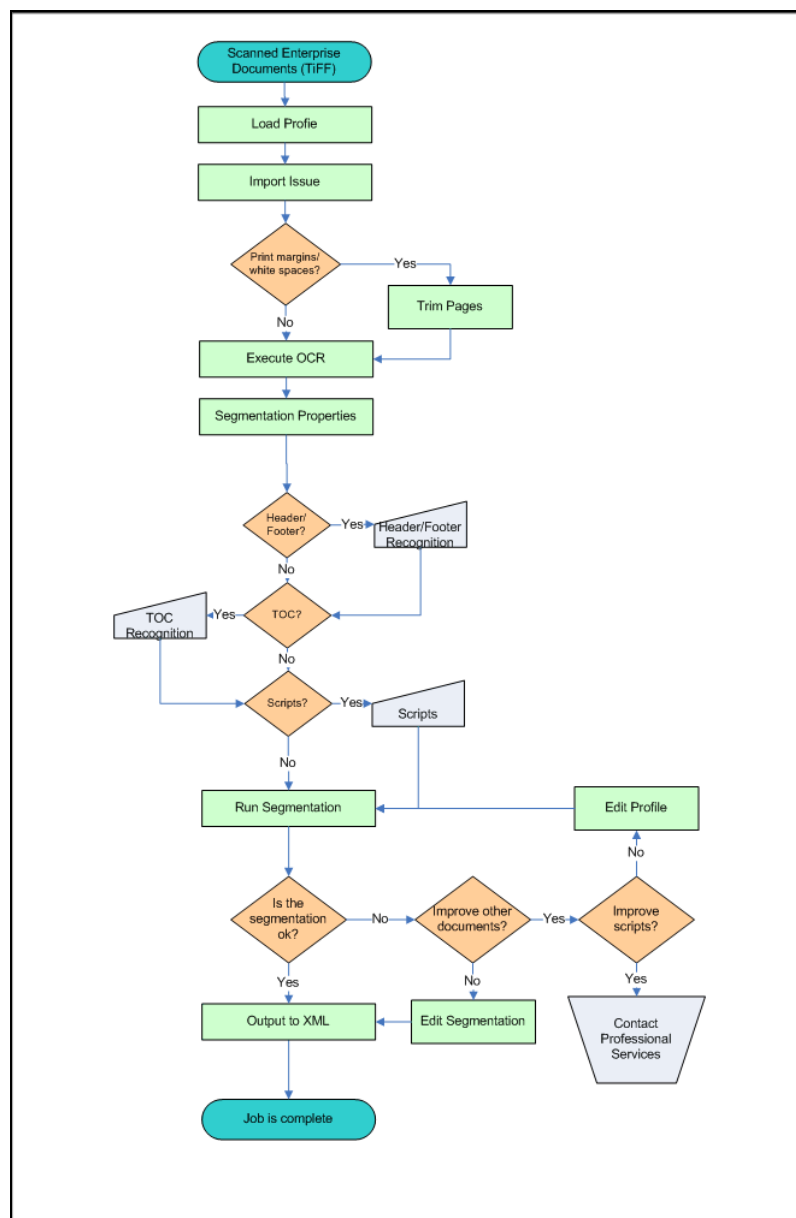


Figure 10 – Scanned Enterprise Documents Workflow

1. Load the appropriate profile using **Load Profile** (page 136). This profile was created by Olive Professional Servicees and already includes the publication's characteristics.
2. Import the page images (TIFF files) using **Import Issue** (page 138).
3. Trim the pages with white spaces or print markings (bleed) so that it does not include them using **Trim Pages** (page 45).
4. Run OCR using **Execute OCR** (page 143).

5. Configure the segmentation properties using **Segmentation Properties** (page 146):
 - Check the **Header/Footer** checkbox if you have headers/footers in your document. XML Distiller will identify the headers and footer and not include them in the segmentation.
 - Check the **TOC** checkbox if the document has a table of contents. XML Distiller identifies the TOC, segments each entry, and links it to the chapter titles.
 - Check the **Scripts** checkboxes if the segmentation process is to use scripts, such as extraction photo credits. The script is activated during the segmentation.
6. Run segmentation using **Run Segmentation** (page 148).
7. If the segmentation is accurate, proceed to output using **Output to XML** (page 151).
8. If the segmentation is not accurate, edit the segmentation using **Edit Segmentation** (pages 70 and 149).
9. Editing segmentation improves the segmentation of the issue itself, but will not improve the segmentation for other issues. If similar issues or the same problems are likely to occur in other issues, improve the profile itself using **Edit Profile** (page 185).
10. Contact Olive's Professional Services department to improve scripts or TOC recognition.
11. Run the segmentation again upon editing the profile, and output to XML (steps 7 and 8 above) if it is accurate.

1.6 XML Distiller Toolbar Reference

XML Distiller adds a number of icons to the **Adobe Acrobat** toolbar:



Button	Tool Name	Description
	Page Size Validation	Show results of the validation of page sizes. See <i>Page Size Validation</i> on page 40.
	Edit Segmentation Tool (Also activated by pressing the <E> key)	Activates the Edit Segmentation Tool. See <i>Segmentation Editing</i> on page 70.
	Page Preparation Tool	Activates the Pre-segmentation features. See <i>Page Preparation</i> on page 58.

	Predefine Areas Tool (Also activated by pressing the <P> key)	Activates the Predefine Areas Tool. See <i>Predefining Areas</i> on page 54.
	Learning Mode (Also activated by the <M> key)	Show/Hide the Learning Mode Console. See <i>Improving Profiles with Learning Mode</i> on page 198
	Zoom sub-mode (Also activated pressing the <Z> key)	Activates the Zoom sub-mode within the Edit Segmentation Tool. See <i>Zooming and Previewing</i> on page 74.
	Primitive Tag sub-mode (Also activated by pressing the <T> key)	Activates the Primitive Tag sub-mode within the Edit Segmentation Tool. See <i>Viewing and Modifying Primitives</i> on page 75.
	View Text Command (Also activated by pressing the <V> key)	Activates the View Text Command when a primitive is highlighted. See <i>Viewing and Modifying Primitives</i> on page 75.
	Recognize Ad Command (Also activated by pressing the <R> key)	Activates the Recognize Ad Command for highlighted ads. See <i>Viewing and Modifying Primitives</i> on page 75.
	QA Tool Command	Activates the QA Tool. See QA Tool for Scanned Periodicals on page 112.

2 Step-by-Step Periodicals Processing

This chapter contains important information about the various processing steps. These steps can be run sequentially via menu commands, and they are also the building blocks of an XML Distiller Job for periodicals. It is recommended that you become familiar with this chapter even if you are planning to do all periodicals processing via the Job Manager.

If you have not yet done so, read the Introduction chapter for a comprehensive comparison between Periodicals and Enterprise Documents.

Refer to the two flowcharts on page 25 in the Introduction that describe the workflows for processing scanned and digital periodicals.

2.1 Overview

There are some publication components that require an operator to use the various XML Distiller tools to manually intervene, as a supplement to the automatic segmentation. This may apply to ads, tables and other periodical elements, in both born-digital and scanned material. Scanned material (such as microfilm) may also require fine-tuning and quality assurance.

One issue mode is used to see how the system processes an issue, one step at a time, and make adjustments at appropriate points in the process. It is equivalent to running a single job in the Job Manager. It allows the user to better understand how the processing works, make corrections to the interim results when necessary, and output the final results.

Sample issues is another use for this mode when a customized profile is used on a small sample, before processing a larger batch using the same settings.

When XML Distiller is run step-by-step, some or all of the chapter's sections will apply:

- | | |
|-------------------------------|--|
| ■ Loading a Profile | ■ Style-Based Segmentation |
| ■ Importing an Issue | ■ OCR for Scanned Periodicals |
| ■ Page Size Validation | ■ Publishing Output |
| ■ Image Structure Recognition | ■ QA Tool for Scanned Periodicals |
| ■ Pre-Segmentation Steps | ■ Removing the Olive Layer (Clean PDF) |



Regular XML Distiller users should review the shortcuts as listed in the **Shortcuts** tab of the Preferences dialog (**Olive** menu >> **Preferences** item).

2.2 Loading a Profile

This step is mandatory in virtually all workflows. It is not needed only when checking the results of using the default settings in the XML Distiller.ini file on your document.

► **To load a profile:**

1. In the **Olive** menu, select **Profile** and then **Load**.

The **Select Profile** dialog box appears. It is identical to the regular Acrobat browse function, but is used to select a profile and not to open a PDF.

The **Select Profile** dialog displays all the profiles (.ini files) saved in the \Acrobat\plug-ins directory. You can select one of the default “starting-point profiles” stored, or browse to any directory in the file system to select a profile.

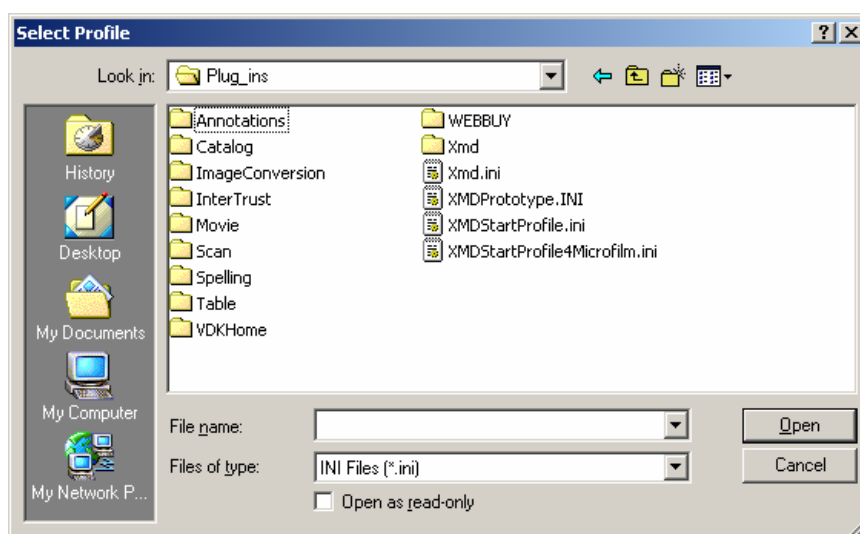


Figure 11 – Load Profile



Select Profile opens to the last folder used, and may not necessarily open to the default directory if you have already loaded a profile (or a PDF, using Acrobat **Browse**) in this session.

2. Select the required segmentation profile (.ini file) and click **Open**.

The name of the loaded profile appears on the title bar, following the issue name.

All the **Olive** menu commands become available.



Upon loading the profile, proceed preparing the pages for segmentation, and then to the segmentation itself. See *Pre-Segmentation Steps* on page 42 and

Step-by-Step Periodicals Processing

Importing an Issue

Style-Based Segmentation on page 61.

You may also edit the loaded profile first. The **Profile** menu provides the optional **Editor** and **Open** features, as described in *Editing/Improving Profiles* on page 185.



For periodicals ("ActivePaper/AP") mode, set the parameter **EnterpriseMode** to 0:

```
[Global]
EnterpriseMode=0
```

You can also update the profile dynamically. This option is in the manual segmentation process and performed in Learning Mode. See *Improving Profiles with Learning Mode* on page 198.

Olive provides the following default Profiles to process periodicals:

Profile	Filename	Description
Scanned Microfilm	XMDStartProfile4Microfilm.ini	Default profile for processing of periodicals stored on microfilm reels.
Scanned Newspaper		Default profile for processing of printed periodicals that need to be scanned
Digital Newspaper		Default profile for processing of born-digital periodicals

2.3 Importing an Issue

This step is relevant for scanned publications. A set of image files are imported to create the PDF file that is processed by XML Distiller. The naming conventions and/or directory structure of the input files can be used to extract relevant metadata such as date and page-sequence information.

► To import files for an issue:

1. Click the **Olive** menu, select the **Import/Export** submenu, and select **Import Issue**. The **Configure Import Issue Parameters** dialog opens.
2. Specify the source file information in the first tab. Specify additional optional import options and page-size validation parameters in the second and third tabs respectively.

2.3.1 Files to Import

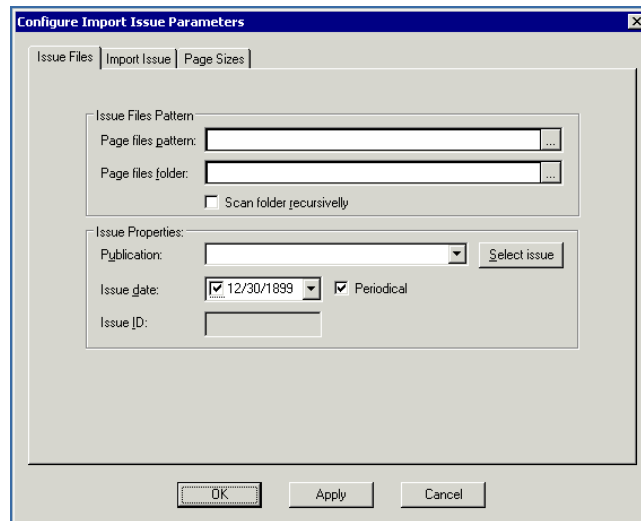


Figure 12 – Import Issue Parameters, Issue Files Tab

This tab contains the following fields and buttons:

Page files pattern	Defines the file naming convention used to find issues in the file system and deduce their meta data.
Page files folder	Defines the folder in which the source files are stored.
Scan folder recursively	Searches all subfolders of the Page files folder for PDFs that conform to the file naming convention.
Select issue	Allows you to select a set of files as an alternative to specifying the information in the Publication and Issue data fields. If a file naming convention is specified in Issue files pattern (the top section) and files are specified here, the "metadata" found in the filenames using the naming convention is used to populate Publication , Issue data and other metadata fields (see Figure 14 below).
Publication	Publication name - typed in or selected from a list.
Issue date	Opens a calendar to specify the issue date.
Periodical	Indicates if the issue is a periodical (should be associated to a date) or not associated to any date.
Issue ID	Relevant for documents receiving an ID not associated with a real date (non-periodical documents).

Step-by-Step Periodicals Processing

Importing an Issue

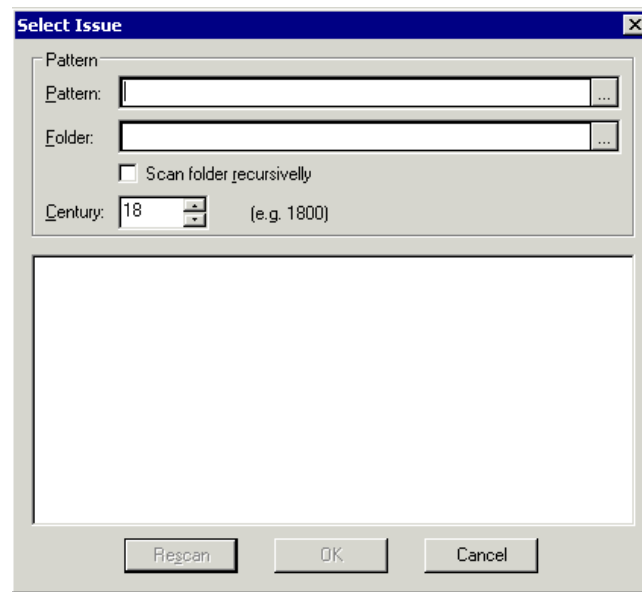


Figure 13 – Select Issue

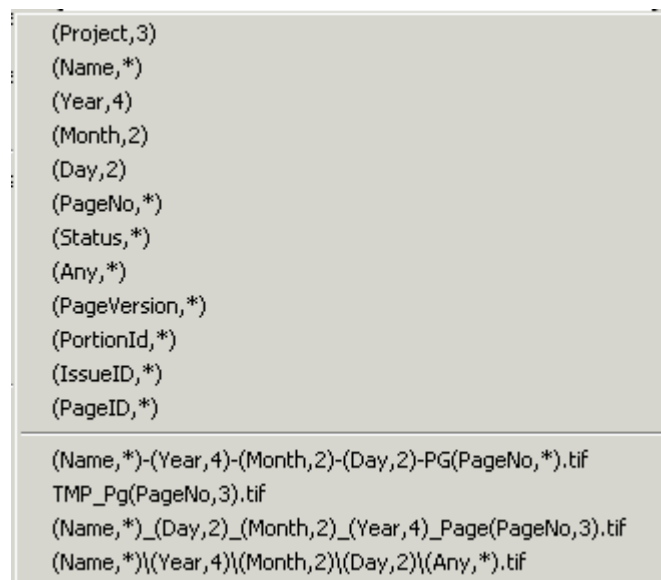


Figure 14 - Filename Patterns and Pattern Components

2.3.2 Import Options

This tab contains the fields customize several import features for a document or periodical issue:

- Page Detection
- Image De-Skewing
- Page Pre-Processing
- Image Cropping

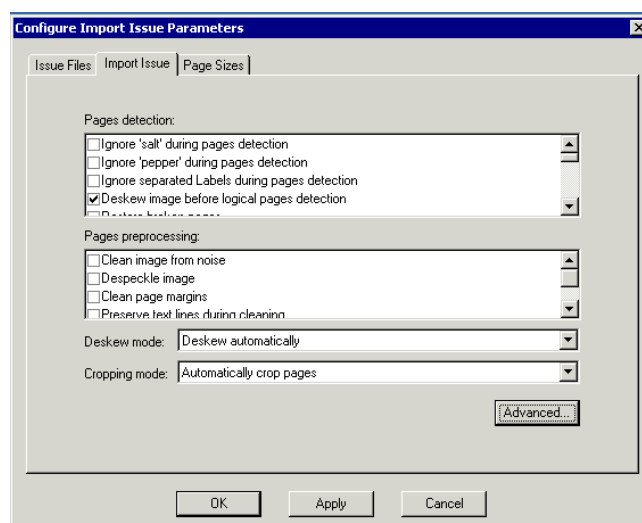


Figure 15 – Import Issue Parameters, Import Issue Tab

2.3.3 Page-Size Validation Parameters

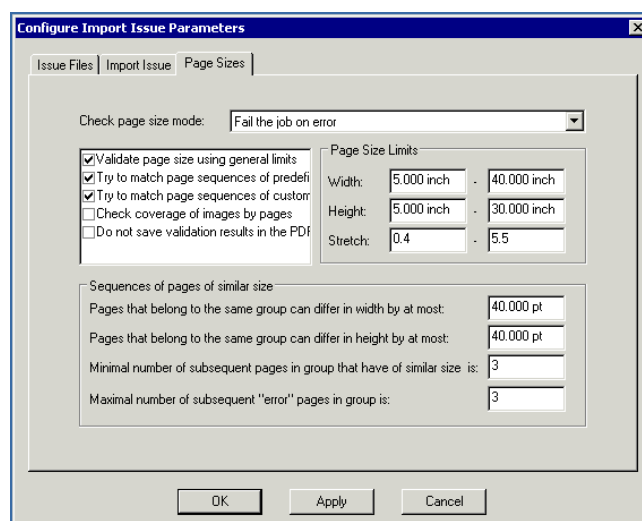


Figure 16 – Import Issue Parameters, Page Sizes Tab

This tab is relevant for publications where the sizes of one or more sets of pages must be consistent, and contains the following fields:

- **Check page size mode** checks for consistent page sizes, and the action to take when encountering errors.
 - Checklist of options on how validation is done and if the results are be saved in the PDF.
- **Page Size Limits** sets the page's width, height and stretch ratio ranges.
- **Sequence of pages of similar size** constrains adjacent similar-sized pages when identified.

2.4 Page Size Validation

Check for determined page size inconsistencies in the imported document.

► **To view page size validation results:**

1. Click the **Olive** menu, select **Image Import/Export**, and select **Validate Issue Page Sizes**.

XML Distiller reviews the page sizes in the current file, without displaying results.

2. Use the toolbar button () , or Click the **Olive** menu, select **Image Import/Export**, and select **View Page Size Validation Log**.

A validation results window opens displaying the components illustrated below.

3. Double-click on a page icon to view the page's image details.

Page	Section	Description
12	1	Page is too narrow
21	1	Page is too wide

Figure 17 – Page Size Validation Summary Table

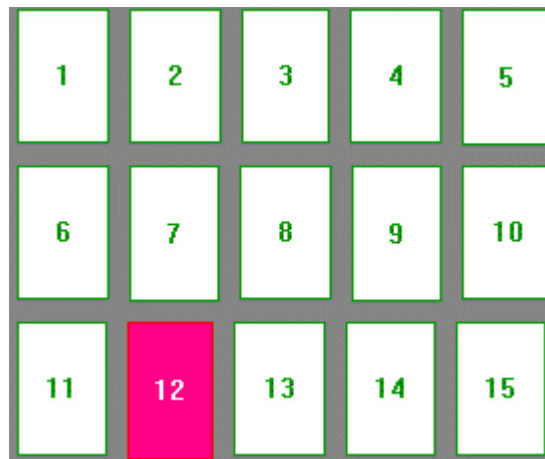


Figure 18 – Page Icons

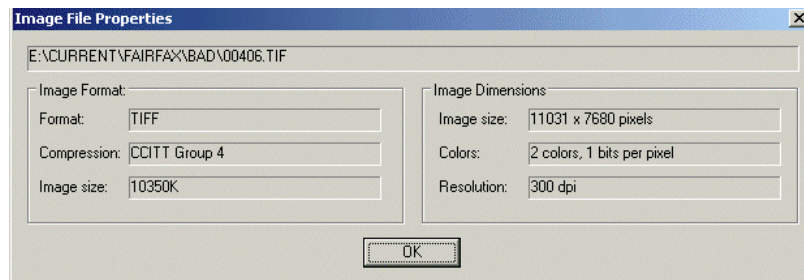


Figure 19 – Page Image Details

2.5 Image Structure Recognition

This stage is part of the process of preparing scanned documents for automatic segmentation.

Users rarely examine the Results of this process. If Segmentation is erroneously recognizing text as images, use this step to check if the problem occurred at this point in processing.

The **Structure Recognition** menu has the following menu options:

- **Analyze Image Layout** runs the structure analysis process to identify lines, text areas and image areas on the document.
- **Visualize Image Layout** displays layout markings as an overlay on the document's contents:
 - **Green:** horizontal or vertical line
 - **Blue:** single line of text
 - **Magenta:** more than one line of text, where specific lines cannot be identified

- **Orange:** high-level image, may also contain smaller **red** sub-images.
- **Customize Image Layout** is currently used by Olive Professional Services only.
- **Edit Image Layout** displays additional component details by double-clicking on any component.
- **Clean Layout Visualization** clear layout markings, leaving only the document contents.

2.6 Pre-Segmentation Steps

When the document has been opened, and the required profile is loaded, the next step is preparing the document's pages for segmentation.

Each part of the Pre-Segmentation process can be accessed by the **Olive** menu's **Pre-Segmentation** menu. The **Pre-Segmentation** menu contains the following items:

- | | |
|---------------------|---------------------------------|
| ■ Combine Pages | ■ Create Ad Frame from Manifest |
| ■ Split Pages | ■ Check Ads from Manifest |
| ■ Trim Pages | ■ Predefine Areas |
| ■ Join Double Pages | ■ OCR Predefine Areas |
| ■ Ignore Pages | ■ Redistill. |
| ■ Import Page IDs | ■ |

See also *Page Preparation* tool on page 58, which offers alternative interaction methods for several functions when preparing a page or pages for segmentation.

2.6.1 Combining and Splitting Pages

After reading this section, refer also to Merging and Splitting Pages using the *Page Preparation* tool (page 59).

Combining Pages

A single PDF containing the entire periodical is required to perform segmentation and other operations. The Combine operation automatically collates several single-page PDF files into a single file containing the entire periodical.

If the entire periodical is available as one PDF file, the Combine operation is not necessary. If, however, the entire periodical is composed of ninety-six single-page PDF files, the Combine operation save the time of manually piecing together the issue with Acrobat's **Insert Pages** function.

The **Combine Pages** function works according to naming conventions defined in the profile. These are the standard naming conventions used to name the single-page PDFs, such as *EVENINGED-170502-PG001-0001*.

► **To combine one-page PDFs into a single file:**

1. Verify that all the single-page PDFs comprising the periodical issue are in the same directory, or in the subdirectory structure associated with the naming convention (see File Pages Pattern on page 37).
2. Open the first page of the sequence in Adobe Acrobat.
3. Click the **Olive** menu, select **Pre-Segmentation**, and select **Combine Pages**.
When the Redistill function is used with the Combine Pages function, each input PDF file is redistilled before being combined.
4. If pages are missing in the sequence, a dialog appears asking to insert blank pages in their place or not. (Blank page insertion can also be handled via parameter from the profile.)



Failing to insert a blank page when there is a gap in the sequence of pages may cause problems at a later stage. For example, if page 3 is missing, and the pages are combined as-is, all pages from 4 onwards will have the wrong page number (4 will be 3, 5 will be 4, and so on). This may interfere with automatic recognition of continuations, which references absolute page numbers.

5. Once the operation has completed, a **browse** dialog appears. Select a **directory** and **filename** for your new PDF that contains the entire periodical issue.

The following are possible reasons for operation failure:

- The program cannot find any pages matching the naming conventions in the directory.
- PDFs are not unique – two or more PDFs contain the same page.

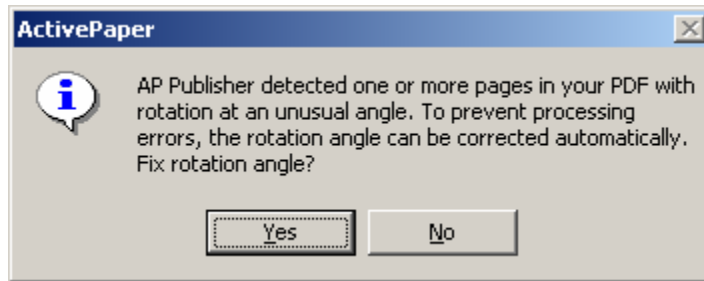
Splitting Pages

ActivePaper has the ability to automatically split and reorder PDF files used for publications such as tabloids, where a single PDF page may contain two tabloid pages that will be ordered as they are printed and not logically. For example, the first PDF page may contain the first page and last page of the tabloid section.



In the current version of XML Distiller, other Pre-Segmentation functions, such as **Trim Pages**, should be done prior to **Split Pages**.

The following alert may appear when opening a document that is to be split:



Click **Yes** to save the original, un-rotated document and show the new, rotated document. The next alert tells where the backup copy has been saved. The new working document is named `<source_name>_Rotated.pdf`.

Auto-rotation may cause problems when multiple PDF pages are combined into a single PDF document. It may be necessary to combine the pages without the ActivePaper plug-in. To do this, open Acrobat while pressing the “**Shift**” key.

► **To split and reorder the PDF, do the following:**

- Click the **Olive** menu, select **Pre-Segmentation**, and select **Split Pages**.

Choosing this option launches the **splitting** procedure, which includes the following operations.

- Creating a new document from an active PDF file. Each page of a source document is split, and two pages are created in a target document.
- Saving the new document in the Working directory under the name `<source_name>_SplitT.pdf`.
- Opening the split files as active in Acrobat.

The size of the pages created can be managed by changing the dedicated parameters in the **Profile.ini** file. By default the pages are split vertically in the center.

The page order in the document created is as follows:

- The right side of page 1 is set as the first page from the beginning.
- The left side of page 1 is set as the first page from the end.
- The left side of page 2 is set as the second page from the beginning.
- The right side of page 2 is set as the second page from the end.

and so on...



An additional method for splitting pages can be used to maintain the original page sequencing: Page 1 for the first page, Page 2 for the second page, etc.

In the profile's `[Split Tabloid]` section, set `bPagesAreInCorrectOrder` to 1.

Arranging Split Pages

In the *Profile.ini* file, the parameter **bPagesAreInCorrectOrder**=0/1 is found in the [SplitTabloid] section.

- **1** means that double pages in the source PDF are created in the exact order as in the *XML Distiller Merge Double Pages* process. Single pages are inserted one after the other in the target PDF file after being split.
- **0** means that the "imposed" structure of double pages is in the source PDF. The default value of the Split Tabloid process is set to be one.

A related parameter is **bEraseEmptyPart**=0/1:

- **1** means that empty pages are not inserted into the target PDF after being split.
- **0** means that all the pages are inserted into the target PDF. The default value is 0.



XML Distiller sets the value for **bEraseEmptyPart** is set to 0 for **bPagesAreInCorrectOrder**=0, in order to provide correct page numbering.

2.6.2 Trimming Pages

Trim Pages

This operation trims the all or selected pages to exclude the print margins and unnecessary white space around the page.



This operation is similar but not identical to the setting of a margin rectangle in the Profile Editor. Setting a margin rectangle does not physically crop the Acrobat pages; rather, it crops the page images created in the Warehouse (Repository) during the Output to XML operation. The Trim Pages operation is different in that it **does** physically crop, or reduce the size, of the Acrobat pages.

► **To trim pages:**

1. Click the **Olive** menu, select **Pre-Segmentation**, and select **Trim Pages**.

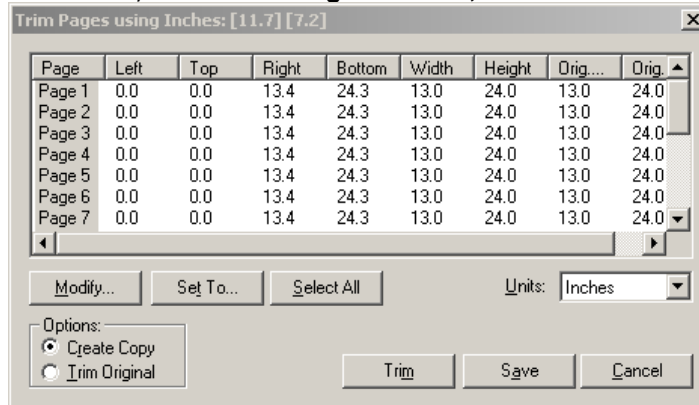


Figure 20 – Trim Pages

The window that opens allows you to select the pages to be trimmed and set the trim frames.

2. Click the **Units** drop-down list and select the measuring units.
3. In the **Options** section, select **Create Copy** to trim a copy of the active document,
or
Select **Trim Original** to trim the current document.

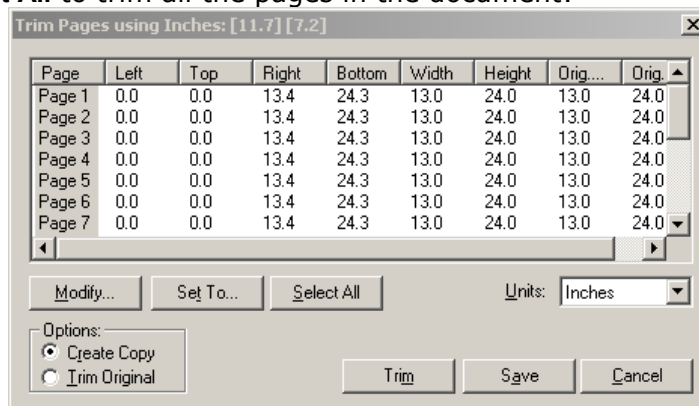


If you made changes to the PDF file, a prompt asks you save the document after pressing **Trim**. The document must be saved for Trim to be performed. Pressing **No** cancels the Trim operation.

4. Select each required page by clicking on it in the **Page** column. Press **Shift** and click to select consecutive pages or **Ctrl** and click to select more than one page.

or

Click **Select All** to trim all the pages in the document:



- Click **Modify** to change the size of the trimming frame.

The following window opens:

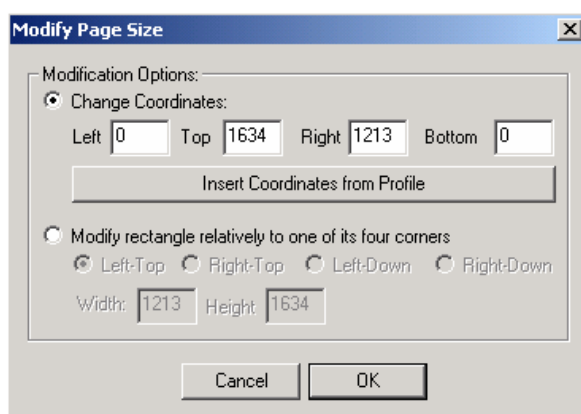


Figure 21 – Modify Page Size

- Select the **Change Coordinates** radio button (default). The outermost Left, Top, Right and Bottom positions of the trimming frame appear in the text boxes.
- To change the trimming frame size to the previous one saved in the profile, click **Insert Coordinates from Profile**. The Left, Top, Right and Bottom values are entered from the profile.
- To modify the frame size, change the values in the Left, Top, Right and/or Bottom text boxes and click **OK**. The frame size changes and the values are updated in the current profile.

To expand the frame, increase the values in the Top and Right boxes and decrease values in the Left and Bottom boxes. Consider the following example:

- A frame with values of:
Left: 0, Top: 2000,
Right: 1000, and Bottom: 0
 - To expand the frame by 20 points in each direction, decrease the Left and Bottom values by 20, and increase the Top and Right values by 20. The result will be:
Left: -20, Top: 2020,
Right: 1020, and Bottom: -20.
- After page selection, the **Set To** button becomes active. Click it to set the trimming frame size to a standard page size. The following list of page sizes appears:

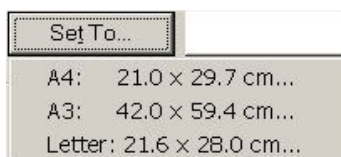


Figure 22 - Set To Standard Page Sizes

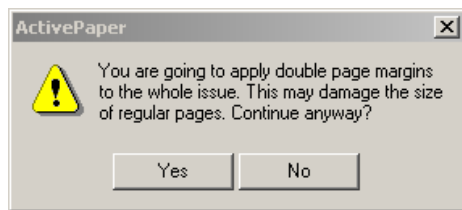
Click the required page size. The **Modify Page Size** dialog box appears with the dimensions of the chosen page size. Click **OK** to confirm or apply further modifications and then click **OK**.

10. To change the size of the trimming frame relative to a corner, click the **Modify rectangle relatively to one of its four corners** radio button.

Select one of the corners (Left-Top, Right-Top, Left-Down or Right-Down) and adjust the rectangle's dimensions by changing the values in the Width and Height text boxes. Confirm with **OK** to apply the changes.



Trimming pages after the Join Double Pages operation also affects the middle of each double page. Be careful **not to trim ALL the pages**. Trimming pages after Join Double Pages returns the following warning message:



Trim Pages operation cannot be performed on PDF files that are read-only.

11. In the **Trim Pages** dialog box, click **Save** to save the updated settings in the profile (note that the **Save** button does not save the PDF), and then click **Trim** to complete the operation.
Or click **Cancel** to abort the operation.



The pages of the current PDF are not trimmed, but rather a new PDF is created with the trimmed pages. Select **Save As** in Acrobat to save the new PDF file.

Trim

The **Trim Box** option can resize a single page or group of pages and can be applied to the active page, a range of pages, all pages, or even/odd pages.

(Units of a mouse cursor position can be set in the **Preferences** dialog **View and Edit** tab. See *Segmentation Visualization Preferences* on page 66.)

Selecting the **Trimming options** by first right clicking the mouse and then clicking on either the **Trim** or **Save** button has the same effect as the **Pre-segmentation** menu selection **Trim Pages**, except that performing trim pages using this method trims all the pages in the document.

2.6.3 Joining Double Pages

After reading this section refer also to Merging and Splitting pages using the *Page Preparation* tool (page 59).

This option joins a pair of pages into a double-page spread in a single Acrobat page, and may be useful when a page layout is spread over two pages, such as when an article heading or a picture starts on one page and continues on the next.

Before joining double pages, first find all the double pages in the document and mark them as such, using the **Double Pages** window.



There is no need to perform this operation on documents with no double pages.

► **To join double pages:**

1. Click the **Olive** menu, select **Pre-Segmentation**, and select **Join Double Pages**.

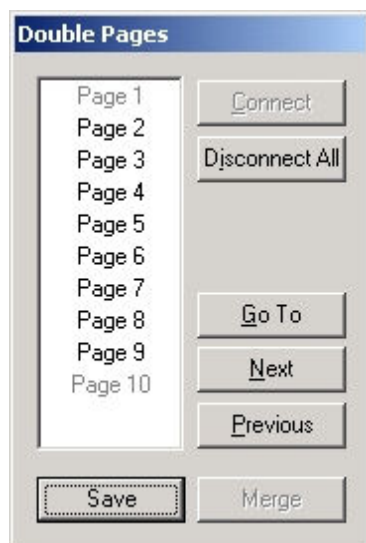


Figure 23 – Double Pages

2. In the window that opens, scroll down through the page list and click pairs of consecutive pages.
3. Click **Go To** to view the selected pair of pages to verify a double page spread.
4. Click **Connect**.
5. Upon connecting several double-page spreads, click **Next** or **Previous** respectively to view the next or the previous spreads.



The connected pages are marked red and blue in the issue and in the page list of the Double Pages window. The **Connect** button changes to **Disconnect**.

To cancel double-page joining, click **Disconnect**.

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To undo all Join Double Pages operations performed, click **Disconnect All** in the **Double Pages** window.

6. Scroll down through the document and repeat steps 2 - 4, until you have marked all the double pages in the document.



To mark an illogical (non-facing) pairs of pages as double pages, particularly with extracted document fragments, press the <Ctrl> key while clicking on a single page. Select both pages of the illogical page pair, and click **Connect**.

7. Click **Save** to close the **Double Pages** window, exit Double mode, and save the double page settings. These settings are saved in the document and will be available next time the PDF file is opened.



Double page processing is irreversible. The **Join Double Pages...** option is no longer available after clicking **Merge**. A new PDF file is created according to the selected double pages. If double-page marking errors are later discovered, you will have to reopen the original file, fix errors using the Page Preparation tool, and rerun double page processing. To save time, verify that all double pages are correctly marked before clicking **Merge**.

8. Click **Merge** to process the double pages.

A progress bar appears at the bottom left of your screen, and XML Distiller executes double page processing on the selected pages.

Each of the marked double pages merges into a single Acrobat page.



If the PDF is **read only**, an error message appears and the operation fails, although double page settings are not lost. Simply **remove** the “**read only**” attribute from the PDF file or copy it to a writeable drive, then open the **Double Pages** dialog again and press **Merge**.

A new PDF file containing merged double pages is saved in the working Distiller (XMD) directory, located directly under the Acrobat program directory. (Another folder can be specified during installation.) The file is named by adding the suffix **_DoublePages** to the original name. We strongly recommend that you review the document once again, to double-check that all pages have been connected properly.

Continue working with the new file in the working directory. You may immediately save the document to another working location using **Save As**.



A periodical issue that has no double pages will continue to work on the original file with the original filename, with no need to save a duplicate file.



Upon merging pages, Acrobat pagination no longer corresponds to pagination in the PDF document. The document contains one page fewer for each merged double page.

2.6.4 Ignoring Pages

This option marks pages to be ignored by automatic segmentation. This operation is optional, and is mainly intended to save time during automatic segmentation for pages that will later undergo manual segmentation.

Pages should be “**ignored**” when:

- Pages that do not contain useful entities.
- Pages that contain very large data tables (such as stock tables). Such tables may contain tens of thousands of text elements, which can influence the performance of the automatic segmentation. Some pages that contain large tables can take up to twenty minutes to segment automatically.

If a page contains a large data table as well as some useful entities, ignore the entire page to save time during automatic segmentation. These entities will not undergo automatic segmentation but can be manually segmented later. Tables can be segmented manually (see **Defining Large Tables** on page 95).



A “**useful entity**” is an entity that you wish to segment so that it will appear properly to the end user. This is a subjective definition that depends on your organization’s internal requirements and procedures. For example, if you have decided not to segment advertisements at all, all page containing only advertisements should be ignored.

► To mark ignored pages using the dialog box:

1. Click the **Olive** menu, select **Pre-Segmentation**, and select **Ignore Pages**.

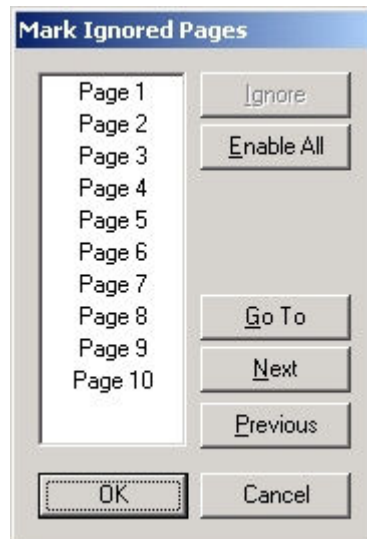


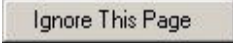
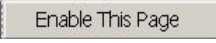
Figure 24 – Mark Ignored Pages

2. In the window that opens, scroll down through the page list and click any page.

Pre-Segmentation Steps

3. Click **Go To** to view the page and check if to exclude this page from automatic segmentation.
4. Click **Ignore** to ignore the selected page.
5. After several pages have been ignored, click **Next** or **Previous** respectively to view the next or the previous ignored page.

► **To mark ignored pages using the context menu:**

- Right-click on the required page.
 - If this page is not marked as ignored,  appears.
 - If the page is marked as ignored,  appears.

2.6.5 Importing Page IDs

When creating Ad Frames using a manifest, import Page IDs if the issue does not already contain them. This requires creating a text file that describes the page IDs for each PDF page in the issue.



Import Issue automatically creates Page IDs (see page 36).

When processing a Manifest, XML Distiller is given a list of PDF files to process in order. XML Distiller then looks for text files associated to these PDF files. These text files are named according to the following naming convention:

log_<name>.txt

Where <name> is the file name of the PDF.

The file contains a series of lines, each assigning a PageID to each page in the PDF file, and follows the following pattern:

Page <PageNumber> <PageId>

Where:

<PageNumber> is the page number in the PDF file (i.e., 01)

<PageID> is the PageID of the issue (e.g., A01_M2_EZ_DAILY_20020827.pdf)

Example

Here is an example of log_<name>.txt:

```
08/27/02 04:34 AM
Page 01 A01_M2_EZ_DAILY_20020827.pdf
Page 02 A06_M2_EZ_DAILY_20020827.pdf
Page 03 A07_M2_EZ_DAILY_20020827.pdf
```

Figure 25 - Caption

2.6.6 Creating Ad Frames from Manifest

This option allows **ActivePaper** to automatically generate Ad Frames from an external manifest file. This external file is created by modifying a file that is produced when your publication produces its ads. The manifest file describes how and where Ad Frames are generated in the publication.



Creating the manifest file is a part of the Preparation steps described below and in [Ad Frames from Manifest](#), on page 221 in the Appendix.

The manifest file contains a CSV (Comma-Separated Value) format table. Each table line represents one advertisement, and contains a unique PageID and coordinates of the Ad Frame in either millimeters, centimeters, or inches.



Before processing a manifest, make sure that you have adequately prepared for manifest processing. (See below and [Ad Frames from Manifest](#), on page 221).

Preparation

Depending on the software used, creating an advertisement manifest is a natural part of the printing process. Complete the following when preparing to process a manifest:

- Configure Manifest
- Update Profile/Settings
- Set PageId via log_<name>.txt File
- Metadata
- Create Manifest Table – Manifest_<name>.txt



Each of the above preparation steps is described in detail in **Ad Frames from manifest**, on page 221 in the Appendix.

Creating Ad Frames

Upon completing preparation, you are ready to create Ad Frames.

► To create Ad Frames from a manifest:

1. Click the **Olive** menu, select **Segmentation**, and select **Create Ad Frame From Manifest**.

A dialog box opens, asking the location of the manifest text file.

2. ActivePaper Manifest processing begins once the manifest file is located.

When complete, a window appears indicating the number of advertisements found (if any). These ads are marked with a colored frame on the PDF file.

Checking Ad Frames

To check if all ad text elements have been placed within an Ad Frame.

Pre-Segmentation Steps

- ▶ **To check Ad Frames from a manifest:**
 - Click the **Olive** menu, select **Segmentation**, and select **Check Ad Frame From Manifest**.

2.6.7 Loading a Manifest XML

An XML file received from an agency or other source can be loaded directly into the Distiller using this function.

- ▶ **To load an XML-formatted manifest file:**
 1. Click the **Olive** menu, select **More**, and select **Load Segmentation Data**.

A dialog box opens, asking the location of the manifest XML file.
 2. Upon specifying the manifest file, the dialog closes and the data is extracted for use by ActivePaper.

2.6.8 Predefining Areas

After reading this section refer also to Table and Picture Areas and

TOC, AdIndex and Page Ref Areas using the Page Preparation tool (page 59).

This option allows you to perform the following operations:

1. **Mark areas to be excluded from the automatic segmentation process.** This is the same operation as **Ignore Pages** (page 51), except that it allows defining part of a page (not an entire page) to be excluded from automatic segmentation.
2. **Predefine display ad areas.** This manually predefines areas for displaying ads, allowing automatic segmentation to recognize internal ad components. *This stage is not necessary when using a manifest.*



Predefining areas is optional. Perform one of the above operations before running automatic segmentation. The automatic segmentation will then account for the predefined areas.

► **To activate Predefine Areas mode:**

- Click the **Olive** menu, select **Pre-Segmentation**, and select **Predefine Areas...**, or click the **Predefine icon**  on the toolbar.

► **To exclude an area from the segmentation process:**

- In **Predefine** mode, draw a rectangle around the area.

The area receives an **Exclude** label and is excluded from automatic segmentation.

► **To include an area within a larger area that has been excluded:**

1. In **Predefine** mode, draw a rectangle around an area within an Exclude area.
2. Right-click anywhere on the frame and choose **Include** from the context menu.

The area receives an **Include** label and is included in automatic segmentation.

► **To manually predefine an area as a Table of Contents area:**

1. In **Predefine** mode, draw a rectangle around a table of contents (TOC).
2. Right-click anywhere on the frame and choose **TOCLocation** from the context menu.

The selection receives the **TOCLocation** label and is used in TOC analysis.

► **To manually predefine an area for an advertisement:**

1. In **Predefine** mode, draw a rectangle around an advertisement.
2. Right-click anywhere on the frame and choose **Ad Frame** from the context menu.

The selection receives the Ad Frame label:



Figure 26 – Ad Frame Labeling



You can split any Ad Frame into a number of separated frames by pressing the <I> key and drawing horizontal or vertical lines inside the selected Ad Frame. To turn off this mode press <I> again.



In Predefine mode, you can right-click any frame and use the context menu to switch any frame from an **Ad Frame** to an **Exclude** frame, or vice versa.

2.6.9 Running OCR on Predefined Areas

After reading this section refer also to *OCR on Predefined Areas* using the Page Preparation tool (page 60).

Upon specifying predefined areas for ads on scanned pages, you can now run Optical Character Recognition (OCR) on the text within the Ad Frames.

Ads often contain text that is included as a graphic or as part of a graphic. Text in graphic form cannot be searched, unless it is converted to searchable text by OCR.

For areas predefined by Ad Frames, OCR Predefine Areas identifying text in graphic form, applies Optical Character Recognition (OCR), and associates the text to the ad.

► **To activate OCR Predefine mode:**

1. Click the **Olive** menu, select **Pre-Segmentation**, and select **OCR Predefine Areas**.
2. A dialog box appears with a progress bar for the OCR process.

Upon completion of the OCR process, text that is in graphic format within any predefined Ad Frame will be selectable and searchable for the end user.



If OCR has already been performed on some of the pages, the document can be managed as a digital document, if enabled by the Profile. The following should appear in the profile:

```
[Global]
ScannedFlag=0
```

XML Distiller uses parameters from [OCROnAreas] to handle OCR text.

2.6.10 Redistilling a Publication

The Redistill function reruns the PDF distilling process. Redistill converts the active PDF file to a postscript file while performing certain changes, such lowering the resolution of high resolution of images in the original PDF file. Redistill then converts the postscript file back to PDF format.

There are two reasons to use Redistill.

- **Errors** – Errors appear when attempting to segment the PDF file.
- **Size** – The graphics in the PDF file are of high resolution, making the PDF file large and increasing the processing time .

► **To Redistill a PDF file:**

1. Click the **Olive** menu, select **Pre-Segmentation**, and select **Redistill**.
A dialog box appears, asking you to verify that you wish to redistill the file.
2. Click **OK**.
A window appears indicating the progress of the redistilling process.

When the redistilling process is complete, the progress window disappears, and you can further process the PDF file.

Redistill as Part of a Job

You can also run the Redistilling process using the Job Manager, by choosing **Redistillation** in the **From step** box, under the **Job** tab. (See *Job Tab* on page 164).

Redistill and Combine Pages

When running Combine Pages – the function to combine multiple single-page PDF files – redistilling is run if the following is specified in the profile:

```
[OnePagePDF]
PerformRedistillation=1
```

(See Combining Pages on page 42).

2.7 Page Preparation

This section provides an overview of the **Page Preparation** tool, and discusses the following facets of page preparation:

- Merging and Splitting Pages
- TOC, AdIndex and Page Ref Areas
- Page Attribute
- OCR on Predefined Areas
- Table and Picture Areas
- Segmentation on Parts of the Publication

2.7.1 Page Preparation Tool Overview

The **Page Preparation** tool  is used during Pre-Segmentation to perform operations on only one page or a subset of pages. It is one of the Olive tools added to the Adobe Acrobat toolbar.

When selecting a page (clicking on it), it becomes the target page. When right-clicking, the following items appear in the context menu:

- Trim
- Attribute
- Merge
- OCR
- Split
- Redistill.

By selecting **Trim**, **Merge** or **Split**, users open a corresponding window.

While one of these dialogs is open, a common right-click context menu is available with the following items:

- **Set Page** sets the visible page as target.
- **Zoom In and Zoom Out** changes the zoom level.
- **Single Page, Continuous or Continuous-Facing** changes the view layout.

2.7.2 Merging and Splitting Pages

Merge

Two distinct pages can be merged to form either a single page or a double page. Double pages cannot be merged with any additional pages.

No page margins are used when double pages are merged using the Page Preparation tool. In the **Join Pages** parameters box, the vertical shift moves the right page up or down relative to the left page.

Selecting the **Merge options** by right-clicking on the page and then clicking on either **Merge** or **Save** is the same as in **Pre-segmentation > Merge Pages**, except that this method merges all the pages in the document.

Split

Individual or groups of pages can be split by setting split separator line. The split pages are inserted into an active copy document instead of into the original document. The new pages are defined as single pages.

The default split separator for a double page is a line between the left and right parts of a page.

Selecting **Split options** by right-clicking and then selecting either **Split** or **Save** is the same as the Olive menu's **Pre-segmentation Split Pages** item, except that this splits all pages in the document and inserts the resulting pages sequentially.

2.7.3 Page Attribute

Right-click a page and select **Attribute**. The submenu contains the following:

- **Set Single** converts a double page to a single page.
- **Set Double** converts a Single Page or Double page to a double page, allowing the user to define location of the separator between left and right parts.
- **Ignore** sets a page that is ignored by the Auto Segmentation process.

2.7.4 Table and Picture Areas

Predefined areas include a "**Table Area**" and a "**Picture Area**".

The Distiller process builds a **Picture** entity in a predefined "Picture Area" and an **Article** entity in a predefined "Table Area".

A new Entity property is set for Articles that are predefined as a "Table Area". This property does not appear in XML Distiller and is not written to the XML file. It is used in ViewPoint when combining Entities after segmentation.

2.7.5 TOC, AdIndex and Page Ref Areas

Three Predefined Areas may be found in the Profile:

- **Ad Index**
- **Page Ref.**
- **TOC**

All three are processed during Segmentation.

```
[PreDefAreas]
AreasTypes=,"Ad Index",TOC,"Page Ref"
PageIndexByAnalyzer="Ad Index",TOC
PageReferByAnalyzer="Page Ref"
```

XML Distiller looks for Page Links inside the three Predefined Areas above. If an **"Ad Index"** or **"TOC"** area contains one or more **"Page Ref"** areas, then the Distiller looks inside internal **"Page Ref"** Areas only. This may be useful if the **"Ad Index"** contains columns with Ad Numbers that cannot be automatically distinguished from Page Links.

If the Page Link is recognized, its box is extended to the left and right sides of the corresponding **"Ad Index"** or **TOC** area.

The following Page Links are to be recognized:

- Text that matches one of the patterns in the [Portion] section.
- Text associated with one of the existing Page Labels in a document.

2.7.6 OCR on Predefined Areas

Selecting **OCR** from the menu performs **Optical Character Recognition** (OCR) on Predefined Areas in the active page or in a range of pages.

Any previous OCR text is removed from these pages.

2.7.7 Segmentation on Parts of the Publication

This function runs Segmentation on an active page or on a range of pages. Previous segmentation is removed from these pages.



This segmentation does not create continuation links between entities.

2.8 Style-Based Segmentation

Before reading this section, familiarize yourself with the segmentation concepts introduced in Chapter 1.

2.8.1 Process Overview

Once all Pre-segmentation has been completed, you can run Segmentation. In this step entities are identified and sequenced, and the set of elements associated with each article are grouped, so that a search can focus on the entire article that satisfies the search criteria.

Automatic segmentation generally takes between five and ten seconds per page. Highly complex or very large pages may take longer.

Automatic segmentation accuracy depends primarily on a periodical's complexity and layout. Segmentation inaccuracies occur when the artificial intelligence engine cannot resolve small ambiguities in the layout of the periodical. These anomalies are corrected by manual segmentation editing (see the subsections *Segmentation Visualization Preferences* and *Segmentation Editing*), though consistently recurring errors can usually be prevented by fine-tuning your profile.

Segmentation information can be saved separately from the PDF contents. See the subsection *Saving and Restoring Segmentation*.

Setting Segmentation Properties

► To select segmentation mode:

1. Click the **Olive** menu, select **Segmentation**, and select **Segmentation Properties**.

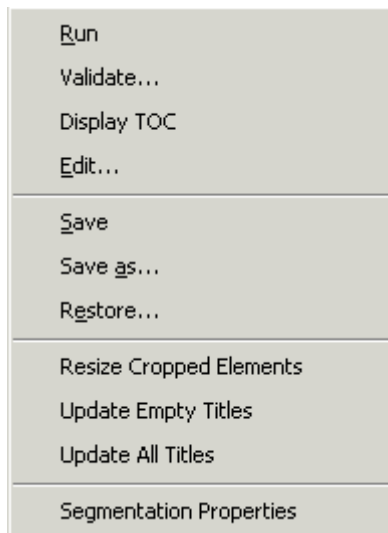


Figure 27 – Segmentation Menu

The **Segmentation Properties** dialog opens, presenting the following options:

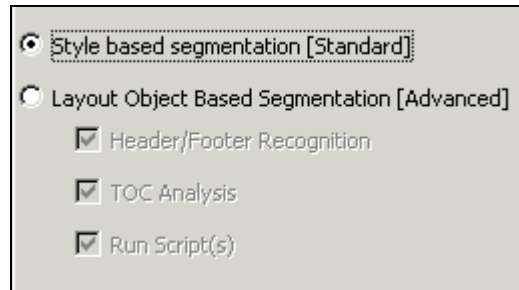


Figure 28 – Segmentation Properties

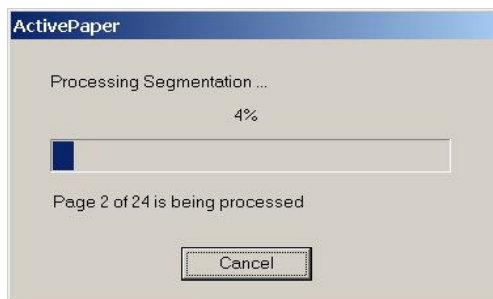
2. Select **Style based segmentation [Standard]** for most periodicals. (See page 145 for details on Layout Objects.)

Running Segmentation

This option starts automatic segmentation of the entire periodical.

- **To run segmentation:**
 - Click the **Olive** menu, select **Segmentation**, and select **Run**.

The following progress bar is displayed:



Note that **Run** erases all existing segmentation, replacing it with the automatically generated segmentation.

Clicking **Cancel** button on the **progress** bar stops the segmentation process. Segmentation results up to that point remain.

Validating Segmentation

This option validates links (URL, Email or See Page links) according to what is checked in the Validation Dialog Box.

This procedure is performed only after segmentation is complete.

► **To run validation:**

- Click the **Olive** menu, select **Segmentation**, and select **Validate**.

The **Validate Segmentation** dialog box appears with the following check boxes:

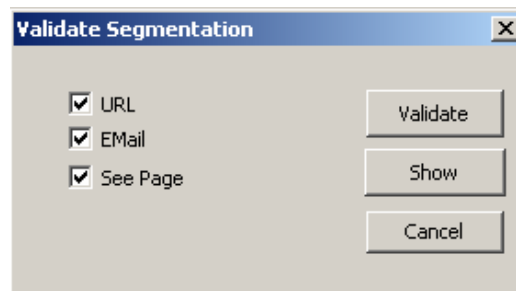


Figure 29 – Validate Segmentation

URL – Check this to validate URLs according to regular expressions and by checking target connections. The latter operation requires an active Internet connection.

Email – Check this to validate email addresses according to regular expressions (one “@” character, legal domain string).

SeePage – Check this to validate “see pages.” Validation is performed when the target page in the document is found.

Validate – Runs link validation according to the check boxes in the dialog box.

Show – Displays a list of invalid links found in the last validation.

Cancel – Cancels the validation process.



Step-by-Step Periodicals Processing

Style-Based Segmentation

► To validate links:

1. In the **Validate Segmentation** Dialog Box, click **Validate**. The system validates all necessary links.

The **Job Execution** window appears:

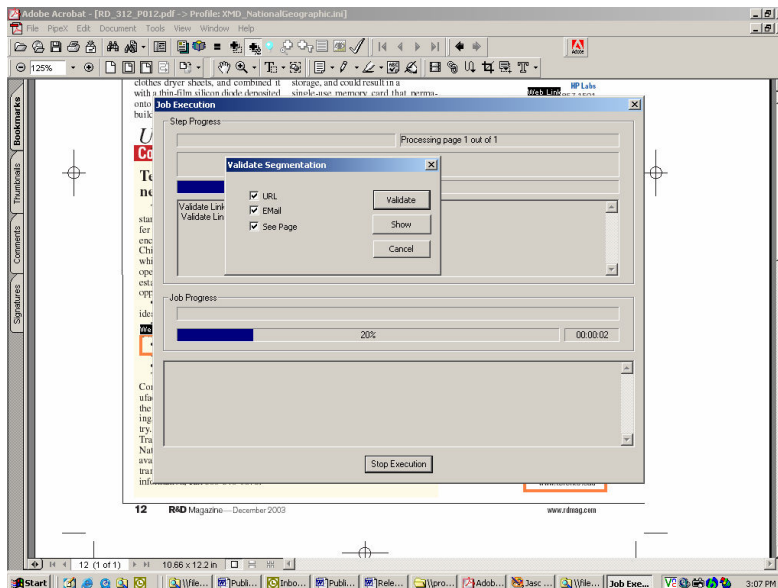


Figure 30 – Job Execution

Any problems with the links that have been validated opens a **Validation Records** Window, showing the problematic links. Invalid links can be edited by double-clicking a link in the **Validation Records** Window.

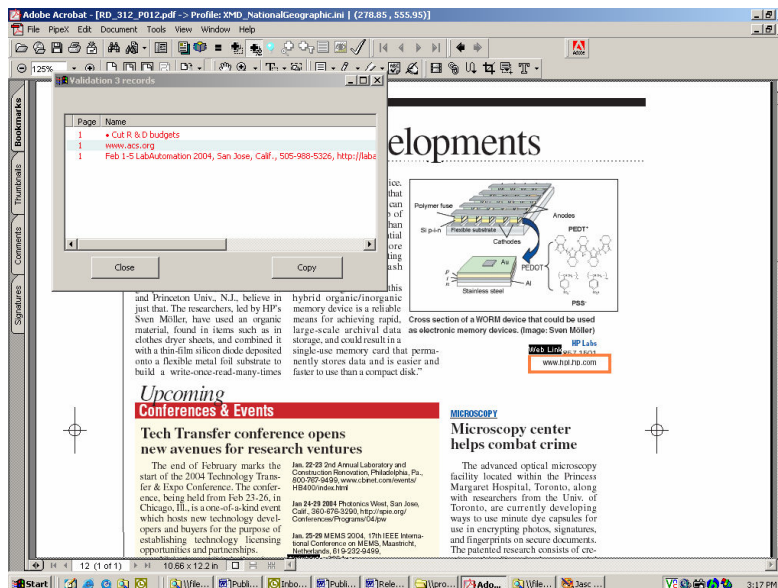


Figure 31 – Validation Records

Selected items in the list are indicated with a "•".

For example:

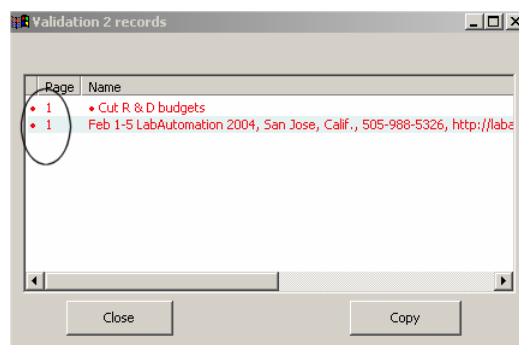


Figure 32 – Validation Records Selection

2. Click **Copy** to copy the links to the Clipboard for editing in MS Word or Notepad.

Displaying the TOC

This option validates Table of Contents links.

A list of Articles and Ads appears. Each record displays a Page Number in the PDF file and a Title.

► To validate TOC links:

1. Click the **Olive** menu, select **Segmentation**, and select **Display TOC**.

The **Display TOC records** window appears.

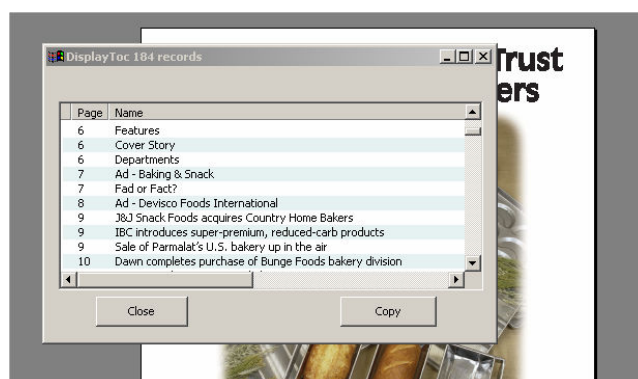


Figure 33 – Display TOC Records

You can navigate to a page by double-clicking its page number or Title in the **Display Toc records** Window.

A "•" marks items in the list that have been clicked.

2. Use the **Copy** button to copy the **TOC** to the Clipboard for editing using any text editor such as Microsoft Word or Note Pad.

Editing Segmentation

Segmented publications can be reviewed to identify and correct segmentation errors. See the section on *Segmentation Editing* (page 70) for detailed instructions.

2.8.2 Segmentation Visualization Preferences

The **Preferences** dialog defines many aspects of segmentation visualization. If the segmentation visualization is difficult to work or straining the eyes, experiment with different options until you find the settings you like best.

Open the **Preferences** dialog using the shortcut key <C>.

General tab

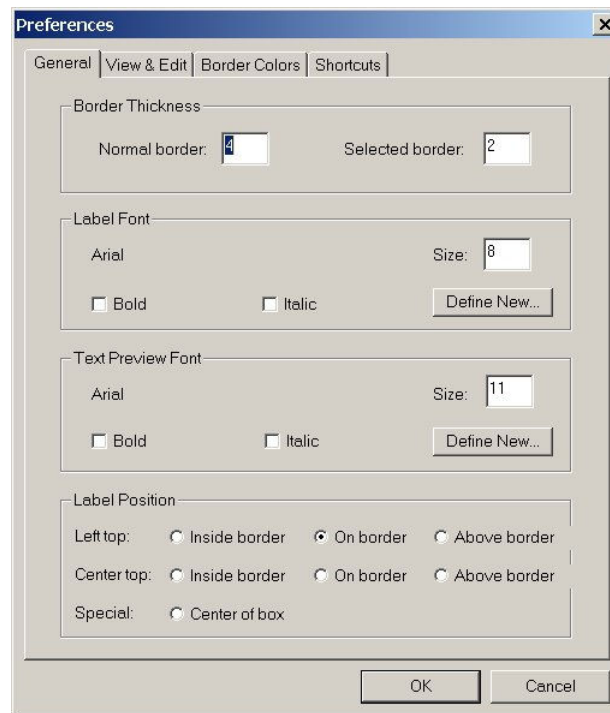


Figure 34 – Preferences: General Tab

Normal border defines the width of the colored borders around primitives (in pixels).

Selected border defines the width of the black inner border on the currently selected primitive.



If the **Selected border** is equal to or larger than the **Normal border**, the color of a selected primitive will not be visible, and its entity will not appear.

Label Font defines the font used for primitive labels.

Label Position defines where labels appear on each primitive.

View & Edit tab

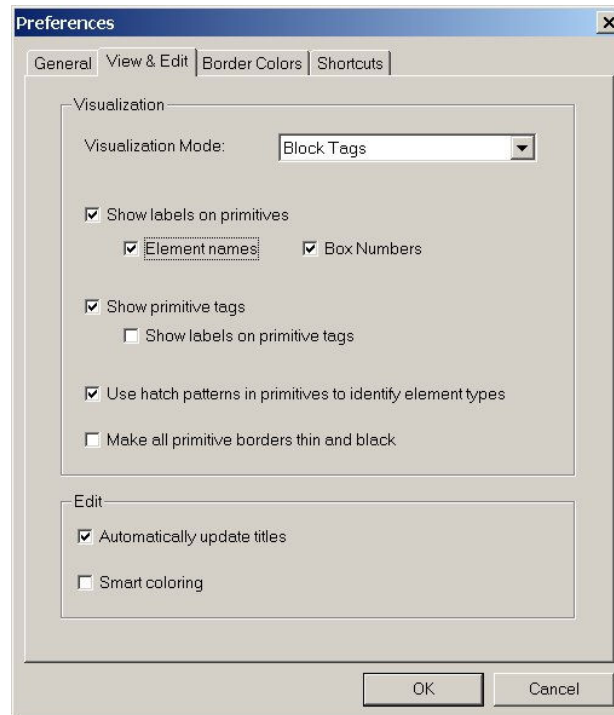


Figure 35 – Preferences: View & Edit Tab

This tab contains a variety of display and editing settings.

Visualization defines the default label settings and turns primitive surface patterns (hatching) On and Off.

Edit defines how segmentation visualization is displayed.

► **To modify viewing and editing preferences:**

1. Choose the required **Visualization Mode** from the corresponding **drop-down** list in the Visualization section: **Entity Structure** or **Block Tags**.
2. Select the required checkboxes of the Visualization section.
3. In the Edit section, choose to update the title automatically, or to mark selected entities with “**smart coloring**” (by selecting the appropriate check box).
Smart coloring prevents adjacent entities having a similar color.

Border Colors tab

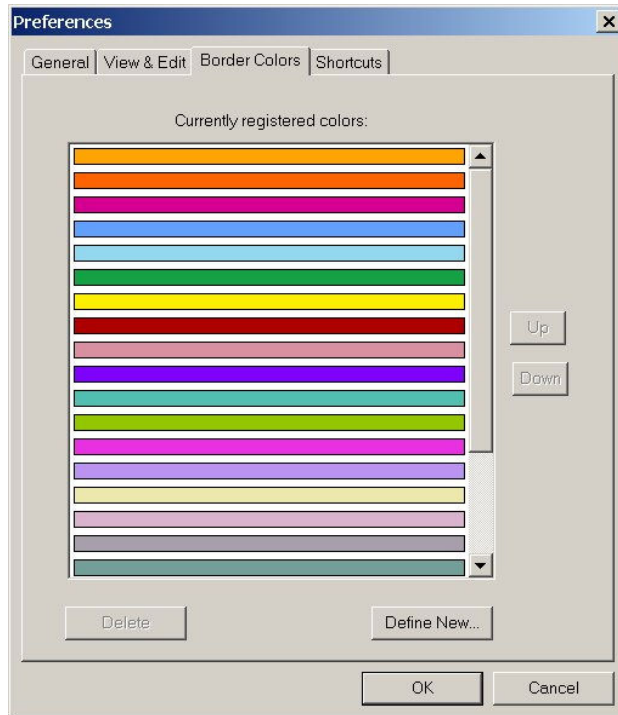


Figure 36 – Preferences: Border Colors Tab

► **To define the colors to be used in primitive borders:**

1. Scroll through the **Currently Registered Colors** list and select **colors** for editing.
2. Define new color settings by using the following buttons:

Button Label	Description
Delete	Deletes a selected color.
Define New...	Opens the following window: 
Up/Down	Change the priority of colors. You can move the colors you find most convenient to work with to the top of the list.

Shortcuts tab

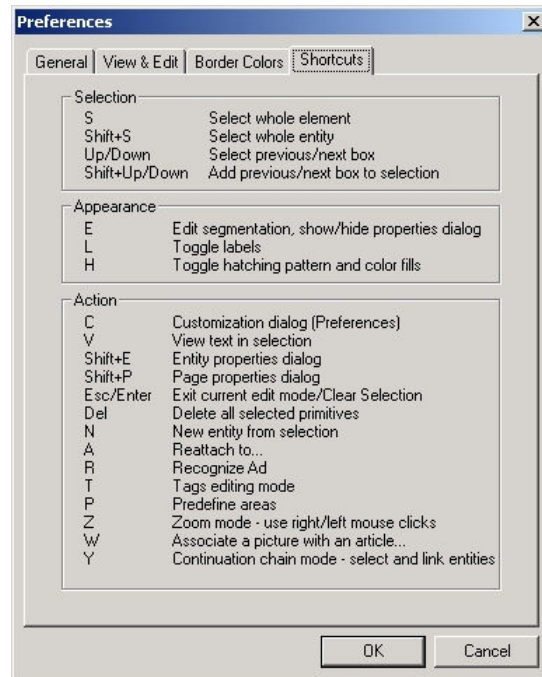


Figure 37 – Preferences: Shortcut Tab

This tab displays a list of keyboard shortcuts in the **Edit Segmentation Tool**.

When **Segmentation Editing** Mode is active, the Olive shortcuts override standard Adobe Acrobat shortcuts that have identical letters. Acrobat shortcuts that are not overridden remain active.

2.8.3 Segmentation Editing

Overview

Reviewing and editing the results of automatic segmentation involve the following topics. The *Edit Segmentation Tool Context Menu* is described at the end of this section.

- Identifying Segmentation Errors
- Segmentation View
- Default Element Patterns
- Zooming and Previewing
- Viewing and Modifying Primitives
- Creating New Primitives, Elements, and Entities
- Continuation Settings
- Working with Links (Annotation Tool)
- Additional Functions:
- Associating Pictures with Articles
- Local Advertisement Recognition
- Defining Large Tables.

Identifying Segmentation Errors

The purpose to reviewing segmentation is to find these inaccuracies:

Garbage articles - When automatic segmentation finds an element that may be a useful entity but appears incomplete or otherwise suspect, it marks it as a **garbage article**. After automatic segmentation, garbage articles are marked by **red labels**. This is the first place to look for segmentation errors.

Garbage articles save precious time during manual segmentation. Parts of a garbage article often belong to a useful entity and can simply be reattached to its logical neighbors. Parts that are not useful are easy to delete.



When defining a garbage article as part of a **“legitimate” entity** using Assign Entity or attaching it to an existing entity, its label becomes blue and it ceases to be a garbage article.



Clear away all garbage articles in the issue before exporting to XML, either by deleting or by defining them as regular primitives. This prevents unsightly material from being seen by the end user.

Note that although the garbage article feature can find some segmentation errors, it cannot be relied on exclusively. The automatic segmentation engine may create segmentation errors that are not defined as garbage articles.

A primitive not marked in red may still be wrongly defined.

- **Ignored document fragments:** The segmentation engine may not draw rectangles around some of the text, an entire element, or even an entire entity.

Correct this by manually creating primitives to include the ignored elements or element fragments, and assign the correct properties to them.

- **Redundant elements or entities:** The engine may draw rectangles around pieces of the periodical that are not relevant to the segmentation process. Anything found on the page which does not belong to an article, a picture, or an ad, should not be segmented.

Correct this by deleting the redundant primitives.

- **Inaccurately-drawn rectangles:** The engine may draw a rectangle that is too large or too small for the text or image that it contains.

Correct this by resizing the relevant primitive.

- **Errors in entity association:** The engine may assign primitives to neighboring entities. Body-type primitives belonging to one article are often mistakenly assigned to another.

correct this by reattaching the primitives to the correct entity.

- **Errors in element type:** The engine may erroneously define the primitive element type.
Corrected this by reattaching the primitive in question to the correct element in the entity, or by creating a new element if it does not exist.
- **Errors in the reading order:** The reading order of primitives, as indicated by the numbers in their titles, must be the same as the logical order of contents within the article. If the article layout is complex, the engine may incorrectly define the reading order.
Correct this by reattaching primitives within the element in order to achieve a logical reading order.
- **Continuations:** Adobe Acrobat page numbers no longer match the periodical page numbers when pages are removed from the PDF file. This causes most continuations in the document to not be automatically defined, and articles must be linked manually.
- **Text as Images:** If the PDF document contains some text as images, automatic segmentation may recognize that text as a picture (see procedure below). These areas require definition as predefined OCR areas.

► **To check if text has been recognized as an image:**

1. Go to a problematic page.
2. Launch Acrobat's **Text Attributes** tool (**Tools -> TouchUp -> Text Attributes**)
3. Select a portion of text.
If the tool cannot display the font attributes, then the text is an image.



If you have not run automatic segmentation or manually defined any segments, the document view does not change when you activate the **Edit Segmentation** Tool, since there is nothing to display. However, the **Selected Segment Properties Console** does appear.

The shortcuts used in the **Edit Segmentation** mode can be found in **Preferences**.
(See *Segmentation Visualization Preferences* on page 66.)

Segmentation View

When activating **Edit Segmentation** as described above, a graphic representation of the document's segmentation is overlaid on the PDF document. This visualization shows rectangles with colored borders, patterned surfaces and labels.

Each rectangle represents a *primitive* whose properties, membership in elements and entities, are indicated by its visual properties. The primitives' viewing options (Color fill) is accessed by toggling the <H> key.

The following options are available:

- | | |
|--------------------------------|-------------------------------|
| ■ Frames with hatch patterns | ■ Frames with FULL color fill |
| ■ Frames with LIGHT color fill | ■ Frames with no fill |

- Frames with MEDIUM color fill



In the **Primitive Tag sub-mode** (activated through the  icon on the toolbar), segmentation is represented by black rectangles.

► **To display segmentation:**

- Click the **Olive** menu, select **Segmentation**, and select **Edit**.

Or simply click **Edit Segmentation Tool** .

A graphic visualization of document segmentation appears (see the description below), as along with the **Selected Segment Properties** window.

Default Element Patterns

Element patterns help identify primitives that belong to the same elements. View them by toggling by pressing the <H> key. Element patterns can be customized or changed by Olive Professional Services as requested by the customer.

► **To change the way labels are displayed:**

- Use the shortcut key <L> to toggle between the following four label modes:
 - Labels displaying link identifiers (if any), element type, and reading order number (default setting)
 - Labels displaying link identifiers only. If there is no continuation defined for the entity, labels are hidden.
 - Labels displaying link identifiers (if any), and element type
 - Labels displaying link identifiers (if any) and reading order number.



Use the **Preferences** dialog box to change label positioning on each primitive.

► **To show/hide the element patterns of primitives and provide color fill for primitives:**

- Use the shortcut key <H> to turn off the element patterns and to show primitives with color fill. The color fill applies to the whole entity. <H> stands for "hatch".

► **To hide segmentation visualization:**

- Deactivate **Edit Segmentation** by clicking any **Acrobat** tool icon, on the **Acrobat** toolbar, that is not associated with XML Distiller.

The Segmentation markings disappear.

► **To customize a segmentation view:**

- Click the **Olive** menu, select **Preferences**.
For more details, see *Segmentation Visualization Preferences* on page 66.
For a more detailed view of the properties of a primitive and of all the higher-level units to which it belongs, use the Selected **Segment Properties** window, which is displayed when the **Edit Segmentation** tool is activated.

Zooming and Previewing

Zooming and Previewing are the two ways to get a view of a PDF page's contents.

► **To zoom in and out:**

1. Press  on the toolbar, or use the shortcut key <Z>.
Your cursor shape now contains the Zoom symbol.
 - In the **Zoom** mode,
 - **Left-Clicking** anywhere in the page zooms in by a factor of two, centered on the clicked point
 - **Right-Clicking** will zoom out by a factor of two.
 - **Click-dragging** specifies the zoom area anywhere in the page.
2. Press  or <Z> again, or press the <Esc> key, to return to the standard edit mode.

► **To preview the text of your selection:**

1. Select one or more primitives of the same element.
2. Press  on the toolbar, or use the shortcut key <V>.
A pop-up window appears in the center of your screen, displaying the text of the entity you selected.
This may be used for small primitives, such as those containing Bylines or Captions, whose text is often obscured by the border or surface pattern of the primitive. It also allows reading the body text from an article while remaining in small zoom.



Text displayed in the preview box is generated by the same program function used to generate the text that in the Warehouse files (Repository). The preview box allows you to preview the actual text the end user will see, when viewing an article in Fast Mode.

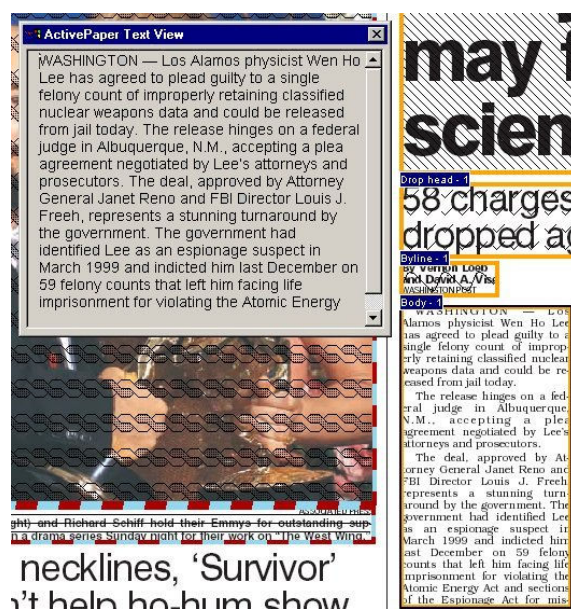


Figure 38 – Text View

3. To return to the standard edit mode, press <V> again or press the <Esc> key.

Before starting to edit, learn how to select, delete, and create the primitives, elements and entities, and how to use these options for editing.

Viewing and Modifying Primitives



This chapter uses the terms “**original**” <object> and “**target**” <object>. **Original** is the primitive selected first to perform an operation that relates the two objects; **target** is the object you subsequently click to make the relation.

Primitives belonging to the same entity have the same border color.



Each entity on the page usually has a unique color and the borders of primitives belonging to this entity all share the same color. The number of easily distinguishable shades is limited, so two entities using the same color may appear on a page with many entities. To prevent this, change the color scheme or apply the *Smart coloring* option (see *Segmentation Visualization Preferences* on page 66 for instructions on these actions).

Picture-type entities connected to an article have a two-color border with the color of the picture entity and the color of the article entity to which it is associated.



If picture primitives are labeled Image - 2, Image - 3, etc. see *Viewing and Modifying Primitives* on page 75.

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Each primitive's element type is indicated by its surface pattern. Each element type has a default pattern.



Surface patterns may be turned On and Off using the shortcut key <H>.

Every primitive also has a label that displays its element type and number. The number indicates the reading order within the element.

If a *continuation* has been defined for an entity, the labels of all the primitives in the entity are marked with an arrow. The arrow -> following the label text (Caption - 1 >) indicates that the entity continues later in the document (**Continuation To** link), whereas the arrow -> preceding the label text (|-> Continuation From - 1) indicates that the entity is continued from a previous page (**Continuation From** link). The ^ symbol at the end of the label text (Image - 1 ^) shows that the entity is associated to another entity (nested within the entity), which is only relevant for picture entities. These symbols are referred to as *link identifiers*.

Red-colored labels belong to Garbage Articles. (See *Identifying Segmentation Errors* on page 71.)



Use the shortcut key <L> to toggle between four label display modes.

The following is an example of segmentation representation:



Figure 39 – Segmentation Primitives

As displayed in the figure above, the segmentation scheme allows you to identify at a glance which entity and element each primitive belongs to.

Each element exists as a part of an entity only once. Each element instance is represented by a different number labeled primitive. The numbering is important to identify the correct reading order of primitives.

Selected Segment Properties

The **Selected Segment Properties** window appears whenever the  icon is activated, though it can be hidden at any time. Press the <E> key on the keyboard or press the  icon again to reopen it.

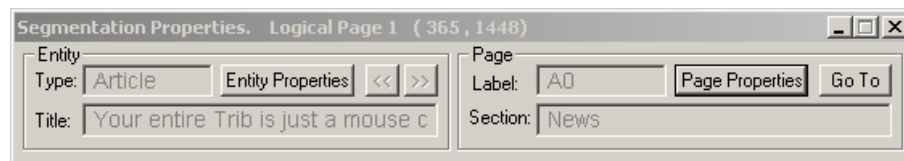


Figure 40 – Selected Segmentation Properties

It displays element, entity and page information about the the selected primitive(s).

The window's **title bar** shows the original page number of the selection; this is particularly useful in documents that have passed double-page processing, where the Acrobat page numbers do not necessarily reflect the actual issue page numbers.

The **Selected Segment Properties** window contains the following fields:

Field Label	Description
Type	Selection's entity type (the current entity).
Title	Current entity's Meta Name (To learn more about Meta Names, see the description of the Entity Properties and Page Properties buttons below.)
Label	Indicates the page's Label. If blank, no label has been assigned to the page.
Section	Name of the section that the current page belongs to.

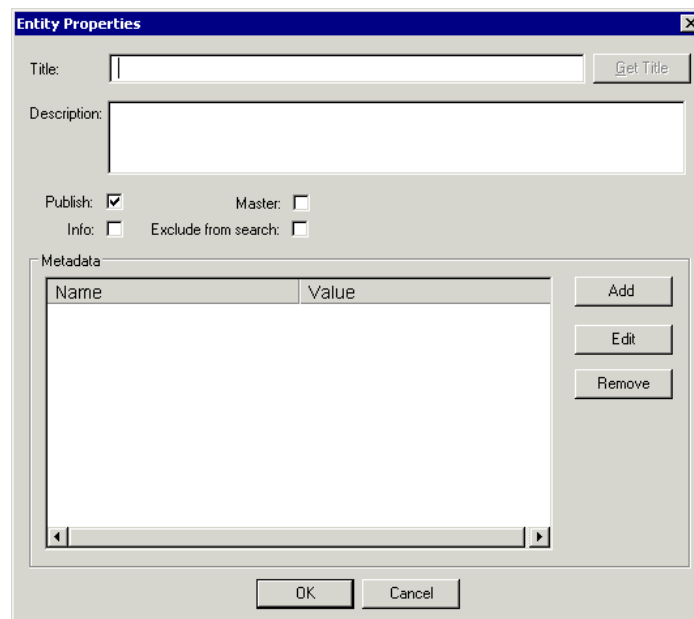
In addition, the **Selected Segment Properties** window contains the following buttons:

Button Label	Description
<< >>	Jump forward to the article continuation, or backwards to start of the article.
(Continuation)	This button is disabled if the current entity (or the entity it is associated to) is not linked to any continuation.
Entity Properties	Entity Properties dialog box.
Page Properties	Page Properties dialog box
Go To	Go To Page Label dialog box

- ▶ **To close the Selected Segment Properties window:**
 - Click the **window control** box (at the top-right corner), or press the shortcut key <E> while the window is open.
- ▶ **To reopen the Selected Segment Properties window:**
 - Do one of the following:
 - In the **Olive** menu point to **Segmentation**, and then choose **Edit....**
 - Activate the **Edit Segmentation Tool** icon  on the toolbar
 - Or
 - Press the shortcut key <E> while the window is closed.

Entity Properties

In the **Selected Segment Properties** window, click **Entity Properties** to open the corresponding window:



The **Entity Properties** dialog box is shown. It has a title bar with the text "Entity Properties" and a close button (X). The dialog contains the following fields and controls:

- Title:** A text input field with a "Get Title" button to its right.
- Description:** A large text area.
- Publish:** A checked checkbox.
- Master:** An unchecked checkbox.
- Info:** An unchecked checkbox.
- Exclude from search:** An unchecked checkbox.
- Metadata:** A section containing a table with two columns: "Name" and "Value". The table is currently empty.
- Buttons:** "Add", "Edit", and "Remove" buttons are located to the right of the metadata table.
- OK/Cancel:** Buttons at the bottom of the dialog.

Figure 41 – Entity Properties

This window contains the following fields and controls:

Label	Description
Title	This field displays the existing entity title that is offered by the system. Enables manual editing of the article's title, or Meta Name. This is the property of each entity in the periodical, and is used as the name of the article in the display area of the search results and in ActivePaper content management modules. The Meta Name is automatically derived from the text of the title element of the article, but you may modify it or enter manually, if the entity not have a title element.  When resizing, moving, or redrawing the title element of an article to include new text, the article's Meta Name is automatically updated. (Manual changes are overwritten.)  Two cases require you to define Meta Names manually: 1. Pictures that are not associated to an article, and therefore do not receive its Meta Name. 2. Articles that do not have a title element. This prevents the untitled pictures and articles from appearing in the search results.
Get Title	This button is used obtain the original article's title from the PDF document.
Description	This field can be used for entering entity description (optional).
Publish	Select this checkbox to include the entity in the XML output.
Info	See page 95, Defining Large Tables.
Master	Select this checkbox to include the entity in the Master output.
Exclude from search	Select this checkbox to remove the entity from search results, even when the search criteria match.
Metadata: Add	Opens a dialog to specify new metadata (name, value) pair
Metadata: Edit	Opens a dialog to modify the highlighted metadata entry.
Metadata: Remove	Removes the highlighted metadata entry.

Page Properties

In the **Selected Segment Properties** window, click **Page Properties** to open the corresponding window:

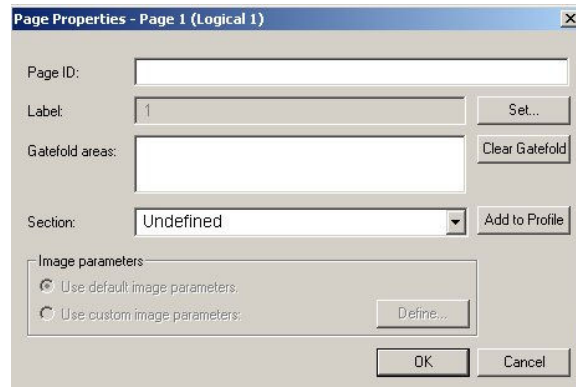


Figure 42 – Page Properties

The **Page Properties** dialog box contains the following fields:

Label	Description
Page ID	Enables manual entry of a string defining the periodical page. PageID is the path within the warehouse where the page is saved.
	 Note: The actual periodical number is shown on the top bar of the console. It may be different from the number of the Acrobat page, especially after double-page processing.
Label	Page identifier, containing section ID and page number. Click Set... to modify.
Gatefold Areas	Only for ActiveMagazine. The window presents information that was automatically produced when the gatefold was created. This cannot be modified manually. Click Clear Gatefold to delete the information in the Gatefold field and remove the gatefold itself.
Section	Section name. Either select from a drop-down list, or enter a new section name and click Add to Profile .

The **Page Properties** dialog box contains the following buttons:

Set...	Click to add or modify the Label for the current page.
Clear Gatefold	Click to delete information in the Gatefold field and to remove the gatefold itself.
Add to Profile	This command becomes available only if the section name is modified or typed in. Click to add the string value that you have typed in the Section text box to the list of section names in the profile. There is no Undo option for this command. To cancel the change, select the old section name from the Section list. The profile must be amended manually.

Assigning Page Labels

The page label is a series of values that describe a page, containing these values:

- **Prefix (string)** Description of the page.
- **Number (integer)** The page number of the page, either the absolute number or the page number within the section.
- **Suffix (string)** Section name where the page resides.
- **Number style (Boolean)** Indicates if the page number is in Roman or Arabic numerals.
- **Label String** String value that contains the Prefix, Number, Suffix, and Number Style.



When XML Distiller sets the first four components of a page label, the user will have the option to automatically label each subsequent, unlabeled page. Each label will be identical except for the page number, which will increment. Pages will label automatically until a non-blank label is encountered.

Automatic Page Labeling

As a part of the segmentation process, ActivePaper attempts to automatically assign a label for each page.

► To specify/edit a Page Label:

1. In the **Segmentation Properties** dialog box, click **Page Properties**.

The **Page Properties** dialog box appears.

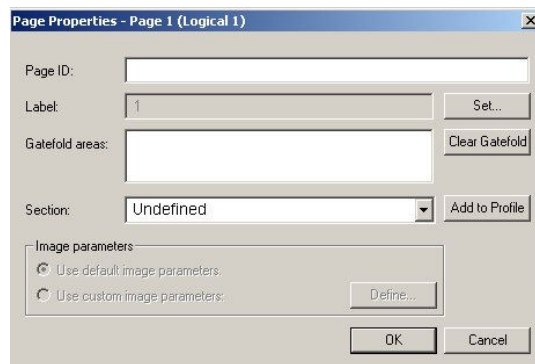


Figure 43 - Segmentation Properties Dialog Box

2. In the **Page ID** text box enter a description of the page.
3. In the **Section** drop-down box, do one of the following:
 - select the appropriate section name
 - or
 - Type the section name and click **Add to Profile**. (From this point forward, this section name will be available from the drop-down list.)
4. Click **Set**.

The **Page Label** dialog box appears.

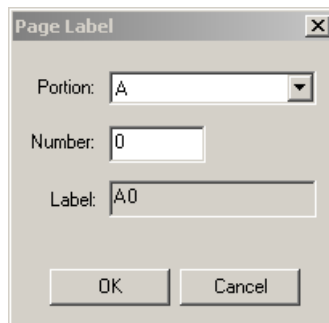


Figure 44 - Page Label

5. In the **Portion** drop box, choose the appropriate selection.

The **Label** text box shows the outcome of your selections.

Normally, the available selections conform to the section names in the publication. If the available selections are inappropriate, open the profile and edit the [Portion] section. (See *Editing and Improving Profiles* on page 185.)
6. In the **Number** text box, enter the page number of the page.

The **Label** text box shows your entry.
7. Click **OK**, the **Page Properties** dialog box reappears.

8. If you have completed making changes to Page Properties, click **OK**.
If you made a change to the page label, you are asked if the new page label settings will be used for subsequent pages.
9. Click either **Yes** or **No** to apply or not to apply the new page label settings to subsequent pages. Clicking **Cancel** returns you to the **Page Properties** dialog box, without applying any changes.

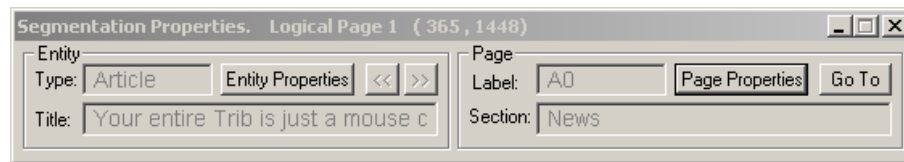


Figure 45 - Page Properties Dialog Box

Working with Page Labels

Go To jumps to a page with a specific label or number.

The Selected group on the panel reflects the properties of the current selection, whereas the Page group refers to the currently visible page, and is empty if more than one page is visible.



When deleting or inserting a page, the labels of pages are preserved.

Page labels are empty by default (no page labels).

Selecting Primitives, Elements, and Entities

In this chapter, the term “**the selection**” refers to the primitive or primitives that are currently selected. “**Current element**” refers to the element that the selection belongs to, and “**current entity**” refers to the entity that the selection is a part of.

► To select a primitive or primitives:

- Use one of the following methods:
 - **Click:** To select a single primitive.
 - **Keyboard Up/Down buttons:** Press <Up> or <Down> on the keyboard to select the previous or next primitive in the current element. Upon reaching the end of the element, continue to the beginning of the next element. This operation only works within the current entity: when reaching the last primitive in the last element of the document (or the first primitive of the first element), you cannot continue to the next entity.
 - **Control-click:** Hold down the <Ctrl> key and click multiple primitives to select them all. Release the <Ctrl> key when done selecting.

- **Shift-click:** Hold down the <Shift> key and click two primitives within the same element to select a range. For example, hold down <Shift>, click the body text primitive numbered 3, then click the primitive numbered 6 (in the same element), to select primitives 3, 4, 5, and 6.
- **Shift-up/down:** Select a primitive, hold down the <Shift> key and press <Up> or <Down> on the keyboard to select a sequence of primitives.
- **Control-drag:** Hold down the <Ctrl> key, then click and drag to draw a rectangle around several primitives. Upon releasing the mouse, all the primitives that intersect with the selection rectangle will be selected.
- **Control-Shift-drag:** Hold down the <Ctrl> key and the <Shift> key, then click-and-drag to draw a rectangle around several primitives. When you release the mouse, all the primitives that are included in the selection rectangle will be selected.
- To deselect, click the document part that does not contain a primitive, or press <Esc>, or press <Enter>.



When moving the mouse over a primitive on the page, a small rectangle highlights it. When clicking to select, the highlighted primitive becomes selected. Note the rectangle that is highlighted when trying to select very small or overlapping primitives.

► **To select a primitive and display the Edit Segmentation Tool context menu:**

- Right-click an unselected primitive to select it and open the context menu.

► **To select the entire current element:**

1. Right-click a primitive to display the **Edit Segmentation** Tool context menu.
2. Choose **Select Whole Element** with the type of the current element (e.g., **Select Whole Body**) displayed in brackets.
Alternatively, use the shortcut key <S>.



Only use this option if the selection does not include primitives from different elements or entities.

Control-Shift-Click any primitive to select the entire element to which it belongs.

► **To select the entire current entity:**

1. Right-click on a selection to display the **Edit Segmentation** Tool context menu.
2. Choose **Select Whole Entity** with the type of the current entity, e.g. (**Select Whole Article**, displayed in brackets).
Or use the shortcut key <Shift-S>.



Only use this option if the current selection does not include primitives from different entities.

Editing and Deleting Primitives

► To resize a primitive:

1. Position the cursor over the primitive's border and its shape changes: .
2. **Click, drag, and release** to resize the primitive. Note that clicking on the sides of the rectangle resizes only in the direction of that side, whereas clicking on a corner allows free resizing.

► To move the selection:

1. Position the cursor over the selected primitives' label or on the center, and its shape changes: .
2. **Hold down** the mouse button, and **drag**. While dragging, a dashed rectangle indicates the new position to which the selection will be moved.
3. Releasing the mouse moves the primitive to this location.

► To delete the selection:



It takes time to redraw an accurate primitive, so it is not advisable to delete existing primitives unless they are totally redundant. An incorrectly assigned primitive can always be reattached.

1. Right-click a primitive to display the context menu.
2. Select **Delete Selection** from the context menu
or use the shortcut key .



Deleting a primitive cannot be undone, so use this command with caution.

Editing Primitive Properties

You can edit the primitives properties in the following ways through the **Edit Segmentation Tool** context menu:

- | | |
|---|--|
| ■ Reattach a selection to any element on the page | ■ Create a new element in the current entity, containing the selection |
| ■ Reattach the selection to another entity, creating a new element (Smart Reattach) | ■ Create a new entity containing the selection. |



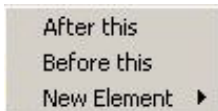
When editing the properties of multiple primitives at the same time, select them in the correct reading order, using the control-click method. When the primitives are moved to another element or entity, the new reading order is the order in which they were selected.

► **To reattach a selection to any element on the page:**

1. Right-click on the selection to display the context menu, and select **Reattach....**
Or use the shortcut key <A>.

The cursor changes its shape (.

2. Click any primitive on the page to reattach the selection after it. The selection is attached to the element of the target primitive (in the target primitive's entity), which follows it in the reading order. The primitives are attached in the order in which they are selected.
3. Right-click the target primitive to display the Reattach context menu:



4. Select one of the following options:

- **After this:** Selecting this command has the same result as a left click.
- **Before this:** Attaches the selection before the clicked primitive. The selection is attached to the target primitive, and precedes it in the reading order. The primitives are attached in the order in which they are selected, so that the last primitive selected will precede the target primitive.



Use this command to change the reading order of primitives within an element. Click a target primitive in the current element to reattach the selection after it, or use the Reattach context menu to attach the selection before it.

► **To reattach the selection to another entity, creating a new element (Smart Reattach):**

1. Select the **Reattach** command.
The cursor changes.
2. Right-click any primitive that is a part of the target entity.
3. In the **Reattach** context menu, select the **New Element** submenu. This submenu lists all the element types that do not presently exist in the target entity.
4. Select an element type.
The selection becomes a new element of the specified type in the target entity.

► **To create a new element in the current entity, containing the selection:**

1. Right-click the selection to display the context menu, and select the **Assign Element** submenu.
The submenu displays all the element types that do not exist in the current entity.
2. Select an element type to create a new element of that type, containing the selection, in the current entity.



Only use this option if the selection contains primitives from the same entity.

► **To create a new entity containing the selection:**

1. Right-click the selection to display the context menu, and select the **Assign Entity** submenu.

The submenu displays all entity types.

2. Click an entity type to create a new entity of that type, containing the selection.



This operation can be performed in one step using the shortcut key **<N>**, for New Entity.

This command can be applied to any selection, even one containing primitives from multiple entities.

If the selection contains primitives from one entity, and the specified entity type is identical to the current entity type, each primitive's element definition remains. For example, selecting one **title**, one **subtitle**, and three **body** primitives from an existing article and performing an **Assign Entity -> Article** operation on them, converts them into a new article that contains a title, subtitle, and body.

Specifying a different entity type (such as **Picture**, for example), reverts all primitives to the default element type of that entity. This also happens if the selection contains primitives from different entities.

Working with Primitive Tags

Primitive tags indicate that specific text within a primitive serves a certain purpose. For instance, the Author's name within a Byline may be tagged, appearing as bold in Text View.

A default list of primitive tags is provided, which can be customized for each publication, and are recognizable in automatic segmentation.

To mark text with a primitive tag, you must enter **Primitive Tag** sub-mode.



Segmentation visualization changes in **Primitive Tag** sub-mode:
 Primitive borders become thin and uniformly black;
 Primitive labels disappear;
 Primitive tags are indicated by green labels.
 Primitive tags appear as highlight over the PDF text;
 The "active" primitive tag appears with a red border.

► **To enter Primitive Tag Sub-mode:**

- Activate the **Primitive Tag sub-mode** icon ();
- OR use the shortcut key **<T>**. (The same key exits the mode, returning you to normal visualization of the segmentation).

► To mark a primitive tag in Primitive Tag Sub-mode:



Only mark a **primitive** tag for text inside an existing primitive.

1. Point at the start of the text to be marked. Click and hold the mouse button.
2. Move the mouse to the end of the text. Make sure the selection includes all the text. (It is sometimes tricky to include the period at the end of a sentence). Release the mouse button.

The primitive tag that has just been marked automatically reverts to the default primitive tag type, which is shown on the label with a green background.



Primitive tags cannot be resized. Highlight the correct text before releasing the mouse. If you made an error, you must delete the tag and re-apply it.

Stop here if this is the primitive tag to be defined. If you want to change the definition, proceed to step 3.

3. Right-click your **primitive** tag to open the context menu that contains the **primitive** tags that are arranged for the publication.



Primitive tags are applied in the profile. For changes regarding primitive tags, please contact Olive Professional Services.

4. Choose the required type. You may change the primitive tag's type at any time.

► To remove an existing primitive tag, in Primitive Tag sub-mode:

1. Click an existing primitive tag to select it.
2. Press key on the keyboard. Alternatively, exit the **Primitive Tag sub-mode** and delete the entire primitive containing the marked **primitive tag**, or resize it to exclude the marked text.

Primitive Tag Element Recognition

To create a separate primitive from a recognized primitive tag, take the primitive tag text, create a separate primitive and change its **Element Type**.

The initial value is the element type that is set in the profile. After Segmentation, the element type is set to the primitive created from the **primitive tag**. (If the value is empty, the ElementType is not changed.)

Creating New Primitives, Elements, and Entities

XML Distiller allows you to:

- add a primitive to an existing element
- add a new element to an existing entity
- create a new entity

- change the type of an entity.

► **To add a primitive to an existing element:**

1. Decide where you want the primitive to appear in the element's reading order. Select the primitive that will precede it in the new reading order.
2. Click at the new origin point in the document, **hold down** the mouse button, and **drag**. A dashed rectangle is drawn. Release the mouse when your new rectangle is of the appropriate size.



You will sometimes find that it is difficult to create new primitives close to existing ones, because the cursor becomes a Resize cursor. To overcome this problem, **hold down** the <Shift> key, and draw a new primitive. This method provides you with the Create Primitive cursor, regardless of the cursor's proximity to existing primitives.

3. Your cursor appears as an exclamation mark. This gives you the option to create more primitives to follow the new primitive in the reading order. To do this, repeat step 2 until you have finished creating new primitives.

When using this method to create picture primitives, you may find pictures that have been labeled Image - 2, Image - 3, etc. However, these images may not appear when the associated article is viewed by the reader. To resolve this issue, do the following:

- Right-click on the picture and select **Assign Entity** from the context menu.
- Select **Picture** from the menu.
- Right click on the picture again, and click **Association > Associate with Article**.
- Select the article that the picture is associated with.

The picture label changes to "Image - 1".

4. Press the <Enter> or <Esc> key, or deselect, to confirm the primitive or primitives you have created. They now belong to the same entity and element as your initial selection (immediately following it in the reading order).

► **To add a new element to an existing entity:**

1. Select any primitive or primitives in the existing entity.



If your picture primitives are labeled **Image - 2**, **Image - 3**, etc. see point 3 in the previous procedure.

2. Click at the new origin point in the document, **drag** and **release** to define the rectangle size.
3. Press the <Enter> or <Esc> key, or deselect.

4. Right-click on the new primitive to invoke the context menu, and select the **Assign Element** submenu. Select the appropriate entity types to which the primitive will be assigned.
5. If you need to create more primitives in this new element, make sure the new primitive is selected, and draw as many primitives as necessary. (They will be attached after it in the reading order.)



There is an alternate way to perform this operation:
Creating a primitive or primitives, and *smart-reattaching* them to another entity.

This creates a new element of the correct type.

To learn how to perform the Reattach operation, see *Viewing and Modifying Primitives* on page 75.

► **To create a new entity:**

1. Make sure nothing has been selected.
2. Click at the new origin point in the document (**hold down <Shift>** to make sure you retain a Create Primitive cursor), and **drag** and **release** to define the rectangle size. Your cursor changes.
3. Create as many primitives as necessary, keeping to the reading order of the element you are defining. Remember that by default, new primitives you draw now will be defined as body primitives within a new article entity.



To save time, define all the primitives in the entity as Body, without taking the time to assign correct element types as you go along. Later, you can go back and change their element types.

4. Press the **<Enter>** or **<Esc>** key, or deselect. The primitive (or primitives) you have selected become new default elements (Body) in a new default entity (Article).
5. If you need to change the element type of some or all of the primitives in this new entity, select them and then select **Assign Element** from the context menu.

► **To change the type of the entire entity:**

1. First select the entire entity, then right-click the selection.
2. Select an entity type from the **Assign Entity** submenu.

Segmentation editing is performed through the Edit Segmentation Tool context menu, which pops up when you right click on any element, both in Edit Segmentation Tool mode and in Learning Mode.

Continuation Settings

See *Edit Segmentation Tool Context Menu* on page 101 for the context of using these settings.

If there is more than one continuation (for example, the article begins on page 2, continues on page 13 and then ends on page 25), you will need to create continuation links between all the articles.

When creating a Continuation link we refer to the **original entity** and the **target entity**. The original entity is the one you click first to perform the continuation. The target entity is the entity you subsequently click, to link the two.

► To link two entities by continuation:

1. Right click any primitive in the original entity. Point to **Continuation** to open the following submenu:



2. If the target entity is a continuation of the original entity (for example, the original entity is on the front page and the target entity continues it on page 14), choose **Continue to...**

If the target entity precedes the original entity in logical order (for example, your original entity is on page 14 and the target entity is the beginning of the article on page 1), choose **Continue from...**

The cursor assumes the following shape: .

3. Navigate to the page containing the target entity, and click one of its primitives.

The two entities are now joined by continuation. The labels of the two entities receive arrow signs. If the entity is connected to another entity earlier in the issue, the arrow appears at the left of the label: . If the entity is connected to another entity later in the issue, the arrow appears on the right.



The target entity must be on another page. If the article is continued on the same page, simply define the entire article as one entity.

► To create continuations between three or more articles:

Create a link between each pair of articles in the chain.

For example, assume you have an article beginning on page 4, continued on page 6, then continued on page 15, then continued on page 41. The following is one way to create a continuation chain in this case:

Style-Based Segmentation

1. Right click the entity on page 6. Use **Continuation From...** to link it to the entity on page 4.
2. Right click the entity on page 6 again, and now use **Continuation To...** to link it to the entity on page 15.
3. Right click the entity on page 15 and use **Continuation To...** to link it to the entity on page 41.

► To break a continuation:

1. Right click any primitive in the original entity. Point to the **Continuation** submenu.
2. If the target entity (with which you want to sever the link) is a continuation of the original entity, choose **Break Continuation to**.

If the target entity precedes the original entity in a logical order, choose **Break Continue from**.

► To replace a continuation:

1. Right click any primitive in the original entity. From the menu, select **Continuation** and **Continue to...** from the submenu.

If the target entity precedes the original entity in logical order, choose **Continue from...** in the submenu.



You do not need to break a continuation to replace a continuation.

2. Navigate to the target entity and click it.

► To cancel a continuation operation:

- Press the <Esc> key before clicking the target entity.

Working with Links (Annotation Tool)

Using the **ActivePaper Annotation Tool**, you can create the following links:

- **See Page** – Link to another page in the same document (default).
- **See other Doc** – Link to another document.
- **Continuation in Page** – Link to the rest of the article on another page.
- **Teaser to Page** – Link to another page in another document.
- **Web Link** – Link to an Internet address.
- **Mail To** – Link to an email address (creating a new message in the default email program).



Regarding links between issues: Because of the production workflow there might be some links that point to an issue that does not exist. Validating if an issue exists must take place in the browser (end-user viewer) implementation code.

If the link is to an existing issue but the page does not exist, then the first page of the issue will be displayed. If an issue that does not exist is requested, the following error will be generated: **“Error: Cannot find requested issue”**.

Examples

The following table shows how each of the links will be shown in the user interface:

Link Type	Resulting End-User Interface Example
See Page	Link to pages: See Page C1 See Page 5
See other Doc	See also: Item 1 Item 2
Web Link	References: http://www.webreference.com
Mail To	Contacts: elisha@OliveSoftware.com

► To create a link using the ActivePaper Annotation Tool:

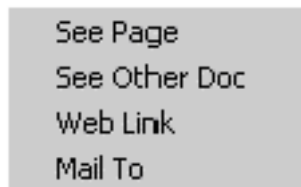
1. Click the **ActivePaper Annotation Tool** button.
2. With the tool, draw a box around the article/location where you want the link inserted.

The box looks like this:



3. To link to a different location, such as an email address, web address, or another document, right-click inside the box.

The following context menu appears:



4. Click one of the menu items.

The tag in the box changes from **See Page**, **See Other Doc**, **Web Link**, or **Mail To**, directing to the selected attribute.

5. Double click inside the box (above).

The following dialog box appears:

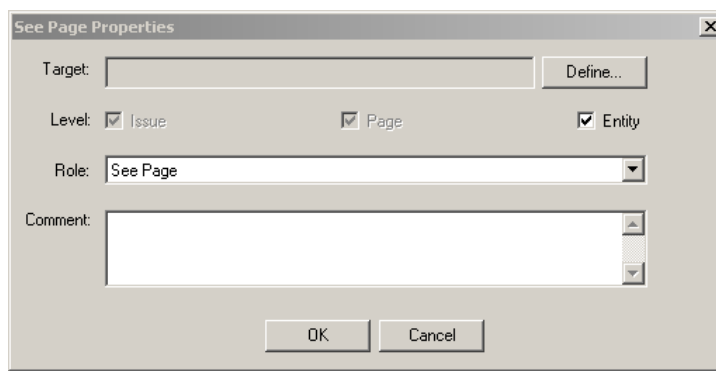


Figure 46 – "See Page" Properties

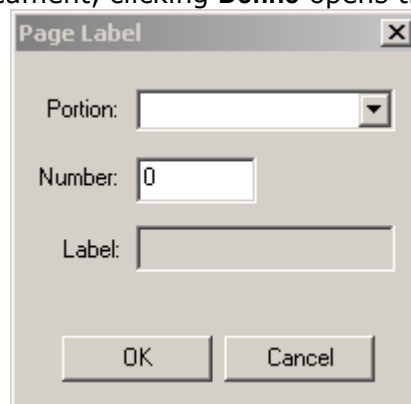


If **See Other Doc**, **Web Link**, or **Mail To** is selected, the dialog box will be different and include relevant information.

6. Click **Define** to set the target of the link.

Depending on the attribute, you may need to type the address of the target directly in the **Target text** box.

For links to pages in document, clicking **Define** opens the following dialog box:



- Select the section from the **Portion drop-down** box and type the page number in the **Number** box.
- The resulting **Page Label** appears in the **Label** box.
- Click **OK**.

7. In the **Role** drop-down box, select the appropriate role for the link and click **OK**.



Teaser to a page is a link to a page in another issue.

Additional Functions

Additional functions performed after Segmentation include:

- Associating Pictures with Articles
- Local Advertisement Recognition
- Defining Large Tables.

Associating Pictures with Articles

The **Association** command is used to associate pictures with articles.

Picture entities can exist independently, in the case of a standalone picture that is not related to any article. A picture supporting an article, however, should be *associated* with that article so that the picture becomes a part of the article entity.

It is important to make this distinction and associate all supporting pictures correctly, to ensure that they are displayed correctly to the end user.

► To associate a picture with an article:

1. Select any primitive of the picture.
2. In the context menu, point to **Association**.

The following sub-menu opens:



3. Choose Associate with article.

The circle sign of the cursor changes its shape to .

4. Click the article to be associated with the selected picture.

The cursor sign changes back to circle, and the label of the associated picture receives the arrow/s pointing to the associated article.



Local Advertisement Recognition

During your manual segmentation session, you can run the recognition engine specifically to recognize a certain advertisement. This function is available from the  icon on the toolbar (described above).

► To recognize the inner elements of a display ad:

1. Create a primitive rectangle around an ad.
2. Use the **Assign Entity** command in the context menu to define the primitive as an advertisement.



Figure 47 – Assigning Entity as Ad

3. To identify different elements inside the Ad Frame automatically, make sure the Ad Frame is selected and click the **Recognize Ad** button  or use the **R** key (shortcut).



Figure 48 – Recognize Ad

The ad's internal components are defined automatically.

4. If you want to define additional elements inside the Ad Frame, or change the automatic segmentation, use regular segmentation editing methods.

Defining Large Tables

Large tables, such as the stock pages and TV listings, cannot presently be processed by the automatic segmentation. It is, however, possible to define them manually using a special procedure and the **Continuation Chain Tool**.

Large tables are defined as articles, and specified with the **Info** property. When you define an article using the **Info** property, the viewer will access the segment in standard (newspaper) view and will not be able to access it in **Quick View** mode.

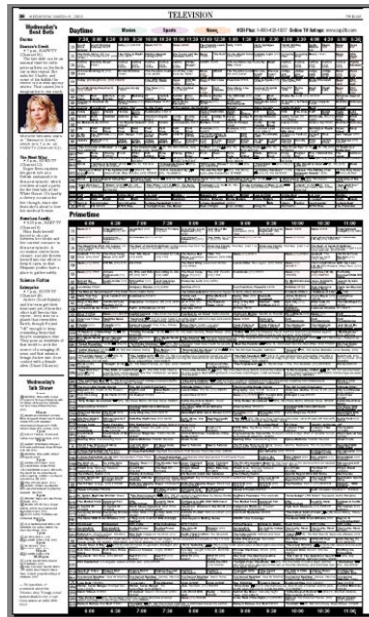


Figure 49 – Large Table

Remember that a large table is an article entity with the **Info** property enabled.



A small, informative table accompanying an article should be defined as a Picture entity associated with the article – *without enabling the info property*. The info property is reserved for large, complex tables that stand on their own.

► To define a large table:

1. Navigate to the page containing the table that you wish to segment.

Segment the table as you would segment an article. However, it is recommended that you define a large table with multiple article entities. You should also organize individual article elements in a way that conforms to the way that a viewer will normally access the table. For example, a table that will be read from left to right should be defined with article elements that will allow the reader to access the table from left to right.

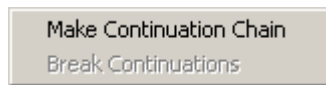


When assigning elements, remember to define table titles and subtitles.

2. Once you have completely segmented the table, press **Y** on the keyboard to activate the **Continuation Chain Tool**.

The cursor changes to: .

3. Holding down the **CTRL** key, click each article entity of the table, in the order that they should be viewed.
4. When you have finished selecting each article entity, release the **CTRL** key and right-click one of the selected entities. The following context menu appears:

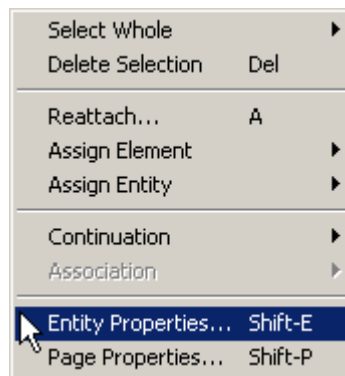


5. In the context menu, select **Make Continuation Chain**.

The menu disappears and each element tag in the table changes to green. Each tag also indicates whether the article entity is at the beginning of the Continuation Chain, at the end of the Continuation Chain, or somewhere in between.

For example, body tags of an article entity at the beginning, middle, and end of the Continuation Chain will be `Body - 1|->`, `|->Body - 1|->`, and `|->Body - 1`, respectively.

6. Once you have created the continuation chain for the table, right-click any article entity and choose **Entity Properties** from the context menu.



The **Entity Properties** window opens:

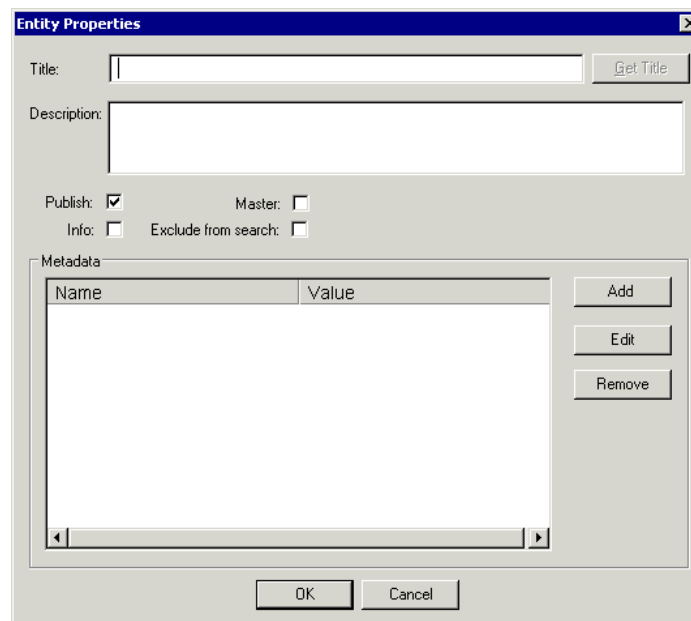


Figure 50 – Entity Properties

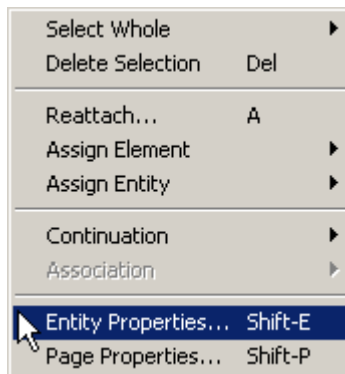
- Click the **Info** checkbox to enable the **Info** property, and click **OK**.
The **Entity Properties** window closes.



You can also open the **Entity Properties** window by pressing **SHIFT+E** on the keyboard, or by clicking the **Entity Properties** button in the **Selected Segment Properties** window.

► **To remove the Info property from the table:**

- Right-click on any element in the table (it should have a green tag) and select **Entity Properties** from the context menu.

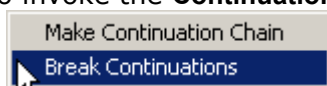


The **Entity Properties** window opens.

- Click the **Info** checkbox to disable the **Info** property and click **OK**.
The **Entity Properties** window closes and the frame tag color of each article element changes from green to blue.
- Repeat for any entities with tags that remain green.

► **To break the continuation chain of the table:**

- On the keyboard, press **Y** to activate the **Continuation Chain** Tool.
- Holding down the **CTRL** key, click each entity in the table. (You only need to click once somewhere in one element of the entity.).
- Release the **CTRL** key and right-click in any of the elements that you have selected, to invoke the **Continuation Chain** context menu (below).



- Select **Break Continuation Chain**. The tags for each element in the continuation chain will go from their continuation form to their regular form:

[->Body - 1]-> -> **Body - 1**

- Exit the **Continuation Chain** Tool by pressing the **<Y>** key.

Edit Segmentation Tool Context Menu

In the **Edit Segmentation** tool, right-clicking an element opens the context menu:



Figure 51 – Edit Segmentation Tool Context Menu

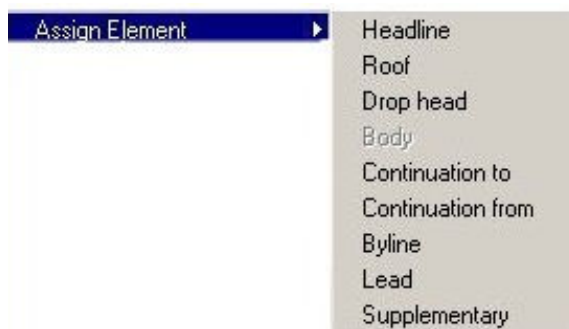
The menu items and sub-menus are described below.

- **Select whole** – Opens the following sub-menu:

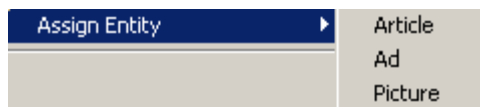


Object	Shortcut Key	Description
Element (<i>current_element</i>)	The S key	Selects the whole element to which the current highlighted primitive belongs. The relevant element name is shown in brackets.
Entity (<i>current_entity</i>)	Shift - S	Selects the whole entity to which the current highlighted primitive belongs. The relevant entity name is shown in brackets.

- **Delete Selection** – This deletes the highlighted segmentation selection. You may also do this by pressing the key.
- **Reattach** – Performs the reattachment of the selection. For instructions regarding the reattachment of primitives, see Viewing and Modifying Primitives on page 75.
- **Assign Element** – Assigns a new element type to the selection. Pointing to this command opens the list of element types, which depends on the corresponding entity:



- **Assign Entity** – Assigns a new entity type to the selection. Pointing to this command opens the following submenu:



For example, if you assign Ad to article element Body1, its label changes to the new default ad element Ad Frame1. If you select a Body element and assign it to an Article, it remains a Body element, but it becomes a new **Article Entity**.

- **Continuation** – The **Continuation** command is used to define jumps, or continuations, in the periodical issue. If an article begins on one page and continues on another page, it is necessary to define the article and its continuation as separate entities, then link them using the **Continuation** command.



Be aware of the difference between this command and the **Continuation to** and **Continuation from** elements. These elements contain the actual text indicating that there is a continuation (for example, "Continued on page 13"). Defining these elements does not create a continuation link between entities. The opposite is also true – defining a continuation link does not mean the **Continuation to** and **Continuation from** links are automatically defined. It is necessary for both the elements to be defined and the continuation link to be in place.

For more on Continuation settings, see

Continuation Settings on page 91.

2.8.4 Saving and Restoring Segmentation

When editing the segmentation of a document, there is often a need to save a document's segmentation information at various stages. For example, you might want to save the original segmentation produced by automatic segmentation; your manually corrected segmentation before it passed QA; and your manually corrected segmentation after it has passed QA.

If you want to save the segmentation of your periodical at various stages, there is no need to save multiple copies of the contents in PDF format. By using the **Save As** feature, you can simply save each version of the segmentation information to a text file. At any point you can restore the entire issue's segmentation from this text file.

It is a good idea to use this feature for backup of your segmentation. If anything goes wrong with your PDF or your segmentation, you will always be able to restore the segmentation from the text file.

► To save your current segmentation to a text file:

1. Save the PDF.
2. Click the **Olive** menu, point to Segmentation and select **Save As....**
3. Choose a location and a filename (keep the extension .txt) and press **OK**.



Label the text file meaningfully. When restoring, you will need to know which periodical issue it belongs to.

► To restore segmentation from a text file:

1. Click the **Olive** menu, point to Segmentation and select **Restore....**
2. Select a text file containing segmentation and press **OK**.

Your segmentation is restored from the text file.



Before restoring segmentation from a text file, make sure it corresponds with the periodical issue you are working on. If your periodical issue has more or less pages than it did before, or if it is not the same issue from which you saved segmentation to begin with, may not work properly.

The **Save** option in the Segmentation submenu saves your segmentation to a default-named text file in the default XML Distiller working directory.

2.9 OCR for Scanned Periodicals

OCR is now performed to digitize the text in each of the segmented entities.

Customization settings can be saved via the **Save** button, and later accessed via the **Load** button. Advanced options, available via the **Advanced** button, are described after the standard options.

Customization options include:

OCR Processor:	Name of the OCR utility
Page layout analysis options:	■ Always use Olive page layout analysis stored in PDF ■ Automatically run OCR page layout analysis refers to the case where there is no Olive page layout analysis. ■ Always run OCR page layout analysis
Text recognition Options:	■ Language of text to be recognized ■ Whether to recognize image zones ■ Whether to collect data on suspicious characters: i.e., those that received a low confidence score by the OCR engine ■ Whether to use the spell checker provided by the utility, if relevant
PDF Text Generation Options:	■ Whether to generate the green overlay ("invisible") text results ■ Whether to generate text that is outside the border of the image ■ Whether to indicate (via color coding) characters that are suspicious (i.e., received a low confidence score) ■ Where relevant, whether to indicate (via color coding) words with spelling errors
QA Options:	■ Whether to check the number of suspicious characters and, if so, warn or fail upon a threshold readability level; ■ Whether to check the number of spelling errors and, if so, warn or fail upon a threshold readability level.

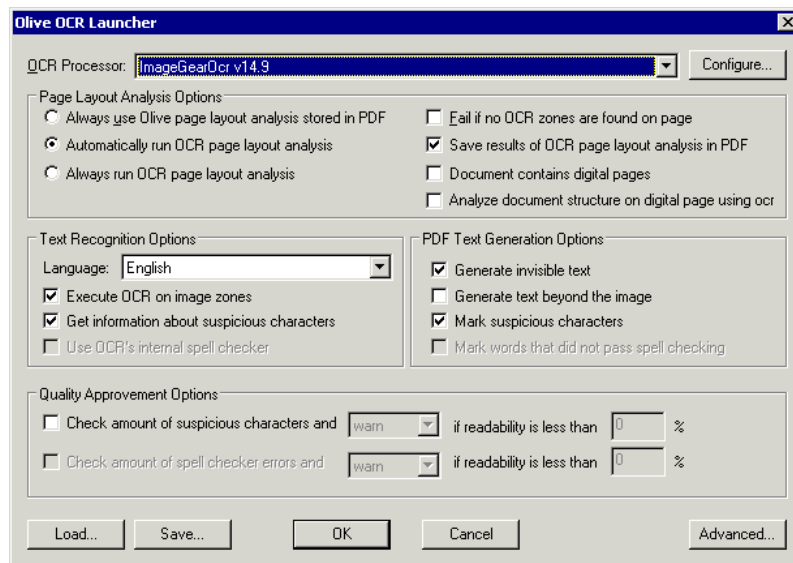


Figure 52 – OCR Launcher

Advanced OCR options include:

- **Trace folder:** location of folder where trace information files (used by Olive Professional Services reps) are to be stored
- **Binarization** options: whether the image is to be made monochrome (black and white) before OCR is run, and if so what level separates the two values. If this option is not checked, the OCR engine automatically attempts to do this but not necessarily with the most effective threshold value.
- **Failure Parameter:** whether the entire job is to fail when one page fails. (If this option is not checked, the job continues after a problematic page is encountered, and OCR results are not obtained for any problematic pages.)

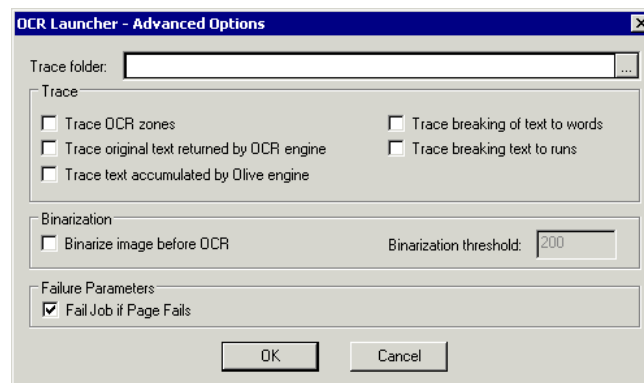


Figure 53 – OCR Launcher Advanced Options

2.10 Publishing Output

Once the segmentation is complete, both automatic and manual, you're ready to publish to XML.

2.10.1 Customizing Publish Properties

Once you have finished editing the segmentation of a document, set the issue date. Entering the correct information here is very important, because the date components determine the periodical issue's location in the XML Warehouse (Repository). After entering the correct date, output the document to XML, completing the conversion process.

By default, the date is set to the current date.

► **To change the issue date:**

1. In the **Olive** menu, point to **Publish**, and then choose the **Properties** command. The **Document Properties** dialog box is displayed:

Name	Value
------	-------

Figure 54 – Document Properties

Name	Value
------	-------

Figure 55 – Add/Edit MetaData

The following table describes the **Document Properties** dialog box parameters and buttons:

Field	Description
Publication	Displays the publication name
Issue Date	Select the Issue Date
Output file identifier	Displays the output file name
Profile	Displays the .ini file
ActivePaper Version	Displays ActivePaper's version
Metadata: Add	Opens a dialog for specifying a new metadata (name, value) pair
Metadata: Edit	Opens a dialog for modifying the highlighted metadata entry.
Metadata: Remove	Removes the highlighted metadata entry.

2. Display the **Issue Date drop-down** list:



Figure 56 – Issue Date Drop-down Calendar

3. Scroll through months using the arrows, and select the appropriate date. Click **OK**.

2.10.2 Individual (Page) Output Parameters

In case the profile is set for two layers (graphic and text) on output (relevant only for Active Magazine), the user is able to set some parameters for every page individually. The parameters can be set in the **Page Properties** dialog.

These settings are stored within the PDF and can be cleaned using the **Clean PDF** dialog, where a special option is added for this. The Republish process can detect changes in these settings; in other words, if the user has created output and since then has changed parameters for some pages, pressing the **Validate** button in the **Republish** dialog marks those pages for republishing.

2.10.3 Outputting to XML

Before performing the **Output to XML** operation, make sure you have sufficient network permissions to write to the disk drive containing the XML Warehouse (Repository). You may want to consult your system administrator.

► To output the document to XML:

1. In the **Olive** menu, point to **Publish**, and then choose the **Output to XML** command.
XML Distiller checks a set of parameters prior to performing the **Output** function. These parameters appear in the **Output Properties** window.

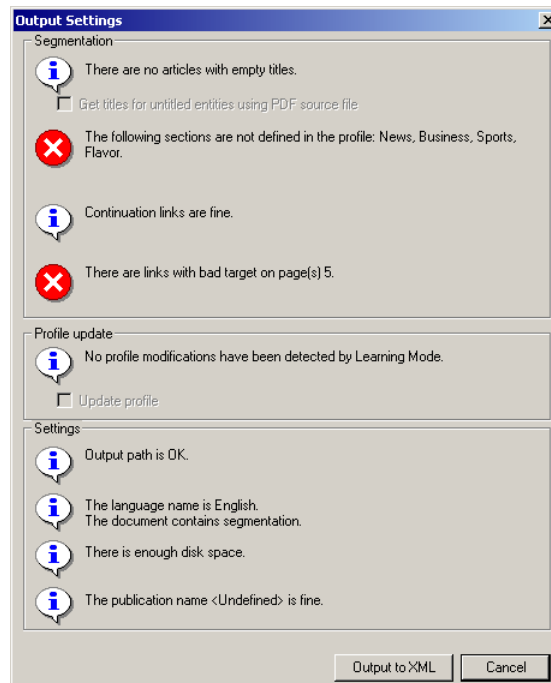


Figure 57 – Output Properties (Status)

2. The is replaced by a in the event of an issue needing resolution.

These issues can be resolved as follows:

Segmentation

Entity Titles:

Check the “**Get titles for untitled entities using PDF source file**” to allow ActivePaper to grab titles for these entities from the PDF.



Untitled entities will be shown as “**Untitled**” during searches done by the end user.

Profile

Profile Update:

If you are using **Learning Mode** and have attempted to output a file without saving changes to the Profile, an alert is indicated. By checking the **Update Profile** checkbox you can request that your changes to be saved to the Profile automatically.

Settings

Output Path:

If the document you are publishing has the same date and publication as an existing document, then an alert is indicated. Continuing with the Output will overwrite data in the existing publication date directory.

If you are updating an existing publication, it is recommended that you delete the original output directory prior to Outputting again. This prevents misnamed and redundant files.



The Output directory is under **Repository**:

[drive]:olive\repository\[Publication]\[yyyy]\[mm]\[dd]\

Content Check:

If no language is specified in the Profile, contact Olive Software Technical Support for assistance.

An alert is also indicated if no segmentation has occurred. In this event, run and edit segmentation prior to output.

Disk Space:

If there is not enough disk space to output this publication, an alert is indicated. See your network administrator to solve this problem.

1. Check the **Keep intermediate data** checkbox when temporary output is required and you intend to continue to work on the same PDF file.
2. Click the **Output to XML** button to output the PDF file to XML, or **Cancel** to cancel the output operation.

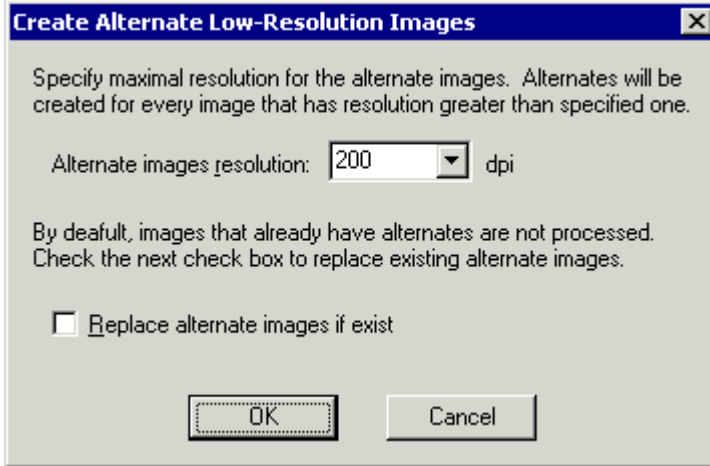
2.10.4 Creating/Viewing Low-Resolution Images

If there is a need to reduce file sizes and quality is not compromised, lower-resolution images can be created.

► To create low-resolution images for the document:

1. In the **Olive** menu, point to **Document**, and then choose **Create Low-resolution Images**.

The **Create Alternate Low-resolution Images** window appears:



2. Specify the desired image resolution (in dots per inch).
3. Check **Replace alternate images if exist** if you do not want to save any existing image files.
4. Press **OK**.

► **To view low-resolution images created for the document:**

- Click the **Olive** menu, select **Document**, and then **View Low-resolution Images**.

The window is refreshed, displaying the document using the new resolution.

2.10.5 Republishing

There are occasions when changes are made to a publication after the user has published the material to the XML Warehouse (Repository). Republishing allows the user to avoid a complete publish of an issue again. With Republishing, the user only needs to publish the pages that were changed.

► **To Republish one or more pages in an issue:**

1. Once changes have been made to an issue that has already been published, select the **Olive** menu, and click **Publish**. In the **Publish** sub-menu, click **Republish**.

The following dialog box appears:

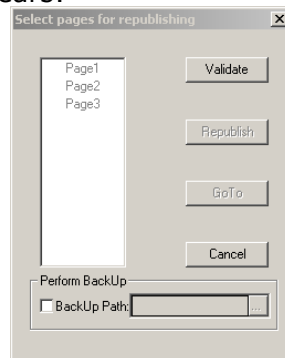
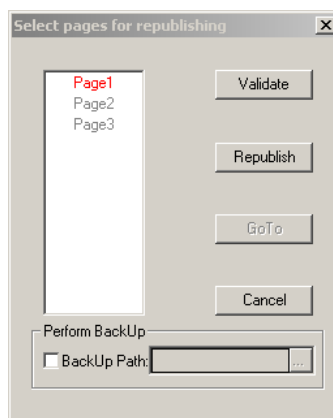
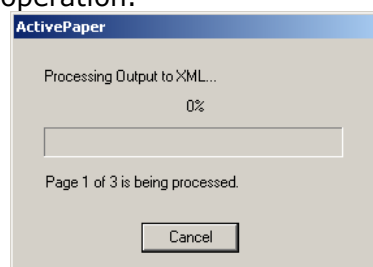


Figure 58 – Pages for Republishing

2. In the **Pages for republishing** dialog box (above), the pages in the published issue are listed in the list box at the left.
3. Click the **Validate** button.
Those pages that have been changed are highlighted.



4. If you wish to view the contents of a certain page, select the page in the box and click the **GoTo** button.
To perform a backup of the original issue being republished, click the **BackUp** check box, and select a path for the backup of the issue.
5. Click **Republish**. The following progress window appears, indicating the status of the Republish operation.



2.10.6 End-User Viewing of the Publication

In the user interface to the publication, clicking on a link to a page opens the requested page in the Page View area (replacing an existing page if present). When appropriate, this operation causes the current issue to be replaced (for example, when an entity is opened from the Search Results listing).

The look and feel of the output varies, depending on the web browser being used:

HTML Clients

In HTML Clients, links are indicated by a single underline below the link. In HTML Client Entity paper view and Page view, the skin determines the line height and color.

In the default skin, **“See Also”** links are located under the **“Links”** menu. The list of links is available in the Article view. The list appearance is determined by the skin.

XML Clients

In XML clients, the list of links is available in Entity view. The list's appearance is determined by the skin.

In the default skin, “**See Also**” links appear in the Info area. A list of links is available in Article view. The list appearance is determined by the skin.

2.11 QA Tool for Scanned Periodicals

2.11.1 Introduction

XML Distiller aims to digitize archived images of periodicals, converting them into structured information objects that can be accessed and searched. To measure the quality of this process, Olive Software has developed an independent quality metric that aggregates the problems of digitization of microfilm and other archival media, and measures the degree to which they have been solved by XML Distiller.

The quality metric is built on three components that make up the overall quality of a digital periodical archive:

- | | |
|----------------------|---|
| Readability | Defined as the ability to read and understand text and images presented on a page. For example, a page with some text blurred out would have lower readability than a page where all the text is clear. |
| Searchability | Defined as the ability to reach relevant content by providing search criteria. The more accurate an archive search is, the higher the level of searchability. |
| Presentation | Defined as the degree to which the periodical's information objects are displayed aesthetically and logically, allowing for comfortable reading. |

For example, let us say an article is mistakenly defined by XML Distiller as two different articles: one containing the top of the original article and the other containing the bottom. An article split in this way cannot display logically to the end user, who is not be able to read it from start to finish without closing one window and opening another.

It is important to point out that while such an article would not be presented perfectly, it could still be completely readable and searchable, according to the definitions above.

Each of the three quality components – readability, searchability and presentation – is measured separately, per issue.

Each issue receives a *Readability Score*, a *Searchability Score* and a *Presentation Score*, on a scale of 0 to 100. Issues start with a perfect score of 100 for each of the three components, and points are deducted from the relevant score upon identification of phenomena known to decrease readability, searchability or presentation, in any page of the issue. Points are deducted for each occurrence of a problem in a page of the issue.

Finally, the three scores are aggregated into an overall *Issue Quality Score*.

Olive Software's engineers have examined the effect of each quality component on the end-user's overall experience in the digital periodical archive. This research has yielded the *Issue Quality Formula*, which defines a weight for each component, allowing you to calculate an *Issue Quality Score*:

Quality component	Weight
Readability	0.6
Searchability	0.25
Presentation	0.15

For example, let us say an issue has achieved a Readability Score of 90, a Searchability Score of 80 and a Presentation Score of 89. The overall Issue Quality Score is calculated as follows:

$$\begin{array}{ccccccc} (90 \times 0.6) & + & (80 \times 0.25) & + & (89 \times 0.15) & = & 87.35 \\ \text{WEIGHTED READABILITY} & & \text{WEIGHTED} & & \text{WEIGHTED} & & \text{ISSUE QUALITY} \\ \text{SCORE} & & \text{SEARCHABILITY} & & \text{PRESENTATION} & & \text{SCORE} \\ & & \text{SCORE} & & \text{SCORE} & & \end{array}$$

The QA Tool allows you to quickly go through a sample issue, identifying and scoring readability, searchability and presentation problems.

2.11.2 QA Workflow

Following is a recommended workflow for the QA Tool. The sequence of steps 2, 3 and 4 can be changed.

Step	Description	Page Reference	
1.	Define the sample in the current issue	In very large issues, mark some pages <i>disabled</i> to exclude them from the sample and from the Issue Quality Report.	114
2.	Identify and score readability problems for the current page	Find readability problems on the page and define their magnitude using the QA Page Problems dialog.	116
3.	Identify and score searchability problems for the current page	Find searchability problems on the page and define their magnitude using the QA Page Problems dialog.	121
4.	Identify and flag presentation problems on the current page	Use the QA Tool Context Menu to flag presentation problems in the page’s segmentation.	124
5.	Repeat steps 2 to 4 for all remaining pages in the issue	(With the exception of pages you have excluded from the sample)	See above
6.	Generate a report for the issue	Generate an Issue Quality Report showing the Issue Quality Score, breakdowns thereof, and details of problems found in the issue.	128

2.11.3 Defining a Set of Pages as the Sample

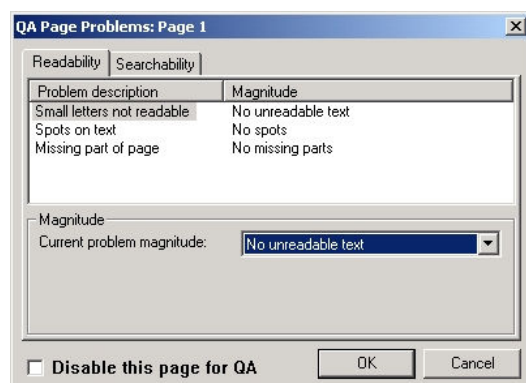
Generally, issues are tested in their entirety using the QA Tool. However, for large newspapers it is recommended to exclude some of the issue's pages from testing and from the Issue Quality Report, effectively sampling only part of an issue.

In general, it is preferable to sample fewer pages from several issues, rather than many pages from a single issue, to increase the chances that the sample is representative of the set of issues.

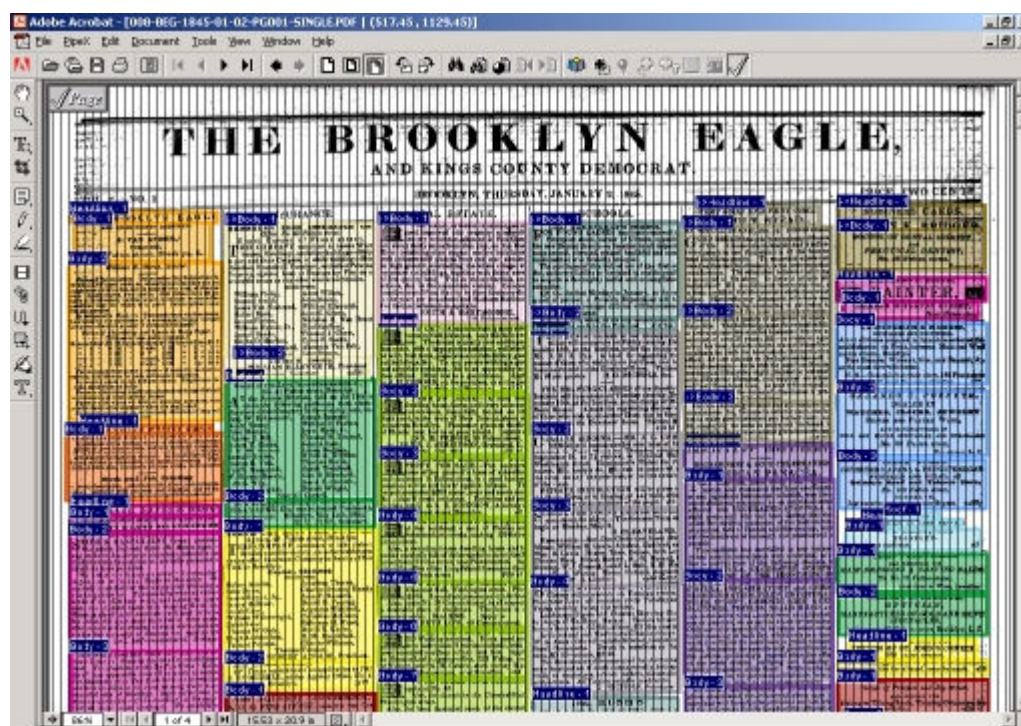
When you start the QA tool, all pages in the PDF are included in the sample by default.

► To exclude some of the issue's pages from the sample:

1. Scroll to a page you want to exclude from the sample.
2. Double click the **✓ Page** button, found at the top left of the page, to launch the **QA Page Problems** dialog.



3. Check the box labeled **Disable this page for QA**, found at the bottom left of the dialog.
4. Press **OK**.



Horizontal black lines are overlaid on the current page, and its entities cannot be selected in the QA Tool. The page is excluded from the sample, and any problems found on it, if they were scored at all, are not reflected in the Issue Quality Report.

5. Scroll to the next page to exclude from the sample and repeat steps 2 to 4.

► **To exclude most of the issue's pages from the sample:**

1. In the **Olive** menu, point to **QA Tool** and select **Disable All Pages**.
All pages in the issue are marked for exclusion from the sample and from the Issue Quality Report. Horizontal black lines obscure the entire issue, and entities cannot be selected.
2. Scroll to a page you want to include in the sample.
3. Double click the ✓ **Page** button, found at the top left of the page, to launch the QA Page Problems dialog.
4. Uncheck the box labeled **Disable this page for QA**, found at the bottom left.
5. Press **OK**.
The horizontal black lines disappear, and the page is marked for inclusion in the sample and in the Issue Quality Report.
6. Repeat steps 2 to 5 for any additional pages you want to include in the sample (keeping in mind that you began by excluding all pages from the sample).

► **To include a page in the sample, after excluding it:**

1. Scroll to a page you have excluded from the sample. You can recognize it by the horizontal black lines obscuring it.
2. Double click the ✓ **Page** button, found at the top left of the page, to launch the QA Page Problems dialog.
3. Uncheck the box labeled **Disable this page for QA**, found at the bottom left.
4. Press **OK**.

The horizontal black lines disappear, and the page is marked for inclusion in the sample and in the Issue Quality Report.

► **To include all issue pages in the sample:**

- In the **Olive** menu, point to **QA Tool** and select **Enable All Pages**.

All pages in the issue are marked for inclusion in the sample and in the Issue Quality Report. The horizontal black lines obscuring excluded pages disappear, and their entities can again be selected.

2.11.4 Readability Scoring

► **To score readability problems in the current page:**

1. Click  in the top toolbar to set the page to "fit in view."
2. Make sure the QA Tool is active. If not, activate it by pressing  in the toolbar.
3. Double click the ✓ **Page** button, found at the top left of the page, to launch the **QA Page Problems** dialog. Click the Readability tab.

Problem description	Magnitude
Small letters not readable	No unreadable text
Spots on text	No spots
Missing part of page	No missing parts

Magnitude
Current problem magnitude: No unreadable text

☐ Disable this page for QA

OK Cancel

The table taking up the top half of the dialog shows readability problems and their current magnitude settings. By default, all problems show up with “zero” magnitude. If you identify a readability problem on the page, you can define its proper magnitude using the **Current problem magnitude** drop-down.

4. Select “Small letters not readable” in the problems table. Now look over the page to see whether this problem occurs, and if it does, how serious it is. This problem exists if segmentation blocks (body text, picture captions, etc.) are missing over some or all of the small text in the page.

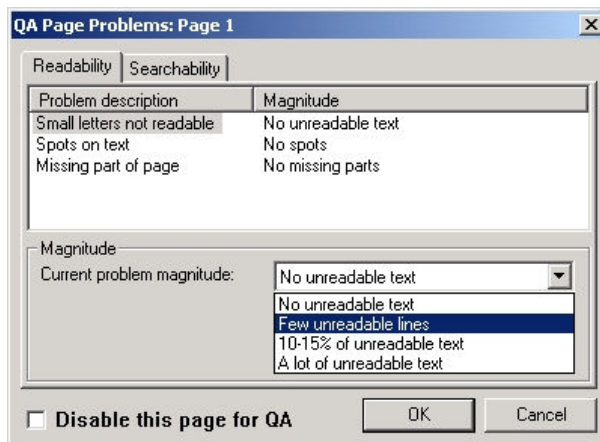
The reason is that if small text on the page was unreadable, it is likely that it was ignored by the segmentation engine.



In the example above, it is easy to see that a part of the extreme right column, which contains tiny body text, is not covered by segmentation. This indicates that small text is not readable in that area, and so you would select “Few unreadable lines” from the **Current problem magnitude** drop-down.

Step-by-Step Periodicals Processing

QA Tool for Scanned Periodicals



- If all small text appears to be covered by segmentation, meaning the problem does not exist in this page, select "No unreadable text" from the **Current problem magnitude** list.
 - If you can only see a few lines of small text not covered by segmentation, as in the example above, select "Few unreadable lines."
 - If you estimate 10-15% of small text on the page is not covered by segmentation, select "10-15% of unreadable text."
 - If you estimate that a larger portion of the page is not covered, select "A lot of unreadable text."
5. Select **Spots on text** in the problems table. Note that the **Current problem magnitude** drop-down changes to show the possible magnitudes for this problem.

Look over the page to see whether this problem occurs, and if it does, how serious it is. The symptom for this problem is spots, stains or noise clouds big enough to obscure entire words on the page.

If you see something that looks like stains, spots or noise, but you cannot be sure in the current zoom level, and with the segmentation overlaying the page, press **OK** to save and close the **Page Problems** dialog, and activate the Acrobat Zoom Tool by clicking the standard Zoom button in Acrobat's toolbar. This turns off the QA Tool and the segmentation visualization, allowing you to see the page without any visual distractions. You can then use the tool to enlarge portions of the page and check spots or noise to see how serious they are. When you are finished examining the page, activate the QA Tool again, launch the **Page Problems** dialog, and continue.



In the image above, it is easy to see that there are large black stains obscuring text on the left and top of the page. These particular stains were caused by moisture on the original printed page. In this case, you would select "A lot of spots" from the **Current problem magnitude** drop-down.

Problem description	Magnitude
Small letters not readable	No unreadable text
Spots on text	No spots
Missing part of page	No missing parts

Magnitude

Current problem magnitude: No spots

No spots

Few spots

A lot of spots

☐ Disable this page for QA

OK Cancel

- If there are no spots or noise on the page, select "No spots" from the **Current problem magnitude** drop-down.
 - If there are only several spots, small stains or minor noise on the page, with not much text obscured, select "Few spots."
 - If there is a large number of spots or heavy noise obscuring considerable portions of the text on the page, as in the example above, select "A lot of spots."
6. Select **Missing part of page** in the problems table. Note that the **Current problem magnitude** drop-down changes to show the possible magnitudes for this problem.

Look over the page to see whether this problem occurs, and if it does, how serious it is. The symptom for this problem is that the page is torn or cut, and as a result, parts of it are missing.

Step-by-Step Periodicals Processing

QA Tool for Scanned Periodicals



The page depicted in the image above is torn. Pieces of it are missing at the top and bottom, and a strip on the left is missing as well. In this case, you would select "Small part is missing" from the **Current problem magnitude** drop-down.

QA Page Problems: Page 5

Readability | Searchability

Problem description	Magnitude
Small letters not readable	No unreadable text
Spots on text	No spots
Missing part of page	No missing parts

Magnitude

Current problem magnitude:

No missing parts
No missing parts
Small part is missing
Large part is missing

☐ Disable this page for QA

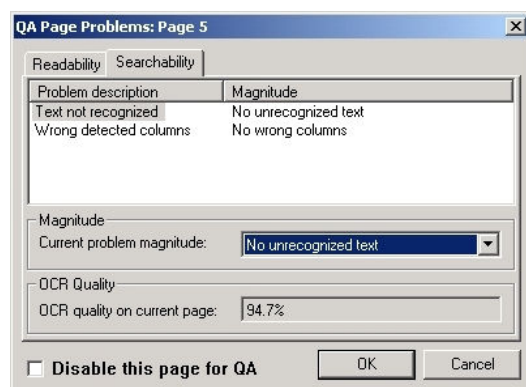
OK Cancel

- If the page is completely intact, select "No missing parts" from the **Current problem magnitude** drop-down.
 - If only a small part of the page is missing, as in the example above, select "Small part is missing."
 - If a big chunk of the page is missing – either because the page was cut or because a large part of it was torn off – select "Large part is missing."
7. To save your changes, press **OK**. Alternatively, click the **Searchability** tab to begin defining searchability problems in the page.

2.11.5 Searchability Scoring

► To score searchability problems in the current page:

1. Click  in the top toolbar to set the page to "fit in view."
2. Make sure the QA Tool is active. If not, activate it by pressing  in the top toolbar.
3. Double click the  **Page** button at the top left of the page, to launch the **QA Page Problems** dialog. Click the Searchability tab.



The table taking up the top half of the dialog shows searchability problems and their currently selected magnitude. By default, all problems show up with "zero" magnitude. If you identify a searchability problem on the page, you can define its proper magnitude using the **Current problem magnitude** drop-down.

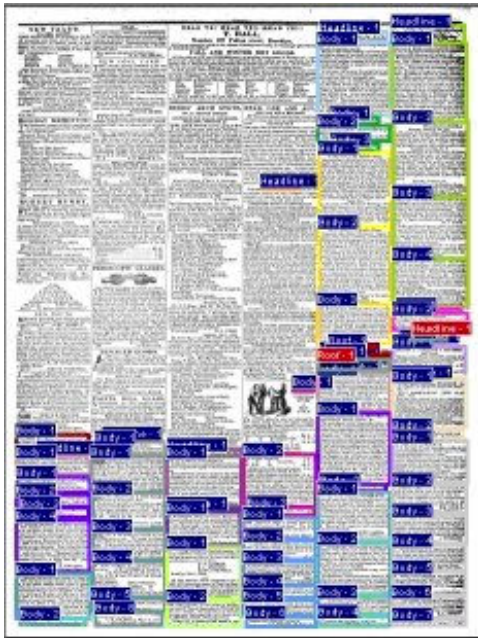
The OCR Quality area at the bottom of the dialog shows the OCR accuracy figure for the current page.

4. Select **Text not recognized** in the problems table, then look over the page to see whether this problem occurs, and if so, how serious it is. The symptom for this problem is that segmentation blocks are missing over pieces of text in the page.

If there is no segmentation defined for a piece of text on the page, this almost certainly means that OCR was not performed on that text, because the segmentation engine defines the text areas on which OCR is performed. If, consequently, the text was not recognized by OCR, it will not be searchable at all for the end user.

Step-by-Step Periodicals Processing

QA Tool for Scanned Periodicals



In the example above, you can see that OCR was not performed on a big portion of the page – the top left quarter, which is not covered by segmentation at all. In this case, you would select “Big area of text” from the **Current problem magnitude** drop-down.

QA Page Problems: Page 3

Readability Searchability

Problem description	Magnitude
Text not recognized	No unrecognized text
Wrong detected columns	No wrong columns

Magnitude

Current problem magnitude: No unrecognized text

OCR Quality

OCR quality on current page: 10-15%

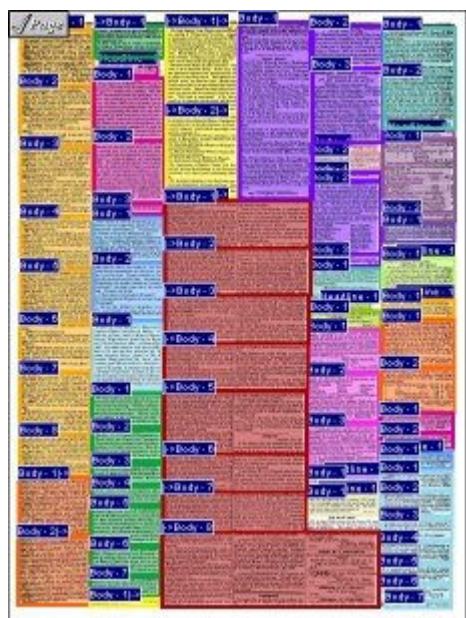
☐ Disable this page for QA

OK Cancel

- If all text appears to be covered by segmentation, select “No unrecognized text” from the **Current problem magnitude** drop-down.
- If you can only see a few words not covered by segmentation, select “Few unrecognized words.”
- If you estimate up to 5% of the page is not covered by segmentation, select “Up to 5%.”
- If you estimate 10-15% of the page is not covered by segmentation, select “10-15%.”
- If you estimate that a larger portion of the page is not covered, as in the example above, select “Big area of text.”

5. Select **Wrong detected columns** in the problems table. Note that the **Current problem magnitude** drop-down changes to show the possible magnitudes for this problem.

Look over the page to see whether this problem occurs, and if it does, how serious it is. The symptom for this problem is that segmentation blocks span two or more columns.



In the example above, most of the blocks comprising the large red entity span two columns, and one of them spans three. This is a sure indicator of erroneous column recognition. In this case, you would select "Big area" from the **Current problem magnitude** drop-down.

QA Page Problems: Page 2

Problem description	Magnitude
Text not recognized	No unrecognized text
Wrong detected columns	No wrong columns

Magnitude

Current problem magnitude: No wrong columns

OCR Quality

OCR quality on current page: No wrong columns
Up to 5% of page
10-15%
Big area

☐ Disable this page for QA

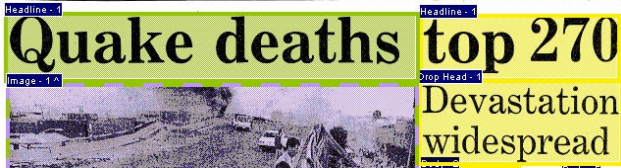
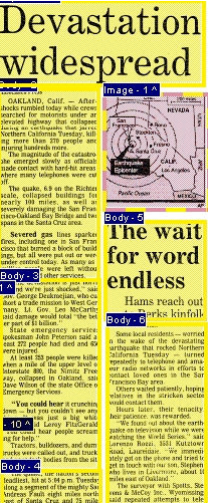
OK Cancel

- If there are no badly recognized columns in the page, select "No wrong columns" from the **Current problem magnitude** drop-down.
- If you estimate that up to 5% of the page is effected by erroneous column recognition, select "Up to 5% of page."
- If you estimate that 10-15% of the page is effected, select "10-15%."

If you estimate that a larger portion of the page contains badly recognized columns, as in the example above, select "Big area."

2.11.6 Presentation Scoring

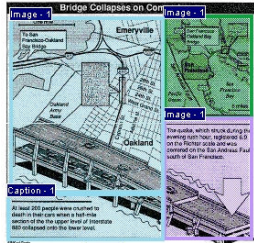
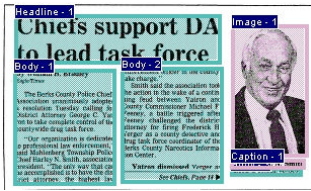
The following table describes and illustrates the various presentation problems you may come across.

Name, Abbreviation and Description	Entities Affected	Example
Title broken (Tbr). One part of a title defined as belonging to one article, with another part defined as belonging to another.	Two or more	
Title not completed (Tnc). Title only partially defined in two or more articles.	Two or more	
Joined articles (JnA). Two or more articles erroneously defined as one entity.	One	

Name, Abbreviation and Description	Entities Affected	Example
Broken article (BrA). Article erroneously defined as two or more article entities.	Two or more	
Advertisement as picture (AdP). Advertisement erroneously defined as picture. Advertisement not recognized (AdW). Advertisement erroneously defined as two or more articles	One Two or more	 
Advertisement split to pieces (AdS). Advertisement erroneously defined as several small ad entities.	Two or more	

Step-by-Step Periodicals Processing

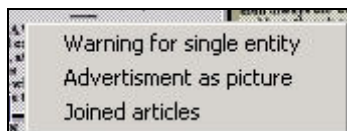
QA Tool for Scanned Periodicals

Name, Abbreviation and Description	Entities Affected	Example
Split image (ImS). Picture erroneously defined as multiple picture entities.	Two or more	
Image not linked to article (Im). Picture not associated to the article it belongs to.	Two or more	
Image not captured (ImE). Picture not defined by segmentation at all.	None (entity not defined)	

► To flag presentation problems in the page:

1. Make sure the QA Tool is active. If not, activate it by pressing  in the top toolbar.
2. Look over the segmentation on the page, and try to find one of the presentation problems detailed above.
3. When you find a presentation problem, check how many entities it effects.

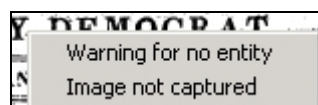
If the problem only effects one entity, right click any of its blocks to invoke the QA Tool context menu.



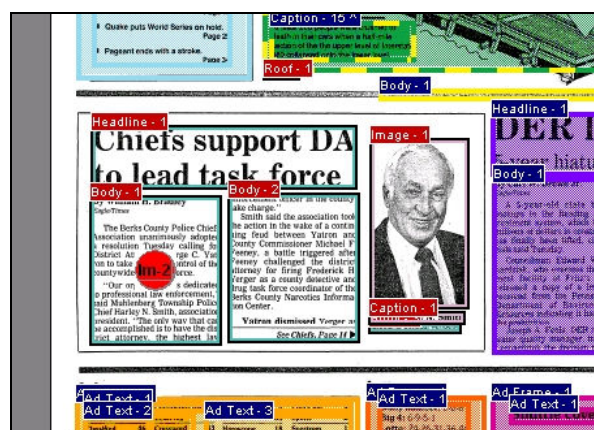
If the problem effects two or more entities, select all the entities it effects by holding down <Ctrl> and clicking a block in each of the entities. Then right-click one of the blocks in your selection to invoke the QA Tool context menu.



If the problem is that an entity has not been defined by segmentation, right click the empty page area where the entity should have been defined, to invoke the QA Tool context menu.



4. Select the presentation problem's name from the Context Menu (for the problem's official name, see the table at the beginning of this chapter).



The entity or entities you selected lose their fill color, becoming transparent, and a red circle showing an abbreviation of the problem name appears in the block you clicked (the official abbreviations appear in brackets in the table at the beginning of this chapter). If you clicked an area with no segmentation, it remains empty, but a red circle bearing an abbreviation of the problem appears where you clicked.

In the screenshot above, the presentation problem is **Image not linked to article**, abbreviated "Im."

5. If the presentation problem you have just flagged is unusual or notable in some way, double click the red circle to launch the **Description** dialog.



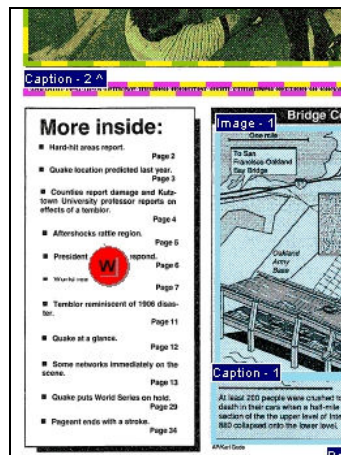
The dialog's title shows, in parentheses, the presentation problem assigned to the entity you clicked. Type in a note or description relating to the problem you have identified. Press **OK** to save your changes or **Cancel** to discard them.

6. Look over the page and try to find additional presentation problems. For each problem you find, repeat steps 3 to 5.

2.11.7 Warnings for Unusual Problems

► ➔ To flag an unusual problem:

1. When you find a new or unusual presentation problem, check how many entities it effects.
 - If the problem only effects one entity, right click any of its blocks to invoke the QA Tool context Menu.
 - If the problem effects two or more entities, select all the entities it effects by holding down Control and clicking a block in each of the entities. Then right click one of the blocks in the selection to invoke the QA Tool context Menu.
 - If the problem is that an entity has not been defined by segmentation, right click the empty page area where the entity should have been defined, to invoke the QA Tool context Menu.
2. Select the **Warning** option from the menu. (Its wording depends on the area you clicked).



The entity or entities you selected lose their fill color, becoming transparent, and a red circle with a 'W' in its center appears in the block you clicked. If you clicked an area with no segmentation, it remains empty, but a red circle with a 'W' in its center appears where you clicked.

In the screenshot above, a warning was flagged on an undefined entity (i.e., on a page area with no segmentation).

3. Double click the red circle to invoke the **Description** dialog.

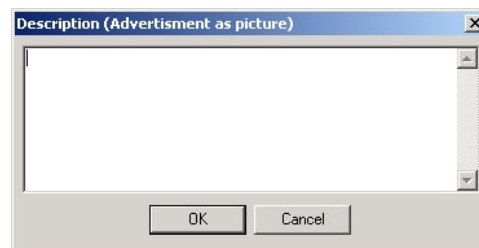


Figure 59 – QA Tool Description

4. Type in a description of the unusual problem you are flagging. Press **OK** to save your changes or **Cancel** to discard them.

A warning is set for the entity, entities or page area you clicked.

2.11.8 Saving a Report

You should only generate an Issue Quality Report once you have finished testing all of the issue's sampled pages.

► To generate an Issue Quality Report for the issue:

1. From the **Olive** menu, select QA Tool > **Make QA Report...**
The **QA Report Properties** dialog opens.

Step-by-Step Periodicals Processing

QA Tool for Scanned Periodicals

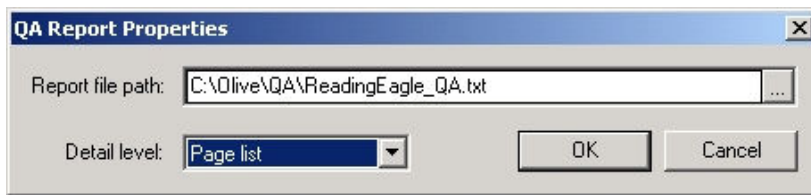
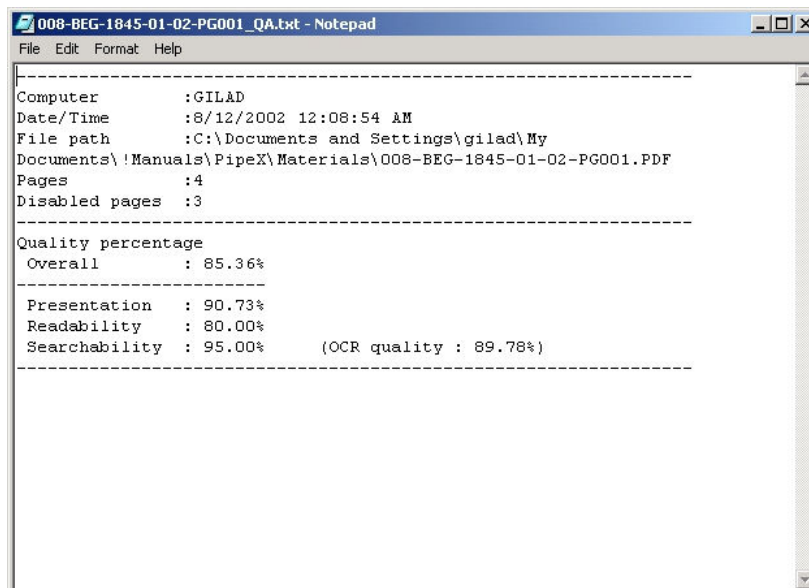


Figure 60 – QA Report Dialog

2. Use the **Report file path** field to enter a name and location for the report you are about to generate. Click ... to browse for a location.
3. Choose from the **Detail level** drop-down menu. Each level of detail builds incrementally on the previous one:
 - At the **No Details** level, the report shows:
 - the name of the computer from which the report was generated
 - the date and time the report was generated
 - the path of the issue tested
 - the number of pages used in the sample
 - the number of pages excluded from the sample
 - the Issue Quality Score
 - the Presentation, Readability and Searchability Scores
 - the average OCR accuracy for the issue.



- At the **Problems List** level, the report additionally includes a table showing the name and total number of occurrences for each presentation problem found in the sample.

```

008-BEG-1845-01-02-PG001_QA.txt - Notepad
File Edit Format Help
-----
Computer      :GILAD
Date/Time     :8/12/2002 12:15:56 AM
File path     :C:\Documents and Settings\gilad\My
Documents\Manuals\PipeX\Materials\008-BEG-1845-01-02-PG001.PDF
Pages        :4
Disabled pages :2
-----
Quality percentage
Overall       : 73.06%
-----
Presentation  : 87.08%
Readability   : 62.50%
Searchability : 90.00%      (OCR quality : 90.33%)
-----
List of errors
-----
-----
Description              Occurances
-----
Small letters not readable      (1)
Spots on text                  (2)
Missing part of page           (1)
-----
Text not recognized            (1)
Wrong detected columns         (1)
-----
Advertisement as picture       (2)
Joined articles                (1)
Broken article                 (2)
Image not linked to article    (1)
Title broken (breaks article)  (1)
Image not captured             (1)
-----

```

- At the **Detailed Problems List** level, the report additionally shows, for each presentation problem, all the pages on which the problem occurred.

```

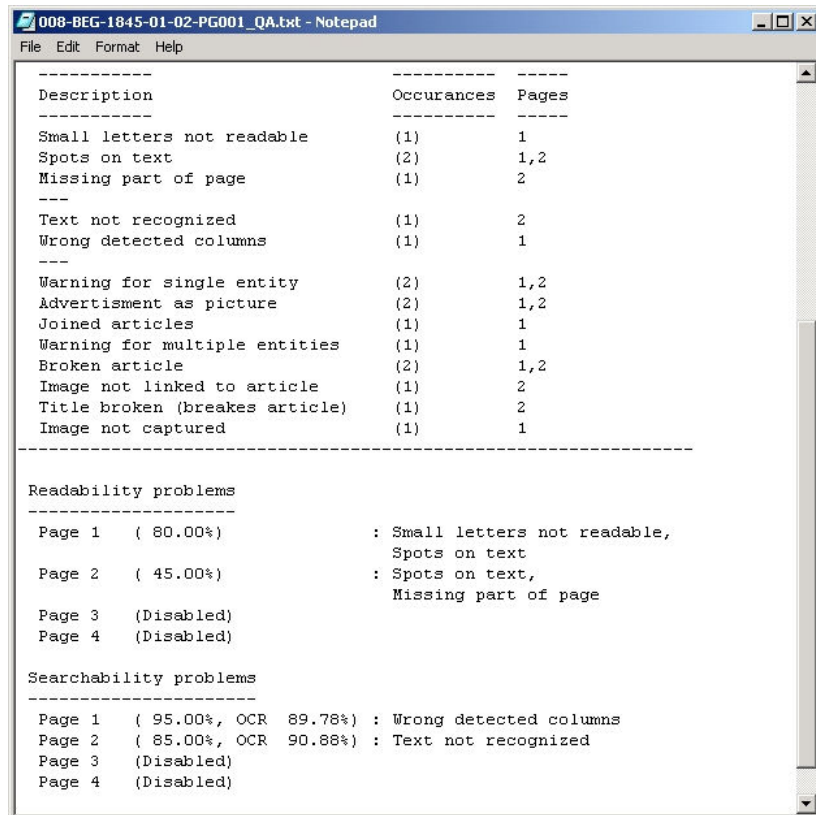
008-BEG-1845-01-02-PG001_QA.txt - Notepad
File Edit Format Help
-----
Computer      :GILAD
Date/Time     :8/12/2002 12:16:49 AM
File path     :C:\Documents and Settings\gilad\My
Documents\Manuals\PipeX\Materials\008-BEG-1845-01-02-PG001.PDF
Pages        :4
Disabled pages :2
-----
Quality percentage
Overall       : 73.06%
-----
Presentation  : 87.08%
Readability   : 62.50%
Searchability : 90.00%      (OCR quality : 90.33%)
-----
List of errors
-----
-----
Description              Occurances  Pages
-----
Small letters not readable      (1)        1
Spots on text                  (2)       1,2
Missing part of page           (1)        2
-----
Text not recognized            (1)        2
Wrong detected columns         (1)        1
-----
Advertisement as picture       (2)       1,2
Joined articles                (1)        1
Broken article                 (2)       1,2
Image not linked to article    (1)        2
Title broken (breaks article)  (1)        2
Image not captured             (1)        1
-----

```

Step-by-Step Periodicals Processing

QA Tool for Scanned Periodicals

- At the **Page List** level, the report additionally shows, for each page in the sample,
 - the Readability Score
 - readability problems
 - Searchability Score
 - OCR accuracy figure
 - searchability problems.



Description	Occurrences	Pages
Small letters not readable	(1)	1
Spots on text	(2)	1,2
Missing part of page	(1)	2
Text not recognized	(1)	2
Wrong detected columns	(1)	1
Warning for single entity	(2)	1,2
Advertisement as picture	(2)	1,2
Joined articles	(1)	1
Warning for multiple entities	(1)	1
Broken article	(2)	1,2
Image not linked to article	(1)	2
Title broken (breaks article)	(1)	2
Image not captured	(1)	1

Readability problems		
Page 1	(80.00%)	: Small letters not readable, Spots on text
Page 2	(45.00%)	: Spots on text, Missing part of page
Page 3	(Disabled)	
Page 4	(Disabled)	

Searchability problems		
Page 1	(95.00%, OCR 89.78%)	: Wrong detected columns
Page 2	(85.00%, OCR 90.88%)	: Text not recognized
Page 3	(Disabled)	
Page 4	(Disabled)	

- At the **All Details** level, the report additionally lists pages excluded from the sample. These pages are not included in quality scores.
4. Press **OK** to create a report using the parameters you entered.
 5. If you have already created a report for this issue, you are prompted to overwrite it. Press **Yes**.

The report is saved to the location you specified, and automatically opens up in Notepad.

2.12 Removing the Olive Layer (Clean PDF)

You can remove one or more of the types of data Olive has added to the input PDF.

To globally remove XML Distiller information such as segmentation, annotations, page labels, etc., open a **PDF** file and select **Clean** from the **Olive** menu.

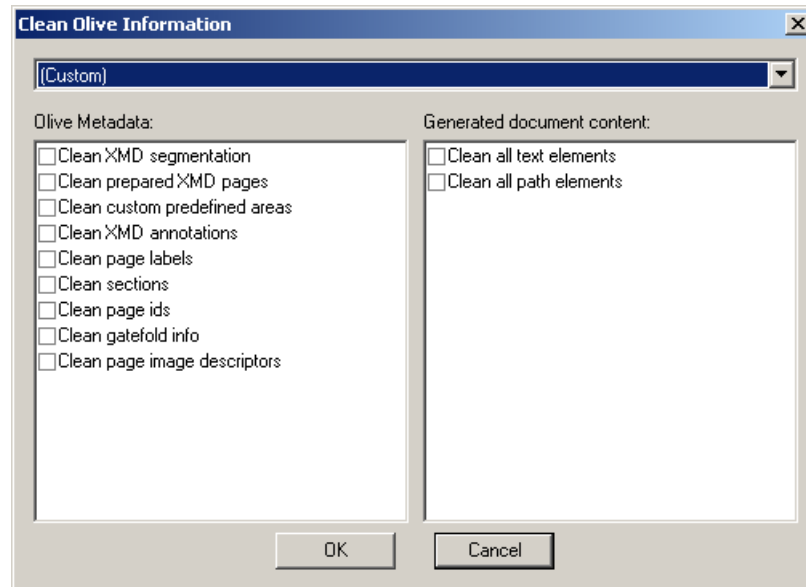


Figure 61 – Clean Olive Information

► To Clean a PDF:

1. By default, the **Clean Olive Information** dialog box appears with (Custom) selected.



Figure 62 – Clean Olive Information Options

2. Choose an option from the drop-down list, or directly select those elements that you wish to remove from the **Olive Metadata** and **Generated Document Content** windows.
3. Click **OK**.

3 Step-by-Step Processing of Enterprise Documents

In addition to processing periodicals as described in Chapter 2, XML Distiller when used as part of the ViewPoint™ platform also handles documents of various kinds that do not have the format of periodicals: academic papers, technical specifications, reports, brochures and many other types of publications.

As opposed to a periodical, which is composed of a set of unrelated articles that have to be identified, an enterprise document typically deals with a single topic and often has a hierarchical structure, with or without a distinct table of contents. ViewPoint attempts to build a table of contents and differentiates between text blocks, images, tables and other elements referred to as *Layout Objects*. After this stage processing continues to transform the objects into XML output.

This chapter contains important information about various document-processing steps. These steps can be run sequentially via menu commands, and they are also the building blocks of a ViewPoint Job for Enterprise documents (see Chapter 4). It is recommended that you become familiar with this chapter even if you are planning to do all document processing via the Job Manager.

If you have not yet done so, read the Introduction chapter for a comprehensive comparison between Periodicals and Enterprise Documents.

Refer to the two flowcharts on page 29 in the Introduction that describe the workflows for processing scanned and digital periodicals.

3.1 Overview

There are some publication components for which it is necessary that, as a supplement to the automatic segmentation done by ViewPoint, an operator intervene manually via the various XML Distiller tools. This can apply to both born-digital and scanned material. In addition, scanned material (such as microfilm) sometimes requires extra fine-tuning and quality assurance.

For a single document only: This mode can be used to see how the system processes a document one step at a time, and make adjustments at the appropriate points in the process. It is equivalent to running a single job in the Job Manager and it allows the user to better understand how the processing works, make corrections to the interim results if necessary, and output the final results.

For sample documents: Another use for this mode is in the case that a small sample is run manually using a customized profile, before running the system on a larger batch using these profile settings.

When ViewPoint is run step-by-step, some or all of the chapter's sections will apply:

- Loading a Profile
- Importing One or More Documents

- Page Size Validation
- Pre-segmentation Steps
- OCR for Scanned Documents
- Layout-object-Based Segmentation
- Publishing Output
-

Loading a Profile

QA Tool for Scanned Enterprise Documents

- Removing the Olive Layer (Clean PDF).



It is advisable that regular users of XML Distiller review the system shortcuts as listed in the **Shortcuts** tab of the Preferences dialog (**Preferences** menu item in the **Olive** menu).

3.2 Loading a Profile

This step is necessary in virtually all workflows. The only case that this is not needed is to check the results of using only the default settings in the XMD.ini file on your document.

► To load a profile:

1. In the **Olive** menu, point to **Profile** and then choose **Load**.

The **Select Profile** dialog box appears. It is identical to the regular acrobat Browse function, but is used to select a profile and not to open a PDF.

By default, the **Select Profile** dialog displays all the profiles (.ini files) saved in the \Acrobat\plug-ins directory. You can select one of the default “starting-point profiles” stored there, or browse to any directory in the file system to select a profile to load.

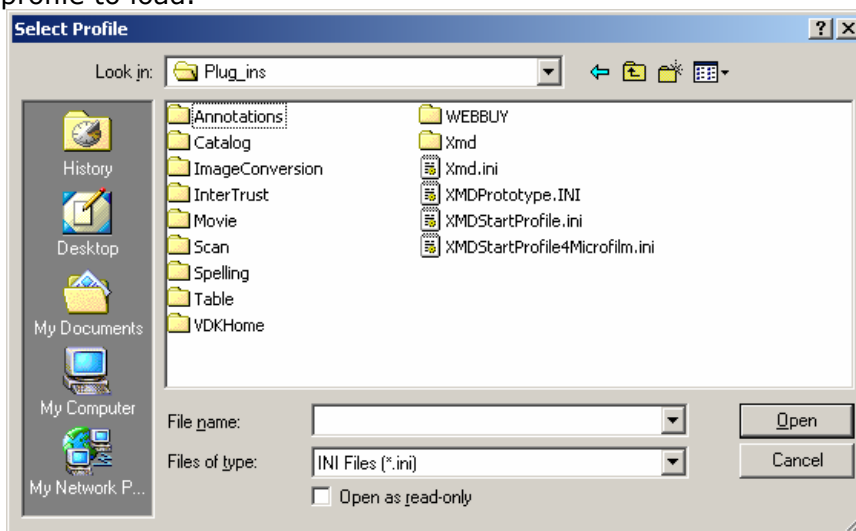


Figure 63 – Load Profile



Select Profile does not necessarily open to the default directory if you have already loaded a profile (or a PDF, using Acrobat **Browse**) in this session. In this case **Select Profile** opens to the last folder used.

2. Select the required segmentation profile (.ini file) and click **Open**.

The name of the loaded profile is displayed on the title bar, following the issue name.

All the **Olive** menu commands become available.



Once the profile is loaded, you can proceed to preparing the pages for the segmentation process, and then to the segmentation itself (See *Pre-segmentation Steps* on page 143 and *Layout-object-Based Segmentation* on page 145).

You may also choose to edit the loaded profile first. For this purpose, the **Profile** menu provides you with the optional features **Editor** and **Open**, described later in this document (See *Editing/Improving Profiles* on page 185).



For enterprise-document ("ViewPoint/EP") mode, the parameter **EnterpriseMode** must be set to 1:

```
[Global]
EnterpriseMode=1
```

You can also perform a dynamic update of the profile. This is an option in the manual segmentation process, and it is performed while working in Learning Mode. (See *Improving Profiles with Learning Mode* on page 198).

Olive provides the following default Profiles for processing of enterprise documents:

Profile	Filename	Description
Enterprise Digital Profile	EP_Digital_Documents.ini	Default profile for born-digital input of enterprise documents.
Enterprise Scanned Profile	EP_Scanned_Documents.ini	Default profile for input of enterprise documents that require scanning and OCR.
PDF Document	AP_PDFDocument.ini	Default profile for born-digital input of enterprise documents that is in PDF format.
Scientific	AP_PDFScientific.ini	Default profile for processing scientific/professional papers.
Books	AP_PDF_Books.ini	Default profile for processing books.
Office Document	AP_OfficeDocument.ini	Default profile for processing Office documents.

3.3 Importing One or More Documents

Many parameters specified in this step are relevant for a set of documents that are scanned. The process is simpler for born-digital publications that are each saved in a single file.

Note that for enterprise documents an ID number for a document is issued by the ViewPoint server and is not associated with a date as it is for periodicals.

► **To import files for a document:**

1. From the **Olive** menu's **Import/Export** submenu, select **Import Issue**. The **Configure Import Issue Parameters** dialog opens.
2. Specify the source file information in the first tab, and optionally specify additional import options and page-size validation parameters in the second and third tabs, respectively.

3.3.1 Files to Import

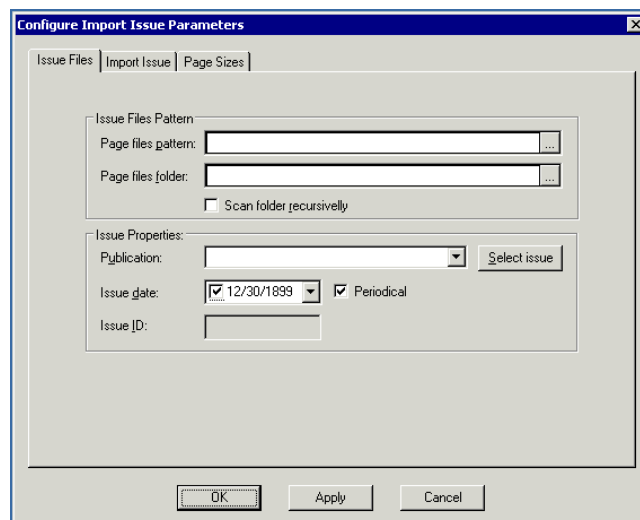


Figure 64 – Import Issue Parameters, Issue Files Tab

This tab contains the following fields and buttons:

- **Page files pattern.** Defines the filename pattern used to find issues in the file system and deduce their meta data.
- **Page files folder.** Defines the folder in which the source files to be processed reside.
- **Scan folder recursively.** Checking this indicates that the software should search all subfolders of the **Page files folder** for PDFs conforming to the filename pattern.

- **Select issue.** This button allows you to select a set of files, and is an alternative to specifying the information in the fields **Publication** and **Issue data**. If a pattern is specified in the top section (Issue files pattern) and files are specified here, the “metadata” found in the filenames using the pattern is used to populate **Publication** and **Issue data** and other metadata (see Figure 66 below).
- **Publication.** Name of the publication: typed in or selected from a list.
- **Issue date.** Opens a calendar for specifying the date of the issue.
- **Periodical.** Indicates whether the issue is a periodical (i.e., should have a date associated with it) or not.
- **Issue ID.** Relevant for documents receiving an ID not associated with a real date.

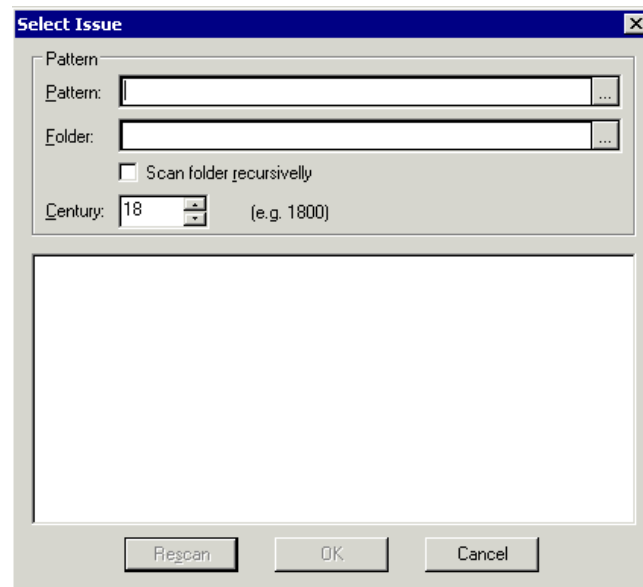


Figure 65 - Caption

Step-by-Step Processing of Enterprise Documents

Importing One or More Documents

(Project,3)
(Name,*)
(Year,4)
(Month,2)
(Day,2)
(PageNo,*)
(Status,*)
(Any,*)
(PageVersion,*)
(PortionId,*)
(IssueID,*)
(PageID,*)
(Name,*)-(Year,4)-(Month,2)-(Day,2)-PG(PageNo,*).tif
TMP_Pg(PageNo,3).tif
(Name,*)_ (Day,2)_ (Month,2)_ (Year,4)_Page(PageNo,3).tif
(Name,*)\ (Year,4)\ (Month,2)\ (Day,2)\ (Any,*).tif

Figure 66 - Filename Patterns and Pattern Components

3.3.2 Import Options

This tab contains the fields for customization of several features when a document or periodical issue is imported:

- Page Detection
- Image Deskewing
- Page Preprocessing
- Image Cropping

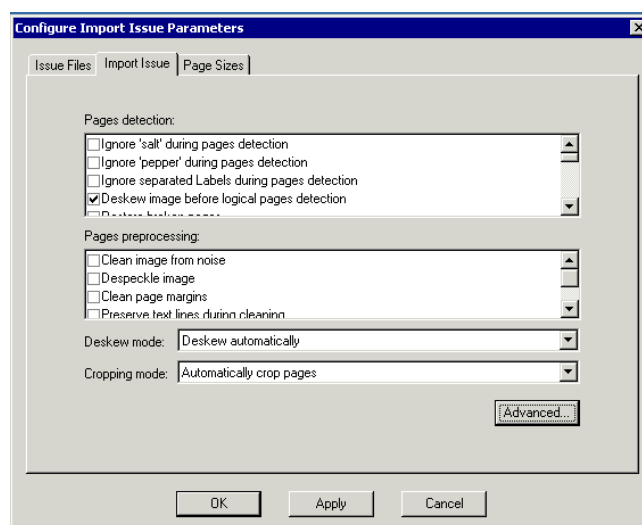


Figure 67 – Import Issue Parameters, Import Issue Tab

3.3.3 Page-Size Validation Parameters

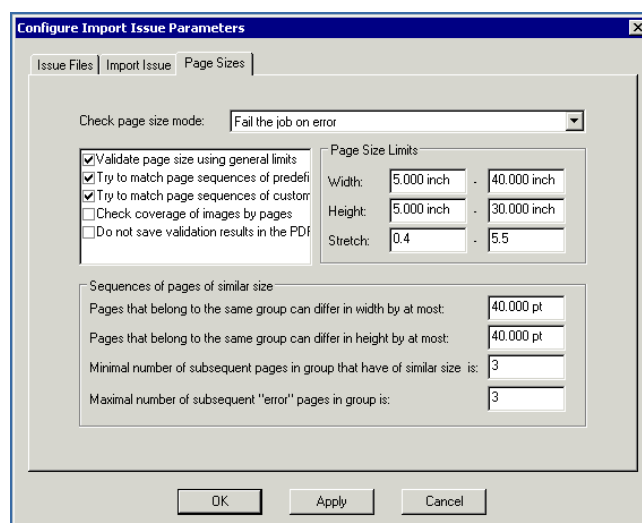


Figure 68 – Import Issue Parameters, Page Sizes Tab

This tab is relevant for publications in which sizes of one or more sets of pages should be consistent, and contains the following fields:

- **Check page size mode.** Whether to check for consistent page sizes, and what action to take if errors are encountered.
 - Checklist of options regarding how the validation is to be done and whether results should be saved in the PDF.
- **Page Size Limits:** ranges for width, height and stretch ratio of a page.
- **Sequence of pages of similar size:** constraints to use when identifying a set of adjacent similar-sized pages.

3.4 Page Size Validation

You can check for inconsistencies in the page sizes determined in the imported document.

► To validate page sizes and view results:

1. Point to **Image Import/Export** in the **Olive** menu and choose **Validate Issue Page Sizes**.

ViewPoint reviews the page sizes in the current file internally; results are not displayed.

2. Use the toolbar button (), or point to **Image Import/Export** in the **Olive** menu and choose **View Page Size Validation Log**.

A validation results window opens displaying the components illustrated below.

3. Double-click on a page icon to view the page's image details.

Page	Section	Description
12	1	Page is too narrow
21	1	Page is too wide

Figure 69 – Page Size Validation Summary Table

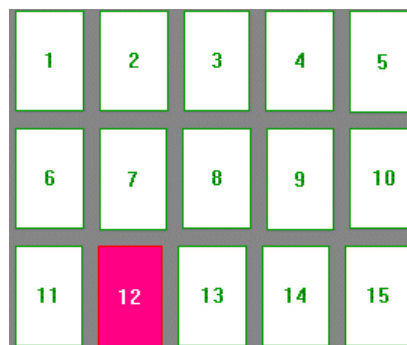


Figure 70 – Page Icons

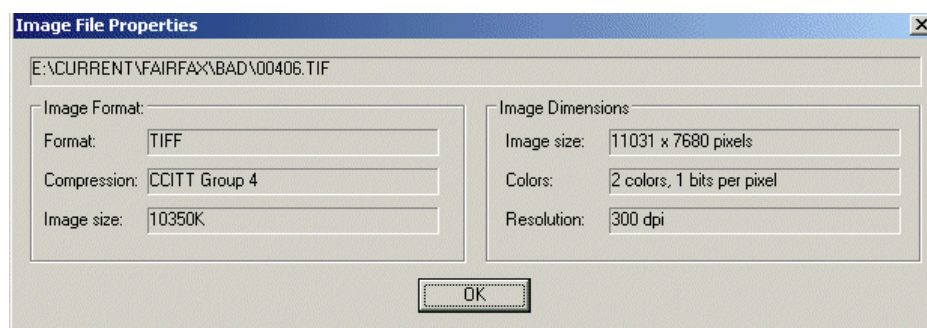


Figure 71 – Page Image Details

3.5 Pre-segmentation Steps

A number of the Pre-segmentation steps that are used for periodicals can also be used for enterprise documents, where necessary. (Manifests for ads are not relevant here.) Refer to the sample Job Types and Templates for enterprise documents to see several possible combinations.

Input in the format of TIFF files undergoes import (Import Issue) and as such does not require pre-segmentation.

Scanned Documents: After files are imported, **Trim Pages** is run when necessary, and Character Recognition is then done to digitize text contents. Other possible steps include **Split Pages** and **Merge Double Pages**.

Digital Documents: It is sometimes necessary to run **Trim Pages** on the documents before proceeding to Segmentation.



Note that in subsequent Layout-object-based Segmentation, any double pages would be ignored.

3.6 OCR for Scanned Documents

This step is done as a pre-requisite for layout-based segmentation.

OCR is now performed to digitize the text on scanned pages.

Customization settings can be saved via the **Save** button, and later accessed via the **Load** button. Advanced options, available via the **Advanced** button, are described after the standard options.

Customization options include:

- **OCR Processor:** the name of the OCR utility
- **Page layout analysis options:**
 - Always use Olive **page layout analysis stored in PDF** (as layout-based segmentation has not been run, the entire page is processed as one segment)

- **Automatically run OCR page layout analysis**
- **Always run OCR page layout analysis**
- **Text recognition Options:**
 - Language of text to be recognized
 - Whether to recognize image zones
 - Whether to collect data on suspicious characters: i.e., those that received a low confidence score by the OCR engine
 - Whether to use the spell checker provided by the utility, if relevant
- **PDF Text Generation Options:**
 - Whether to generate the green overlay ("invisible") text results
 - Whether to generate text that is outside the border of the image
 - Whether to indicate (via color coding) characters that are suspicious (i.e., received a low confidence score)
 - Where relevant, whether to indicate (via color coding) words with spelling errors
- **QA Options:**
 - Whether to check the number of suspicious characters and, if so, warn or fail upon a threshold readability level;
 - Whether to check the number of spelling errors and, if so, warn or fail upon a threshold readability level.

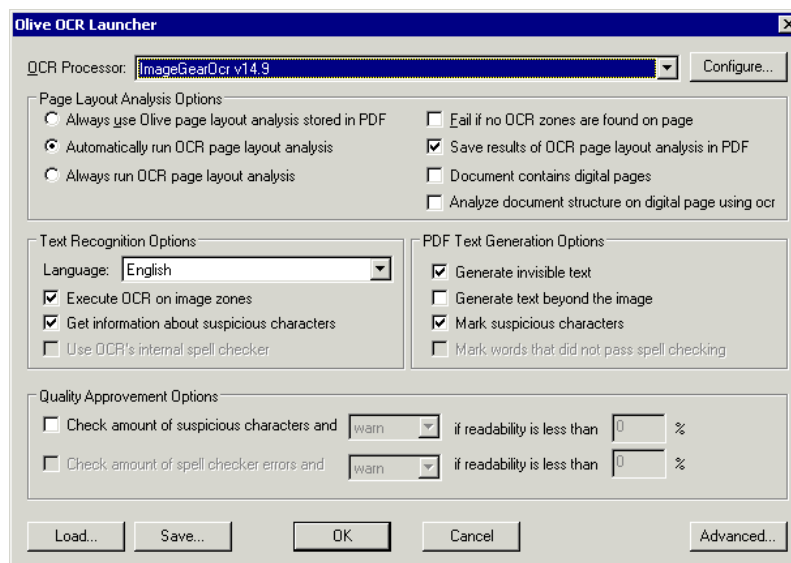


Figure 72 – OCR Launcher

Advanced OCR options include:

- **Trace folder:** location of folder where trace information files (used by Olive Professional Services reps) are to be stored
- **Binarization** options: whether the image is to be made monochrome (black and white) before OCR is run, and if so what level separates the two values. If this option is not checked, the OCR engine automatically attempts to do this but not necessarily with the most effective threshold value.
- **Failure Parameter:** whether the entire job is to fail when one page fails. (If this option is not checked, the job continues after a problematic page is encountered, and OCR results are not obtained for any problematic pages.)

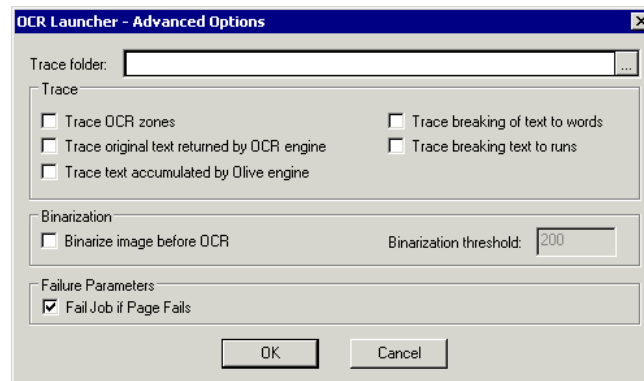


Figure 73 - OCR Launcher Advanced Options

3.7 Layout-object-Based Segmentation

Although the standard (style-based) mode of segmentation described for periodicals is effective for some types of enterprise publications, Olive has designed an alternative mode that is customized for typical enterprise documents. It is assumed that each of these documents contains a single “article” (such as a professional paper, a form, or a report) and as such does not require issue-based segmentation for clustering dispersed text blocks into a set of articles, or associating pictures with articles.

For each document ViewPoint looks for components called *Layout Objects*. Each layout object has a location and additional data used by API programming and for scripting. Layout objects can be of the following types:

- Text block, whose structure contains data such as the block’s font and contents
- Graph blocks (images and diagrams)
- Horizontal and vertical separators
- Tables, including data pertaining to the number of cells, their locations, etc.

Segmentation of a document based on Layout objects has several purposes:

- Identification of the structure of a single document (for example, a three-level outline or a flat list).
- Construction of a hyperlinked Table of Contents if relevant (see page 150 regarding two navigation modes);
- Improved table segmentation;
- Reading order specification;
- Metadata extraction;
- Optional customization of aspects of the final presentation via scripts.

Scanned documents: For non-digital material, it is necessary for OCR processing to be performed before this step.

See page 61 for details on standard Style-based segmentation if you wish to consider using that method on a document.

Reading Order: The **Reading Order** parameter determines how the text blocks are sequenced. This is significant when words are split between text blocks. For example, a hyphenated word that is partially at the bottom of the left column of text and partially at the top of the right column can be found in a search because the reading order determines that the left bottom block is followed by the right upper block.

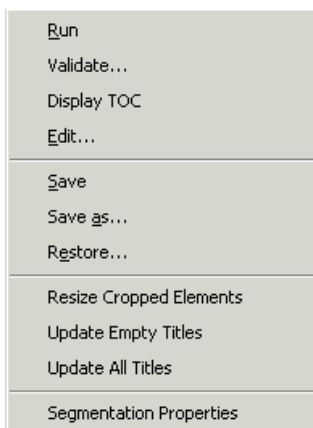
3.7.1 Full Segmentation

Typical ViewPoint users configure and run Layout-object-based Segmentation in its entirety from Olive's **Segmentation** menu.

Setting Segmentation Properties

► To select the segmentation mode:

1. In the **Olive** menu, point to **Segmentation**, and then choose **Segmentation Properties**.



The **Segmentation Properties** dialog opens, presenting the following options:

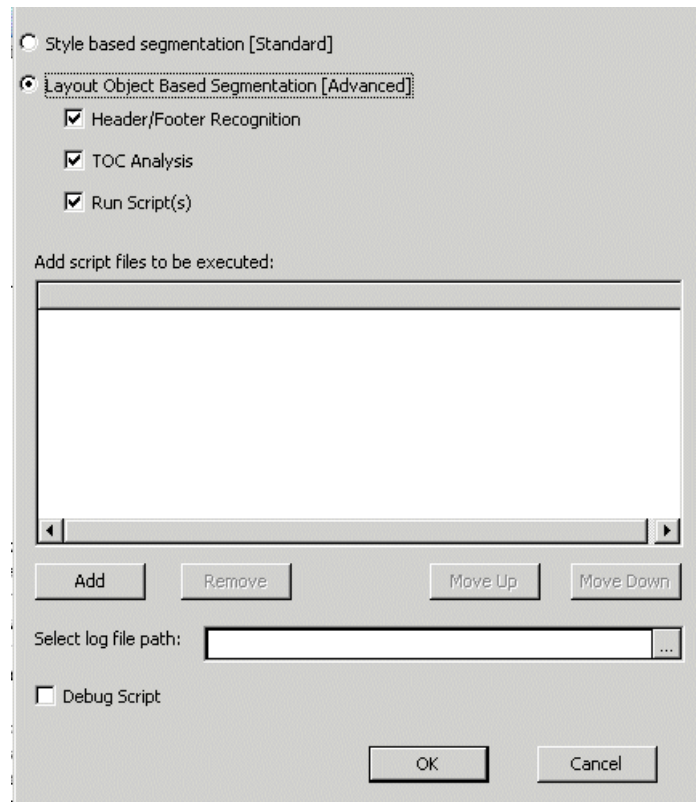


Figure 74 – Segmentation Properties (Advanced)

2. Select **Layout Object based segmentation [Advanced]**.

The fields specific to this mode are displayed.

3. Check **Header/Footer Recognition** to enable the step in which Olive attempts to identify headers and footers and remove them from the segmentation.

Check **TOC Analysis** if the document has a recognizable hierarchical structure (or a simple one-level list structure), and if you wish to create a Table of Contents in the output document containing links to the various parts of the document. Page numbers in the TOC are not used; Distiller attempts to locate the section associated with each TOC entry on its own.

Check **Run Script(s)** if there is one or more Javascript files to specify that are to be run as part of the segmentation process.

4. If you wish to add a script, press **Add** and specify a .js location and filename. Repeat for any additional script files.
5. If there are two or more scripts in the resulting list, change the sequence if necessary by selecting a script in the incorrect position and using the **Move Up** and **Move Down** buttons.
6. In **Select log file path**, specify a folder location in which the log files are to be saved.

Running Full Segmentation

► **To run layout-object-based full segmentation:**

- From the **Olive** menu, point to **Segmentation** and select **Run**.

The progress of the internal segmentation processing is indicated by a progress bar, and problematic areas (coordinates) are listed when processing is complete.

3.7.2 Basic Segmentation

Basic segmentation results are generally produced and viewed only by Olive Professional Services.

Basic segmentation takes the results of the previous step (Character Recognition for scanned material, or imported digital files after any pre-segmentation is done) and identifies separate areas of text or images, but does not differentiate between headers/footers and other parts of the page and does not analyze the text itself to attempt to link Table of Contents entries to the associated sections in the document,.

Some text blocks such as a table of contents are also broken down into smaller components such as specific lines of text, so that the user can add tags to a specific line. (For example, TOC entries at different levels could have different tags.)

This step is only done when an advanced user wants to perform intermediate functions interactively (see below).

► **To run layout-object-based basic segmentation:**

- From the **Olive** menu, point to **Layout Objects** and select **Build Layout Objects**.

The progress of the internal segmentation processing is indicated by a progress bar, and problematic areas (coordinates) are listed when processing is complete. See the following section for instructions on viewing basic segmentation results.

3.7.3 Display Layout Objects

This function is usually performed only by Olive Professional Services.

► **To display layout-object-based segmentation:**

- From the **Olive** menu, point to **Layout Objects** and select **Display Layout Objects**.

The Layout Objects resulting from the segmentation are displayed:

- **Table of Contents** (if relevant)
- **Text Blocks** (later separated into headings and their content blocks)
- **Tables**
- **Images**
- **Horizontal and vertical separators**
- **Headers/Footers** (if relevant)

3.7.4 Object, Page, Document Properties

Once basic segmentation is done, the document's segmentation information is comprised of document-level, page-level and object-level details that are accessible for scripting. This information is available after the Layout Objects are displayed (see previous procedure).

► **To view a page/document description:**

1. After following the above procedure to display segmentation objects, right-click anywhere on the active page.

The context menu is displayed.

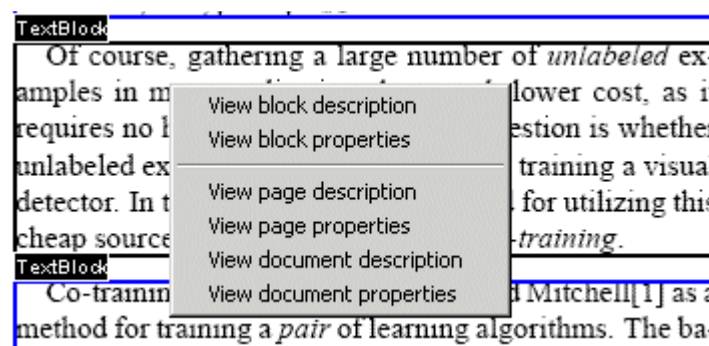


Figure 75 – Layout Object Context Menu

2. Select **View page description** or **View document description**.

The requested information is displayed in a new window.

► **To view page/document properties:**

1. After following the above procedure to display segmentation objects, right-click anywhere on the active page.

The context menu is displayed.

2. Select **View page properties** or **View document properties**.

The requested information is displayed in a new window.

► **To view an object (block or table) description:**

- After following the above procedure to display segmentation objects, double-click on the block/table or right-click and select **View Block Description** from the menu.

The requested information is displayed in a new window.

► **To view object (block or table) properties:**

- After following the above procedure to display segmentation objects, double-click on the block/table or right-click and select **View Block Properties** from the menu.

The requested information is displayed in a new window.

3.7.5 Intermediate Segmentation Functions

Launching of these functions individually (via the **Olive** menu's **Layout Objects** sub-menu) is usually performed only by Olive Professional Services.

Three intermediate segmentation functions make changes to properties of one or more layout objects:

- Header/Footer identification: separates headers and footers from the remainder of the page to be segmented;
- TOC analysis, after Headers/Footers are identified: labels the titles of each hierarchy element – for example, "Unit", "Chapter", and "Section" (using the desired number of levels, as specified in the Settings/Profile);
- Scripts (JavaScript code in .js files).

3.7.6 Final Segmentation

Results of basic segmentation are blocks identified by type (text vs. image vs. table), that are not sequentially numbered.

Final segmentation sequences the various elements identified in basic segmentation, taking into account the property changes made after basic segmentation. As in Style-based Segmentation, this process also creates a set of entities (here, from the Layout Objects) as well as a set of links (including "Web", "See Page", and "e-mail").

3.8 Publishing Output

In this stage of the ViewPoint process, the segmented material is converted to XML by XML Distiller and prepared for end-user viewing.

In most cases, a navigational aid is included to facilitate access to specific parts of the document. There are two presentation formats available:

- view by Table of Contents, where the Table of Contents contains hyperlinks to specific chapters and sub-chapters;
- view a group of pages at a time (for example: First five pages, next five pages, and so on.)

A parameter in the Settings file or Profile is used to specify which of these is to be used for the document:

`HierarchyOutputMode=0` (default) for the segmentation-based hierarchy;

`HierarchyOutputMode=1` for page grouping.

3.8.1 Customizing Publish Properties

Once you have finished editing the segmentation of a document, several properties can be specified before Output to XML.

► **To change the publication properties:**

- In the **Olive** menu, point to **Publish**, and then choose the **Properties** command. The **Document Properties** dialog box is displayed.

The following table describes the **Document Properties** dialog box parameters and buttons:

Field	Description
Publication	Displays the publication name
Issue Date	Select the Issue Date
Output file identifier	Displays the output file name
Profile	Displays the .ini file
ActivePaper Version	Displays ActivePaper/ViewPoint 's version
Metadata: Add	Opens a dialog for specifying a new metadata (name, value) pair
Metadata: Edit	Opens a dialog for modifying the highlighted metadata entry.
Metadata: Remove	Removes the highlighted metadata entry.

3.8.2 Outputting to XML

Before performing the **Output to XML** operation, make sure you have sufficient network permissions to write to the disk drive containing the XML Warehouse (Repository). You may want to consult your system administrator.

► **To output the document to XML:**

1. In the **Olive** menu, point to **Publish**, and then choose the **Output to XML** command. XML Distiller checks a set of parameters prior to performing the **Output** function. These parameters appear in the **Output Properties** window.

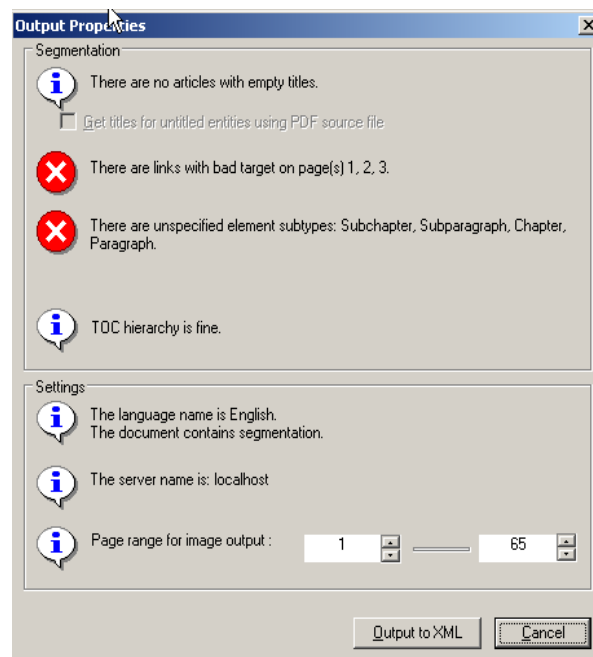


Figure 76 – Output Properties (Status)

2. The  is replaced by a  in the event of an issue needing resolution.

These issues can be resolved as follows:

Segmentation

Entity Titles:

Check the “**Get titles for untitled entities using PDF source file**” to allow XML Distiller to grab titles for these entities from the PDF.



Untitled entities will be shown as “**Untitled**” during searches done by the end user.

Profile

Profile Update:

If you are using **Learning Mode** and have attempted to output a file without saving changes to the Profile, an alert is indicated. By checking the **Update Profile** checkbox you can request that your changes to be saved to the Profile automatically.

Settings

Output Path:

If the document you are publishing has the same ID and publication as an existing document, then an alert is indicated. Continuing with the Output will overwrite data in the existing publication date directory. This case is relevant only when ActivePaper mode is on: [Global] EnterpriseMode=0).

If you are updating an existing publication, it is recommended that you delete the original output directory prior to Outputting again. This prevents misnamed and redundant files.



The Output directory is under **Repository**:

[drive]:olive\repository\[Publication]\[yyyy]\[mm]\[dd]\

Content Check:

If no language is specified in the Profile, contact Olive Software Technical Support for assistance.

An alert is also indicated if no segmentation has occurred. In this event, run and edit segmentation prior to output.

Disk Space:

If there is not enough disk space to output this publication, an alert is indicated. See your network administrator to solve this problem.

3. Check the **Keep intermediate data** checkbox when temporary output is required and you intend to continue to work on the same PDF file.
4. Click the **Output to XML** button to output the PDF file to XML, or **Cancel** to cancel the output operation.

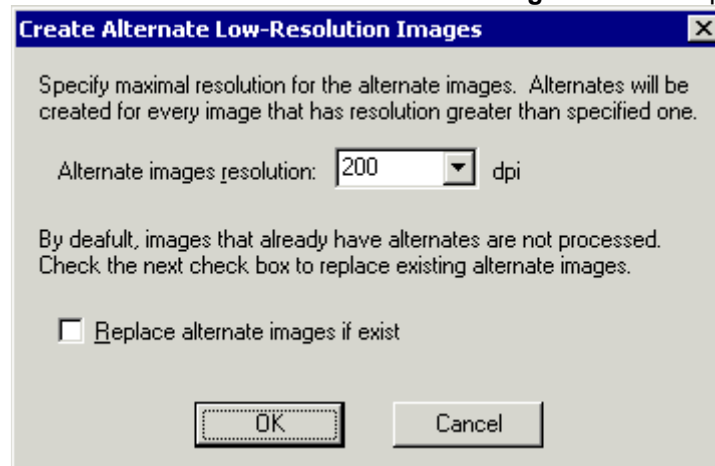
3.8.3 Creating/Viewing Low-Resolution Images

If there is a need to reduce file sizes and quality is not compromised, lower-resolution images can be created.

► To create low-resolution images for the document:

1. In the **Olive** menu, point to **Document**, and then choose **Create Low-resolution Images**.

The **Create Alternate Low-resolution Images** window appears:



Publishing Output

2. Specify the desired image resolution (in dots per inch).
3. Check **Replace alternate images if exist** if you do not want to save any existing image files.
4. Press **OK**.

► **To view low-resolution images created for the document:**

- In the **Olive** menu, point to **Document**, and then choose **View Low-resolution Images**.

The window is refreshed, displaying the document using the new resolution.

3.8.4 Republishing

There are occasions when changes are made to a publication after the user has published the material to the XML Warehouse (Repository). Republishing allows the user to avoid a complete publish of a document again. With Republishing, the user only needs to publish the pages that were changed.



For enterprise documents, Republishing is only possible if
[XML_Output]: **IndependentOfServerMode** is set to 1 in the Profile.

► **To Republish one or more pages in a document:**

1. Once changes have been made to a document that has already been published, select the **Olive** menu, and click **Publish**. In the **Publish** sub-menu, click **Republish**.

The following dialog box appears:

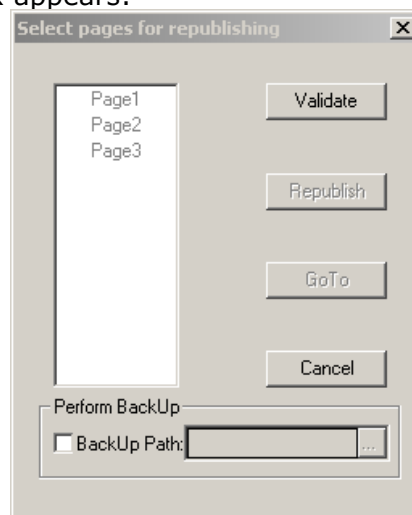
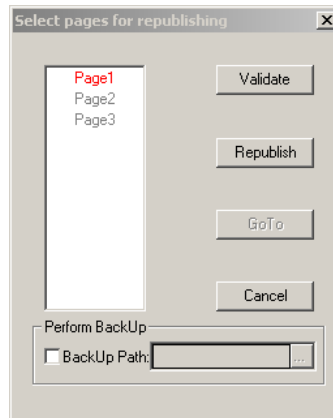


Figure 77 – Pages for Republishing

2. In the **Pages for republishing** dialog box (above), the pages in the published document are listed in the list box at the left.

3. Click the **Validate** button.

Those pages that have been changed are highlighted.



4. If you wish to view the contents of a certain page, select the page in the box and click the **GoTo** button.

To perform a backup of the original document being republished, click the **BackUp** check box, and select a path for the backup of the document.

5. Click **Republish**. A progress window appears, indicating the status of the Republish operation.

3.9 QA Tool for Scanned Enterprise Documents

The QA Tool is designed for checking the quality of large numbers of scanned periodicals, such as newspaper archives stored on microfilm. See *QA Tool for Scanned Periodicals* on page 112 if your enterprise is scanning periodicals of this type and quantity.



The QA tool can be disabled via a flag in the Settings file:

EnableQATool=0.

See *Settings and Default Values* on page 182.

3.10 Removing the Olive Layer (Clean PDF)

You can remove one or more of the types of data Olive has added to the input PDF.

To globally remove XML Distiller information such as segmentation, annotations, page labels, etc., open a **PDF** file and select **Clean** from the **Olive** menu.

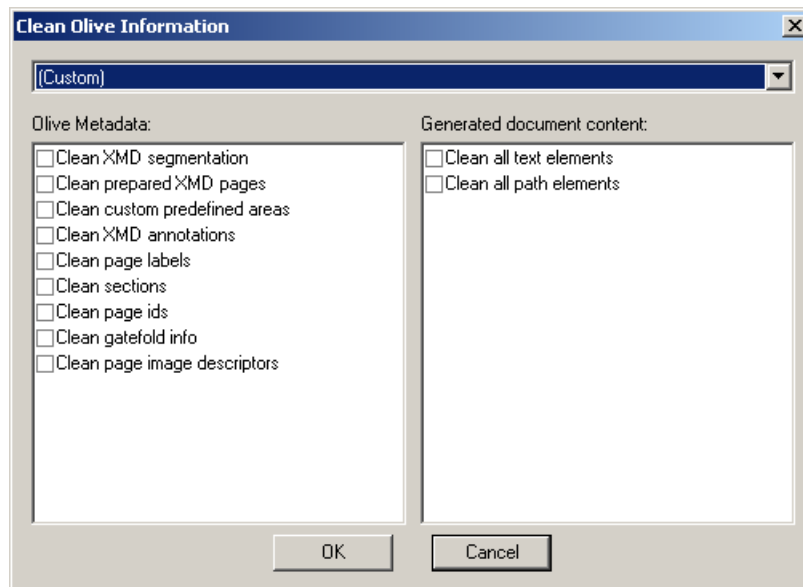


Figure 78 – Clean Olive Information

► **To Clean a PDF:**

1. By default, the **Clean Olive Information** dialog box appears with (Custom) selected.

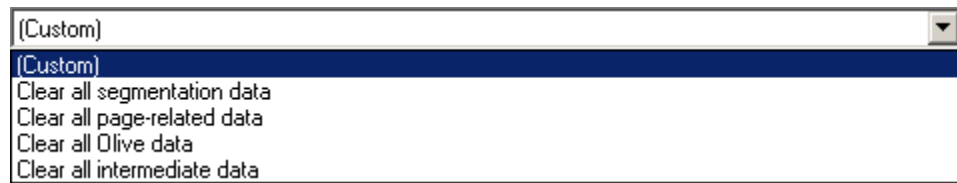


Figure 79 – Clean Olive Information Options

2. Choose an option from the drop-down list, or directly select those elements that you wish to remove from the **Olive Metadata** and **Generated Document Content** windows.
3. Click **OK**.

4 Job Management

4.1 Job Manager Overview

The Job Manager allows you to perform batch processing of a set of documents such as court proceedings and newspaper archives. Various ActivePaper/ViewPoint functions (such as Combine Pages and automatic segmentation) are performed in a preset order, on one or more documents, without the need for an operator to intervene (via menu selections or dialog boxes) between steps.

Several important Job Manager concepts are explained in this overview. Make sure you understand them before proceeding to the operating instructions in the following sections.

4.1.1 Job Descriptors

In the **Olive Job Manager**, a *Job Descriptor* is defined as a file carrying the extension DSC – descriptor, which describes exactly what the system should do with one specific document. The Job Descriptor file is a list of operating instructions, accompanied by various parameters, which the system follows to automatically process one document.

A *job* is a processing operation carried out on a document according to a certain Job Descriptor.

If it would be necessary, for example, to do automatic processing on 120 documents, then 120 Job Descriptor files would have to be created, and one job executed according to each of these files. This is not as tedious as it sounds, because of the use of templates.

Job Descriptor Templates

If you want to process 120 documents, you can simply open up a standard Job Descriptor that contains the appropriate instructions for what you want to do – a *Job Descriptor Template* – and copy it 120 times, inserting specific information for the document, such as its date and its number of pages, into each copy.

To use a simplified example, the Job Descriptor Template might say the following:

“Run automatic segmentation on the issue, then output it to XML.”

One of the Job Descriptors you create from the template would then say:

“Run automatic segmentation on the Evening Gazette issue dated June 22, 2002, which contains 54 pages, then output it to XML.”

The text in bold is specific information about the document, which is added to the copied template to form the Job Descriptor for that publication. You could do this manually, by copying and editing the template DSC file in Notepad; instead, it is done quickly and automatically by the Job Manager.

Patterns

The Job Manager uses filename *Patterns* you define to find documents in the file system and get information about them.



The underlying assumption is that all the PDF files we work with in batch processing conform to strict filename conventions. **You cannot perform batch processing on files if they do not conform to clearly defined filename conventions.**

For example, consider the filename “**EveningGazette-07-26-2002-Page001.pdf**”.

Assuming this filename was created using a strict naming convention, the naming Pattern could be something like this:

“Publication name, any number of characters; dash; month, two characters; dash; day, two characters; dash; year, four characters; dash; the word ‘Page’; page number, three characters; the extension ‘.pdf’.”

In the Job Manager’s standard syntax, the Pattern looks like this:

(Name, *) - (Month, 2) - (Day, 2) - (Year, 4) - Page (PageNo, 3) . pdf

Once you define this Pattern, the software can go through a folder or a tree of folders, look for files conforming to the Pattern, and immediately determine:

- How many documents there are;
- Where they are located;
- What publications they belong to;
- Their dates;
- The number of pages they contain.

This applies even in the case where these documents exist only as a set of one-page PDFs, and not in collated form.

Once the software knows all this, it can automatically add this information to the Job Descriptor Template, to create Job Descriptors for each document.

4.1.2 Steps and Job Types

The Job Manager contains a number of hard-coded *steps*. These define what the software should do automatically in different stages of processing.

For example, one hard-coded step is “**Combine Pages**,” and it defines how the software should do the Combine Pages operation on a document, without human intervention.

The steps provided by Olive are summarized in Appendix A.

A *Job Type* is simply a list of steps, in the order they should be performed. It is called a Job Type because it defines what the job should do, or what type of automatic processing will take place.

The Job Types provided by Olive are summarized in Appendix B.

Each Job Descriptor Template and each individual Job Descriptor points to a certain Job Type, from which it inherits its processing instructions. As you will see upon reviewing the template, it is important to distinguish between the Job Types designed for digital input files and those designed for scanned material, as there are significant differences in processing required by the two input types. In addition, as discussed in previous chapters, periodicals and other publications are generally processed differently.

When editing the Job Descriptor, you cannot change what is done in each step of the process (except for parameter settings), and you cannot change the order of the steps (as this is defined in the Job Type). You can, however, determine what step to start on and what step to end on.

To use a simplified example, if the Job Type says:

“Perform Combine Pages, insert issue date to the PDF, run automatic segmentation on the issue, then output it to XML”

then a Job Descriptor using that Job Type cannot change the order of the steps or what is done in each step, but it can define where to start and where to end.

For example:

“Insert issue date to the PDF, then run automatic segmentation”

Or in other words, **“Start on step 2 and end on step 3 of the steps defined in the Job Type.”**

4.1.3 DIGITAL Job Types

Several DIGITAL job types are provided. For example, the default job type in ActivePaper Daily and ActivePaper Archive is DIGITAL.

Following are the steps defined in the DIGITAL Job Type, in their correct order:

Combine issue from pages. This is an automated version of the **Combine Pages** operation, in which one-page PDFs are collated to form a single PDF containing the entire document.

Write issue properties to the PDF. This is an automated version of the Properties operation, in which the date of the document is defined in the PDF.

Clean Olive information from PDF. This step clears all segmentation from the PDF, if any exists. In batch processing, this is a just-in-case step intended to prevent problems arising from running automatic segmentation on top of existing segmentation.

Analyze issue articles. This is the **Run** (automatic segmentation) operation.

Output issue to XML. This is the **Output to XML** operation.

4.1.4 Batch Folders

The Job Manager executes jobs using several file system directories known as **Batch Folders**. These consist of a **Watching Folder**, which the software watches expecting job descriptors to appear, and destination folders to which job descriptors are sent after execution begins.

The process is as follows:

1. Job Descriptors are created in the **Watching** Folder, or existing Job Descriptors are moved to the **Watching** Folder, so that the folder contains one or more DSC files.
2. **Jobs Watching** is turned on.
The system starts processing the jobs in the Watching Folder.
3. When job execution stops, the Job Descriptor file is sent to a destination folder (Failed, Successful or Recovery) based on the reason it was stopped and the definitions in the Job Descriptor.



Throughout this process it is Job Descriptors, or DSC files, that are moved from folder to folder. The Batch Folders do not at any time host the actual PDF files being processed; these PDF files simply remain in their original location.

4.1.5 Alternative Work Modes

If you want to set up batch processing for one document, run **Define and Execute Job** (page 177).

If you want to set up batch processing for more than one document, run the **Jobs Generator Wizard** (page 161).

If you have set up a batch process and would like to execute it, run **Batch Execution** (page 176).

To gain a full understanding of the workings of the Job Manager, it is recommended that you read the upcoming sections in the order they are presented.

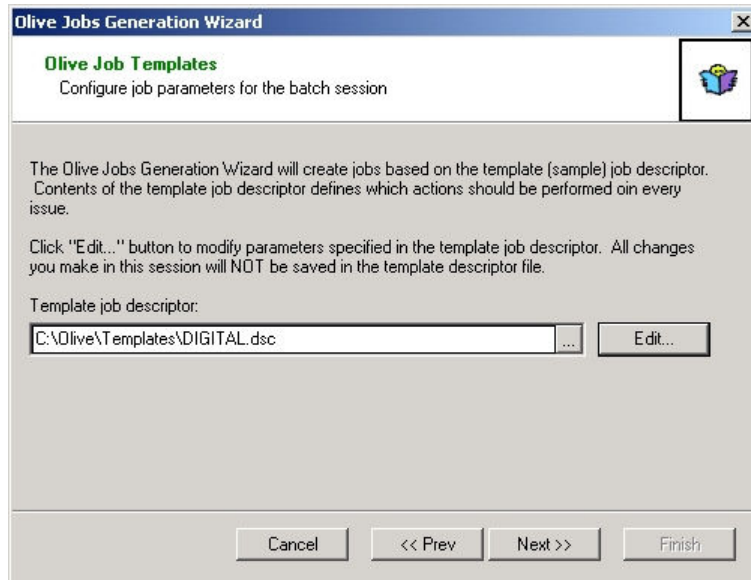
4.2 Jobs Generation Wizard

The Jobs Generator is a wizard that assists you in quickly creating Job Descriptors for multiple documents. You do this by:

- selecting a Job Descriptor Template,
- editing it (if necessary),
- making temporary changes to it (if necessary), and
- choosing the documents you want to process.

The end result is that one or more Job Descriptors are created in the **Watching** Folder.

4.2.1 Choosing a Job Descriptor Template



In this step of the Wizard, you select a **Job Descriptor Template** on which your individual Job Descriptors are to be based.

► **To select a template:**

- Enter its path and file name in the **Template job descriptor** field.

or

- Click the ellipsis (...) button to browse the file system, select a template file, and press **OK**.

► **To edit a template:**

- Press the **Edit** button to open the **Job Descriptor Editor**.



This is an optional step; if the template fits your needs you do not need to use the Job Descriptor Editor, and you can simply proceed to the next step of the wizard.



Contrary to the explanatory text in the dialog, editing will change the **Job Descriptor Template** if you **save** your changes.

4.2.2 Creating and Editing a Job Descriptor Template

In practice, the standard template is often not exactly suited to your purposes. If this is the case, you can change the template, or create a new one.

► **To create a new template:**

1. Open **Windows Explorer** and find the original template you want to base your new template on. Templates are generally stored in **..\Olive\Templates**.
2. Copy the **DSC** file to a new name.
3. Return to the **Jobs Generation Wizard** and select your new template (currently identical to the original) using the **Template job descriptor** field.
4. Press the **Edit** button and define settings for your new template.



The **Job Descriptor Editor** can only be launched from within the **Jobs Generation Wizard**. After you edit the Job Descriptor and press the **Close** button, you will return to the first step of the wizard.

Job Descriptor Editor Toolbar Buttons



Tool	Description
	Saves your changes to the Job Descriptor Template.
	Opens the last saved version of the Job Descriptor Template in Notepad.
	Prints the last saved version of the Job Descriptor Template.
Other buttons are not active.	



Using the Job Descriptor Editor involves numerous settings which are displayed in a set of tabs. The first tab – the **Job** tab – is common to all jobs. Each of the other tabs relates to a step in the job, and the set of tabs displayed is determined by the Job Type of a particular job.

For example, the **Republishing** job below entails only two steps with settings to customize, while other types entail additional customizable steps (see additional tab descriptions in this section).

Job Management

Jobs Generation Wizard

The screenshot shows the 'Olive Job Descriptor Editor' window with the 'Job' tab selected. The 'Job Type' is set to 'REPUBLISHING'. The 'Execution Options' section includes: 'From step' set to 'Write issue properties to the PDF', 'To step' set to 'Output issue to XML', 'Working folder' set to 'C:\Olive\Batch\Working', and 'Error Folder' set to 'C:\Olive\Batch\Recovery'. The 'Logging Options' section includes: 'Activity log DB' set to 'OliveLog', 'Log Folder' set to 'C:\Olive\Log', and 'Trace Level' set to 'Full logging'. The window has a menu bar with 'File' and 'Edit', a toolbar with icons for file operations, and a status bar at the bottom with 'Close' and 'Cancel' buttons.

Figure 80 – Job Manager Job Tab (Republishing)

Job Tab

The screenshot shows the 'Olive Job Descriptor Editor' window with the 'Job' tab selected. The 'Job Type' is set to 'DIGITAL'. The 'Execution Options' section includes: 'From step' set to 'Write issue properties to the PDF', 'To step' set to 'Output issue to XML', 'Working folder' set to 'C:\Olive\Batch\Working\', and 'Error Folder' set to 'C:\Olive\Batch\Recovery\'. The 'Logging Options' section includes: 'Activity log DB' set to 'OliveLog', 'Log Folder' set to 'C:\Olive\Log\', and 'Trace Level' set to 'Full logging'. The window has a menu bar with 'File' and 'Edit', a toolbar with icons for file operations, and a status bar at the bottom with 'Close' and 'Cancel' buttons.

Figure 81 – Job Manager Job Tab (Digital)

This tab contains the following fields:

- **Job Type.** As discussed earlier, each Job Descriptor points to a **Job Type** that defines which steps are to be performed and in what order. This field cannot be changed, as it is an inherent part of the template.
- **“From” step.** This **drop-down** menu allows you to select the processing step to start with, from the steps defined in the **Job Type**. The steps in the **drop-down** are the steps defined in the **Job Type** the template points to, and you cannot add or remove steps or change their order.
- **“To” step.** Allows you to select the processing step to end with, from the steps defined in the **Job Type**. This can be the same as the **“From step”** – in this case only one step will be performed in processing.
- **Working folder.** Defines the **Working** Folder to be used for the **Job Descriptors**. When **Job Descriptors** are picked up from the **Watching** Folder they are moved to the **Working** Folder, and remain there while being executed. It is recommended that you keep this as the default **Working** Folder location: **..\Olive\Batch\Working**.
- **Error Folder.** When a job is stopped due to an error, the **Job Descriptors** are moved from the **Working** Folder to the folder defined here, indicating the step completed just before the step in which an error was encountered.
- If you define the **Failed Jobs** folder, by default **..\Olive\Batch\FailedJobs**, jobs that result in errors are **“trashed”** in this folder and are not executed again.
- If you define the **Recovery** folder, by default **..\Olive\Batch\Recovery**, jobs that result in errors are run again when **Jobs Watching** is turned on.
- **Activity log DB.** Defines the name of the database used for logs. Do not change it unless requested to by **Olive Professional Services**.
- **Log Folder.** Defines the folder to which log files are sent. Log files provide detailed descriptions of how a specific job fared, and what if anything went wrong. The default log folder is **..\Olive\Logs**.
- **Trace Level.** Defines the level of detail for logs.

Issue Tab

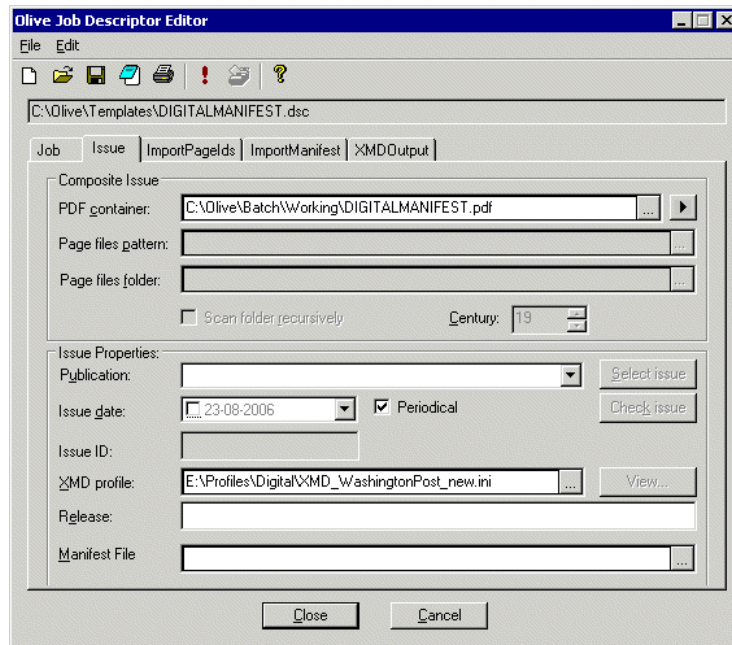


Figure 82 – Job Manager Issue Tab

This tab is relevant for scanned periodicals or sets of other documents that are scanned, and it contains the following fields and buttons:

- **PDF container.** Specifies the target location and filename for the issue or document PDF file to be processed: an existing one, or one that will be created from a set of smaller input files if the **Combine Pages** step is used.
- **Page files pattern.** Defines the filename pattern used to find issues or documents in the file system and deduce their meta data. (This field is disabled if you skipped the **Combine Pages** step in the Job tab.)
- **Page files folder.** Defines the folder in which the source files to be processed reside. (This field is disabled if you skipped the **Combine Pages** step in the Job tab.)
- **Scan folder recursively.** Checking this indicates that the software should search all subfolders of the **Page files folder** for PDFs conforming to the filename pattern.
- **Select issue.** This button allows you to specify one or more files, and is an alternative to specifying the information in the fields **Publication** and **Issue data**. If a pattern is specified in the top section (Page files pattern) and files are specified here, the “metadata” found in the filename using the pattern is used to populate **Publication** and **Issue data**, and other metadata.
- **Check issue.** This button is not currently active.

- **Century.** If your filename pattern uses only two characters for years, this field fills in the most significant two digits. It uses "real" centuries and not the accepted count, i.e. 1972 is in century 19. If your filename pattern uses four characters for years, this field has no effect.
- **Publication.** Name of the publication: typed in or selected from a list.
- **Issue date.** Opens a calendar for specifying the date of the issue. For enterprise documents this is not used and an 8-digit document ID is received from the ViewPoint Server.
- **Issue ID.** Relevant for documents receiving an ID not associated with a real date, usually for enterprise documents.
- **XML Distiller (XMD) Profile.** Sets the path for the segmentation profile to be used with the documents you intend to process.



It is essential that you select a valid profile here, even if you do not intend to perform automatic segmentation. Many critical parameters in Batch Processing are derived from this profile.

- **View.** Opens the selected profile in Notepad (as text). If the **View** button is disabled, there is no profile at the path entered.
- **Release.** Not currently used.
- **Manifest file.** Defines the path(s) and filename(s) of the ad's manifest file(s) (e.g., those received from an ad agency).

Import Issue Tab

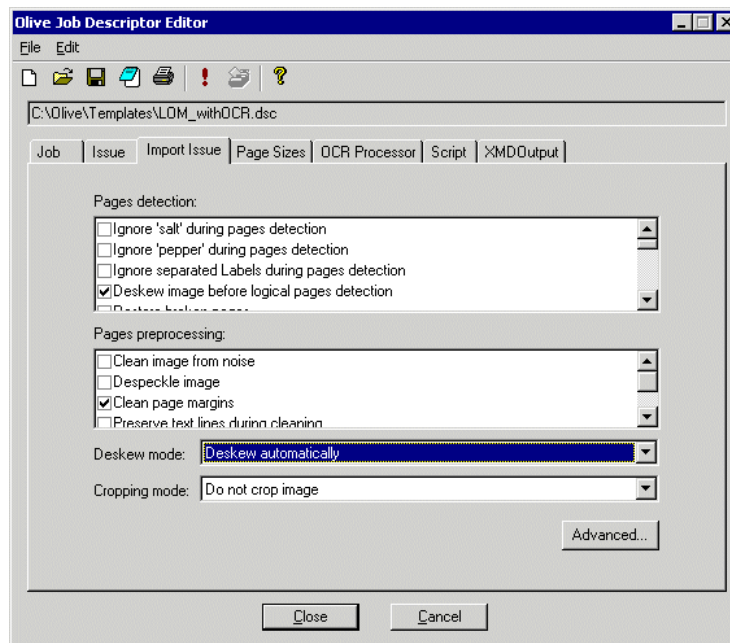


Figure 83 – Job Manager Import Issue Tab

This tab contains the fields for customization of several features when a document or periodical issue is imported:

- Page Detection
- Page Preprocessing
- Image Deskewing
- Image Cropping.

Combine Pages Tab

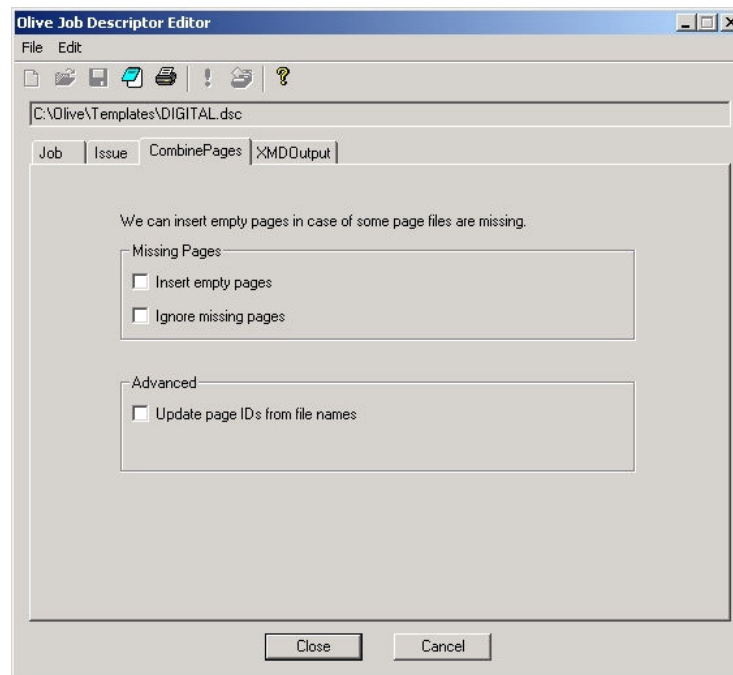
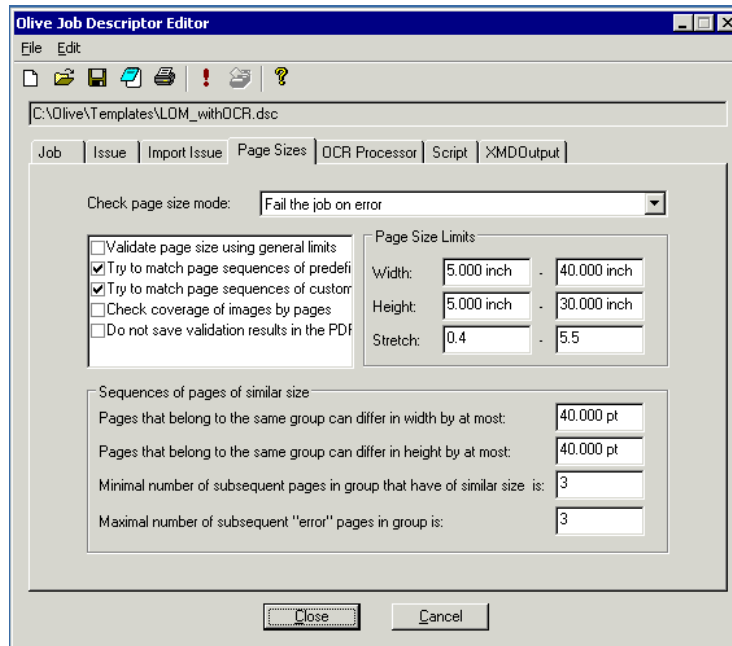


Figure 84 – Job Manager Combine Pages Tab

This tab is relevant for jobs containing a **Combine Pages** step, and it contains the following fields:

- **Insert empty pages.** Indicates whether to insert empty PDF pages when there are gaps in the sequence of pages.
- **Ignore missing pages.** Indicates whether to ignore missing pages, and not insert empty pages when there are gaps in the sequence of pages. (Cannot be checked when **Insert empty pages** is checked).
- **Update page IDs from file names.** A page ID defines the “real name” for each page in the periodical, which may be different from the Acrobat page number (for example, page 34 in Acrobat may be page C15 in a newspaper). Checking this dialog box indicates that page IDs should be populated with page numbers found in the filenames of the one-page PDFs.

Page Sizes Tab*Figure 85 – Job Manager Page Sizes Tab*

This tab is relevant for jobs in which sizes of one or more sets of pages should be consistent, and contains the following fields:

- **Check page size mode.** Whether to check for consistent page sizes, and what action to take if errors are encountered.
 - Checklist of options regarding how the validation is to be done and whether results should be saved in the PDF.
- **Page Size Limits:** ranges for width, height and stretch ratio of a page.
- **Sequence of pages of similar size:** constraints to use when identifying a set of adjacent similar-sized pages.

OCR Processor Tab

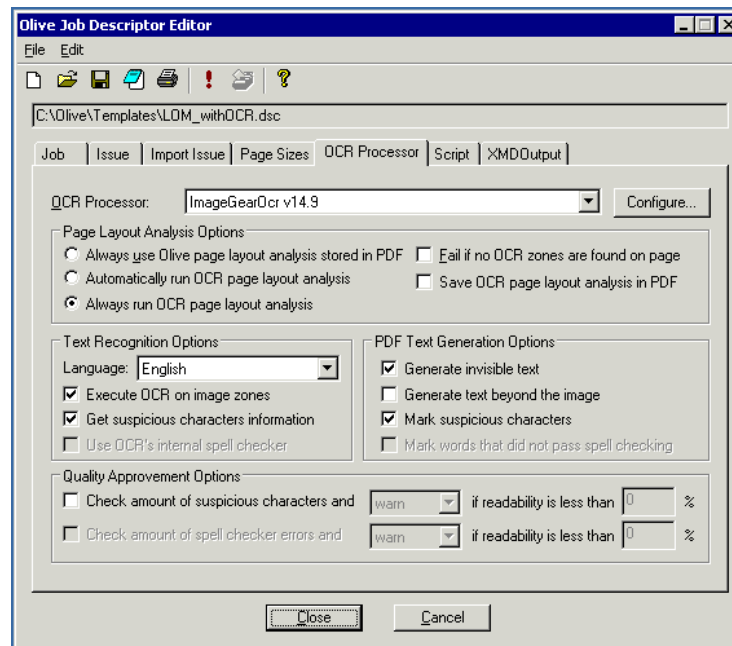


Figure 86 – Job Manager OCR Processor Tab

This tab, relevant for scanned material, allows you to customize OCR and QA settings.

Script Tab

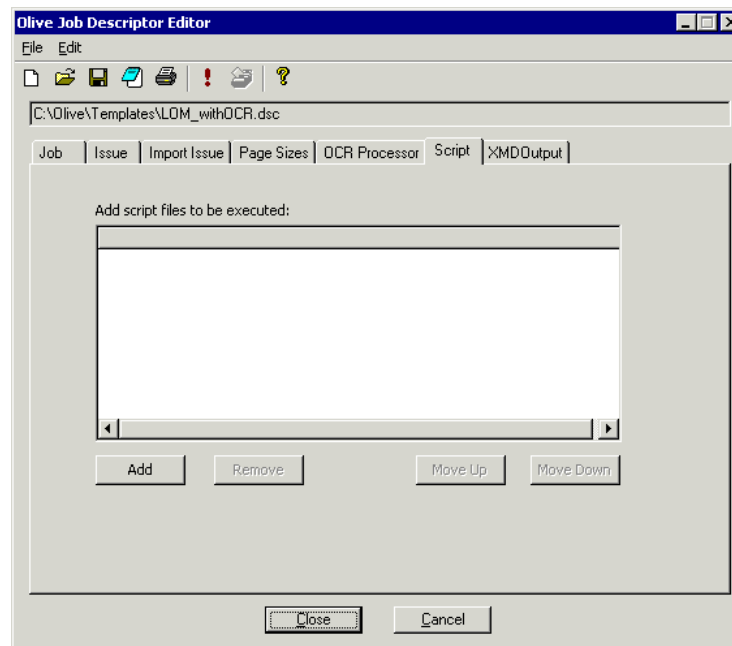


Figure 87 – Job Manager Script Tab

In this tab, relevant for scanned material and Advanced (layout-object) Segmentation, you can specify one or more Javascript files to be run as part of the segmentation process:

1. If you wish to add a script, press **Add** and specify a .js location and filename. Repeat for any additional script files.
2. If there are two or more scripts in the resulting list, change the sequence if necessary by selecting a script in the incorrect position and using the **Move Up** and **Move Down** buttons.

XMDOutput Tab

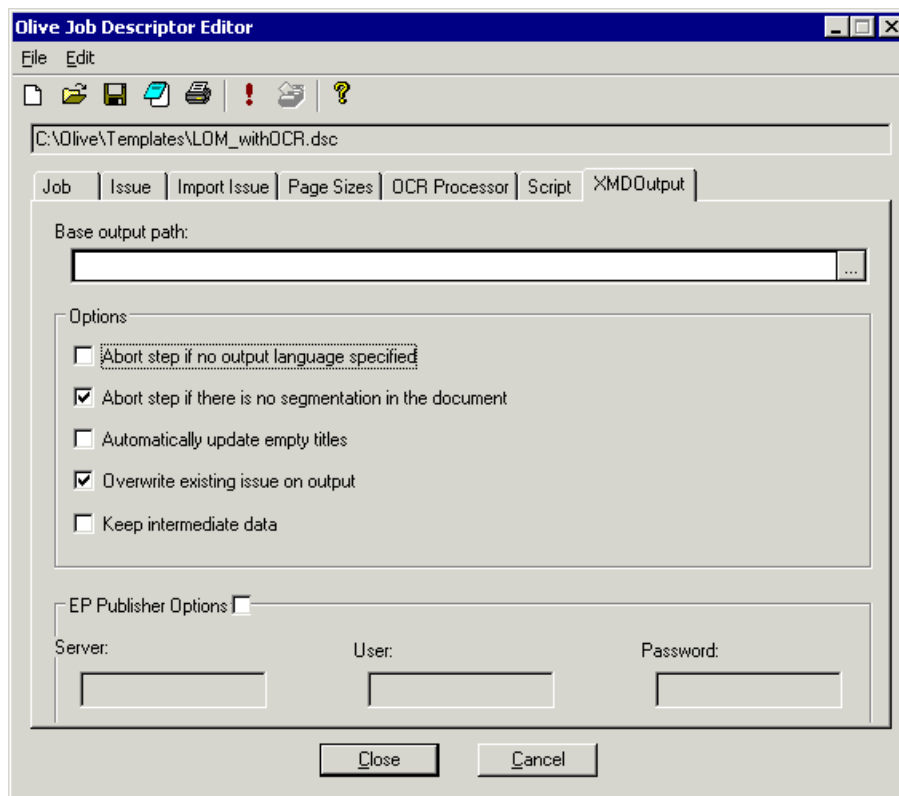


Figure 88 – Job Manager XMD Output Tab

This tab is relevant for any job containing an Output step, and contains the following fields:

- **Base output path.** Defines a location for **XML** Output. The full hierarchy of Warehouse (Repository) folders will be created automatically under this path. *It is recommended that you leave this empty;* in this case the output location is the Warehouse (Repository) directory indicated in the profile.
- **Abort step if no output language specified.**
- **Abort step if there is no segmentation in the document.**

- **Automatically update empty titles.** When checked, indicates that empty meta names for entities should be automatically updated from their headline elements. Meta names, stored as meta data, are the entity names that show up in search results, and need not be identical to the headline.
- **Overwrite existing issue on output.**
- **Keep intermediate data.** When checked, indicates that an internal cache within the PDF, which speeds access to segmentation information, should be maintained. Keeping this cache increases **PDF** size but speeds up work on the PDF later on.
- In addition, when the ViewPoint Server is used, three fields are provided for specifying server login information:
 - **Server.** Name of the server machine.
 - **User.** Username for client access to the server.
 - **Password.** Password associated with the user.

4.2.3 Selecting Issues for Automatic Processing

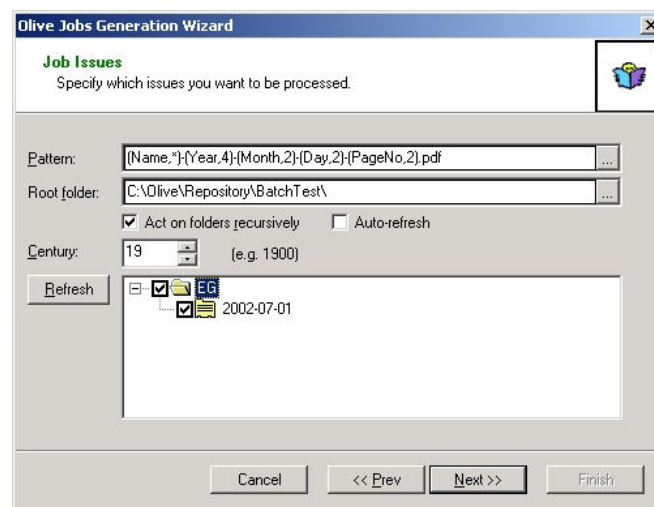


Figure 89 – Jobs Generation Wizard: Job Issues Stage

In this step of the wizard, you search for page files corresponding to a defined filename pattern, which make up periodical issues or documents.

The Wizard displays the following fields:

- **Pattern.** This field defines the filename pattern to be used when searching for page **PDFs**. By default it shows the filename pattern found in the **Job Descriptor Template**.

The ellipsis (...) button opens up the following context menu:

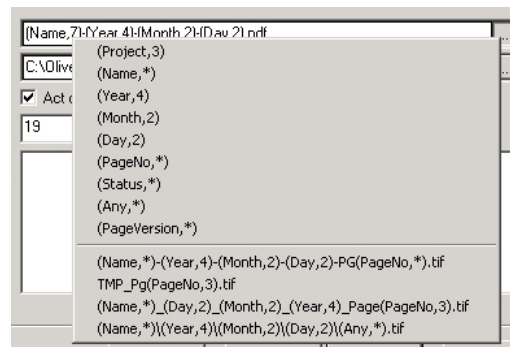


Figure 90 – File Patterns and Components

Above the divider are typical components of filename patterns: information type and number of characters; below it are several typical filename patterns, in full. Clicking any of them inserts it, as is, at the cursor position.

- **Root folder.** Defines the folder in which the issues or documents reside. By default it shows the root folder defined in the **Job Descriptor Template**.
- **Act on folder recursively.** Checking this indicates that the software should search all subfolders of the **Root** folder for PDFs conforming to the filename pattern. By default it inherits its state from the **Job Descriptor Template**.
- **Century.** If your filename pattern uses only two characters for years, this field fills in the most significant two digits. It uses “real” centuries and not the accepted count, i.e. 1972 is in century 19. If your filename pattern uses four characters for years, this field has no effect. By default this field shows the value defined in the **Job Descriptor Template**.



If you make changes to these four fields, be aware that **these changes will not be reflected in the template**, though they **will affect the Job Descriptions created by the wizard**. If there is a permanent change in the filename pattern, root folder or century, edit the template and change it there. Otherwise, you must make the changes each time you run the wizard.

- **Refresh.** This button scans the directory and its subdirectories (if required) for files conforming to the filename pattern.
- **Display window** (to the right of the **Refresh** button). Once **Refresh** has been pressed, displays all issues/documents that have been found in the **Root folder**. They are presented as dates nested under a respective publication in a tree configuration. Each item in the tree has a checkbox, which can be unchecked if the item should not undergo batch processing.

4.2.4 Defining Batch Folders

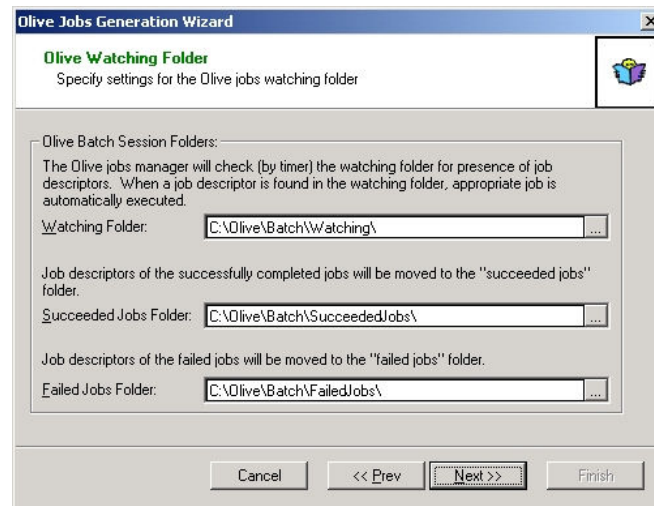


Figure 91 - Jobs Generation Wizard: Watching Folder Stage

In this step of the wizard, folder paths are specified in the following fields:

- **Watching Folder.** Defines the **Watching Folder** to be used for the Job Descriptors created by the wizard. Jobs being executed will be moved here.
- **Succeeded Jobs Folder.** Defines the **Succeeded Jobs** Folder to be used for the Job Descriptors created by the wizard. Jobs completed successfully will be moved here.
- **Failed Jobs Folder.** Defines the **Failed Jobs** Folder to be used for the Job Descriptors created by the wizard. Jobs stopped due to an error will be moved here.



It is recommended that you keep the default settings for these folders.

4.2.5 Completing the Wizard

Once you complete this step you have come to the end of the wizard. Pressing **Finish** launches **Batch Execution** (page 176).

At this point, if you check your **Watching** Folder, you will find that the Job Descriptors you requested have been created there.

4.3 Batch Execution

You can make last-minute changes to **Batch Folders**, and then start executing the **Job Descriptors** in the **Watching Folder**.

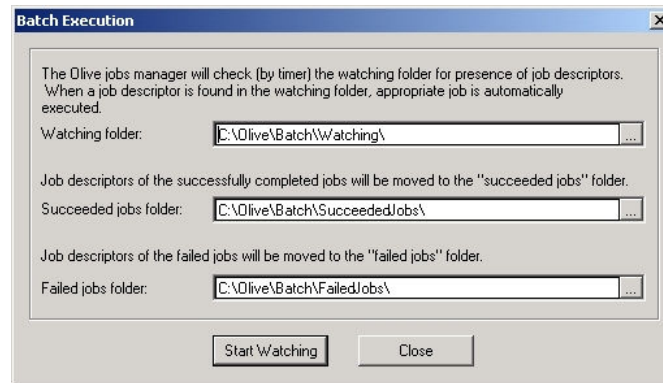


Figure 92 – Batch Execution

► To run Batch Execution on Job Descriptors in the Watching Folder:

1. In the **Olive** menu, point at **Olive Job Manager** and select **Jobs Watching (Batch)....**
Or run through the **Jobs Generator** wizard and press **Finish**.
2. Press **Start Watching**.

The **Batch Progressing** panel displays the status of jobs being executed.

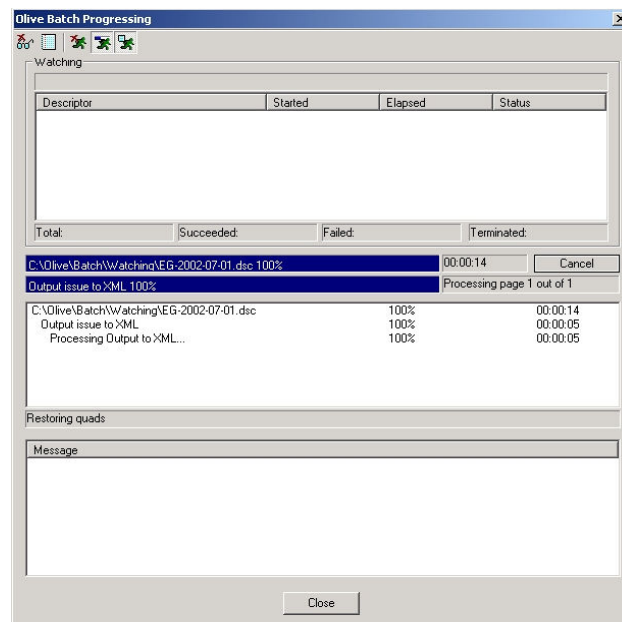


Figure 93 – Batch Progressing

Batch Progressing Buttons



Stops watching



Opens a general system log



Shows/hides the Progress Stages pane



Shows/hides the Job Log pane

Cancel

Cancels the job currently running

Batch Progressing Panes

Top pane	Shows status of Job Descriptors run from the Watching Folder in this session. Double-clicking a Job Descriptor opens its Job Log.
Middle pane	Shows name and progress of processing steps for the current job.
Bottom pane	Shows error and warning messages from the current session.

4.4 Defining and Executing a Single Job

In this mode, you open a **Job Descriptor Template**, make quick changes to it if necessary, select a single periodical issue or document and execute a single job on it.

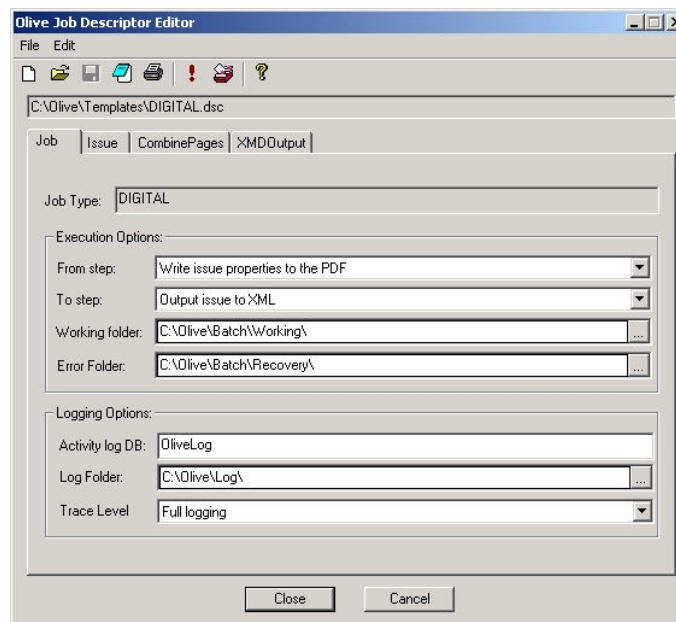


Figure 94 – Job Descriptor Editor: Job Tab

4.4.1 Opening and Editing the Template

In this respect, the **Define and Execute Job** dialog is identical to the Olive Job Descriptor Editor (see *Editing a Job Descriptor Template* on page 97).

Use the **Job**, **Issue**, **CombinePages** and **XMDOutput** tabs to modify the template, and press the  button to save it.

4.4.2 Selecting a Single Issue to Process

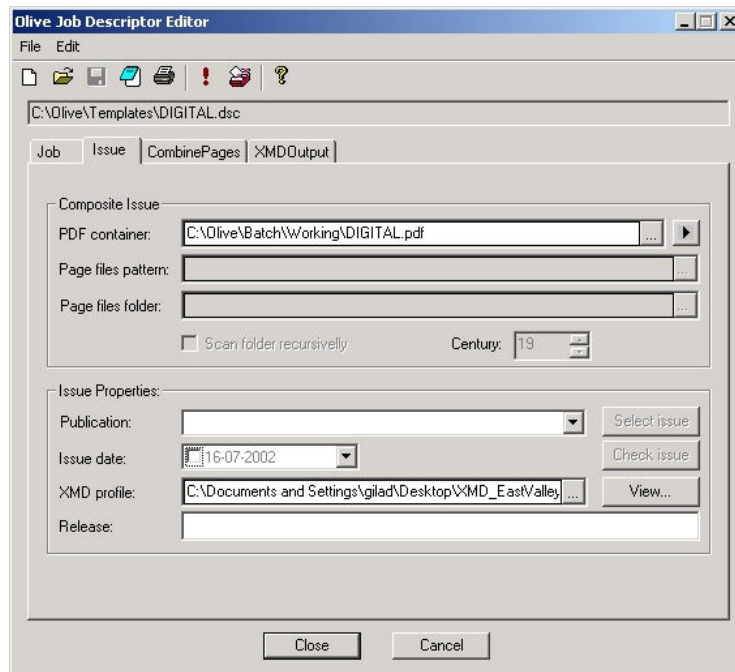


Figure 95 – Job Descriptor Editor: Issue Tab

In the **Issue** tab, use the **PDF Container** field to specify a periodical issue or document for automatic processing:

- You can enter a free path for the issue/document or use the ... button to browse for it.
- If you have a PDF open and would like to perform automatic processing on it, press the  button, then select **Use active PDF** from the menu that appears.



This assumes that your issue/document exists as one Combined PDF and not as several page PDFs. If it does exist as several page PDFs, make sure you start from the Combine Pages step in the Job tab, and then enter the pattern and folder containing the files in the relevant fields in the **Issue** tab.

4.4.3 Running Batch Processing on the Document

To run batch processing, press the  button at the top of the dialog.

4.5 Job Management Logs

The Job Manager keeps detailed logs of its activities. These are particularly important when jobs fail or when problems occur. In many cases consulting the log can yield the exact cause of trouble.

There are two types of logs generated by the Job Manager: **Job Logs** and **General System Logs**. General System Logs are intended for use by Olive Professional Services and so will not be discussed here.

A job log is created for every Job Descriptor run by the **Job Manager**. The log is saved in a location specified in the Job Descriptor Template (and subsequently in the Job Descriptor itself), with the default location being **..\Olive\Log**.

The job log displays three types of information:

- Start, end and duration for each step.
- Start, end and duration for the entire job.
- Errors and Warnings.

An error is a problem that caused a task to fail. For this reason a job log generally does not show more than one error.

A warning is a problem that did not cause the task to fail, but may have interfered with its activities.

Both errors and warnings can be the result of system problems (such as low disk space, insufficient write permissions, etc.); problems with Acrobat or with the PDF being worked on; or problems with the Job Manager software.

Most errors and warnings are accompanied by clear explanations of what went wrong. If you come across an error or warning you don't understand, please contact **Olive Professional Services**.

Example: Log for a Successful Job

```
-----  
Computer : PAMELA-2K  
File      : d:\Olive\Log\GENEVE-2002-06-14.txt  
-----  
[01-Aug-2002 15:21:29] Job 'd:\Olive\Batch\Watching\GENEVE-2002-06-  
14.dsc' started  
[01-Aug-2002 15:21:29] Starting 'Combines issue pages (PDF files) to the  
single PDF' step [CombinePagesStep].  
[01-Aug-2002 15:21:46] Step [CombinePagesStep] completed, elapsed time:  
00:00:13, step succeeded.  
[01-Aug-2002 15:21:46] Starting 'Writes issue properties to the PDF  
document properties' step [DocPropertiesStep].  
[01-Aug-2002 15:21:48] Step [DocPropertiesStep] completed, elapsed time:  
00:00:00, step succeeded.  
[01-Aug-2002 15:21:48] Starting 'Removes all information generated by  
Olive products from the PDF' step [COLvPdfCleanerStep].  
[01-Aug-2002 15:21:51] Step [COLvPdfCleanerStep] completed, elapsed  
time: 00:00:01, step succeeded.  
[01-Aug-2002 15:21:51] Starting 'Analyze issue articles' step  
[XMDAnalyzerStep].  
[01-Aug-2002 15:25:03] Step [XMDAnalyzerStep] completed, elapsed time:  
00:03:06, step succeeded.  
[01-Aug-2002 15:25:03] Starting 'Output issue to XML' step  
[XMDOutputJobStep].  
[01-Aug-2002 15:31:46] Step [XMDOutputJobStep] completed, elapsed time:  
00:06:18, step succeeded.  
[01-Aug-2002 15:31:46] Job 'd:\Olive\Batch\Watching\GENEVE-2002-06-  
14.dsc' completed, elapsed time: 00:10:18, job succeeded.
```

Figure 96 – Successful Job Log

Example: Log for a Failed Job

```
-----  
Computer : PAMELA-2K  
File      : d:\Olive\Log\DIGITAL.txt  
-----  
[17-Aug-2002 15:23:07] Job 'D:\Olive\Templates\DIGITAL.dsc' started  
[17-Aug-2002 15:23:07] Starting 'Combines issue pages (PDF files) to the  
single PDF' step [CombinePagesStep].  
[17-Aug-2002 15:23:08] ERROR: (Step [CombinePagesStep])  
Error on step 1 [CombinePagesStep], while executing step:  
Page 3 is absent. Process is cancelled.  
[17-Aug-2002 15:23:08] Step [CombinePagesStep] completed, elapsed time:  
00:00:01, step failed on Combine PDF pages, job aborted.  
[17-Aug-2002 15:23:09] ERROR: (Step [CombinePagesStep])  
Error on job 'DIGITAL', while executing steps:  
Step 1 [CombinePagesStep] failed  
[17-Aug-2002 15:23:09] Job 'D:\Olive\Templates\DIGITAL.dsc' completed,  
elapsed time: 00:00:02, job failed
```

Figure 97 – Failed Job Log

4.6 Settings

In addition to customization via the Profile Editor (see page 185) you can modify Job Manager settings and other global ActivePaper/ViewPoint settings via Notepad.

4.6.1 Job Manager Settings

► **To modify Job Manager settings:**

- In the **Olive** menu, point to the **Olive Job Manager** menu and select **Edit Settings**.

The Job Manager settings file is opened in Notepad, and editing of the settings in the [JM_SETTINGS] section is possible.

```
[JM_SETTINGS]
LOG_FOLDER=C:\Olive\Log
WORKING_FOLDER=C:\Olive\Batch\Working
TEMPLATES_FOLDER=C:\Olive\Templates
BATCH_WATCHING_FOLDER=C:\Olive\Batch\Watching
BATCH_SUCCEEDED_JOBS_FOLDER=C:\Olive\Batch\SucceededJobs
BATCH_FAILED_JOBS_FOLDER=C:\Olive\Batch\FailedJobs
RECENT_BATCH_TEMPLATE=
LOG_EXPIRATION_PERIOD=2
TEMP_CLEANUP_WILDCARD=Acr*.tmp
POST_JOB_BATCH_PATH=D:\PipeX\Bin\InitCopy.exe
AUTORELOAD_LOG_MODE=
;BATCH_RELOAD_ACROBAT_AFTER_EVERY_JOB=No/Yes
LAST_DESCRIPTOR_PATH=
```

Figure 98 – Job Manager Settings

4.6.2 Settings and Default Values

The xmd.ini file contains application-level configuration parameters (such as EnterpriseMode to specify if the publications are periodicals or enterprise documents), as well as default values for all parameters that can be specified in a profile (see Editing/Improving Profiles on page 185).

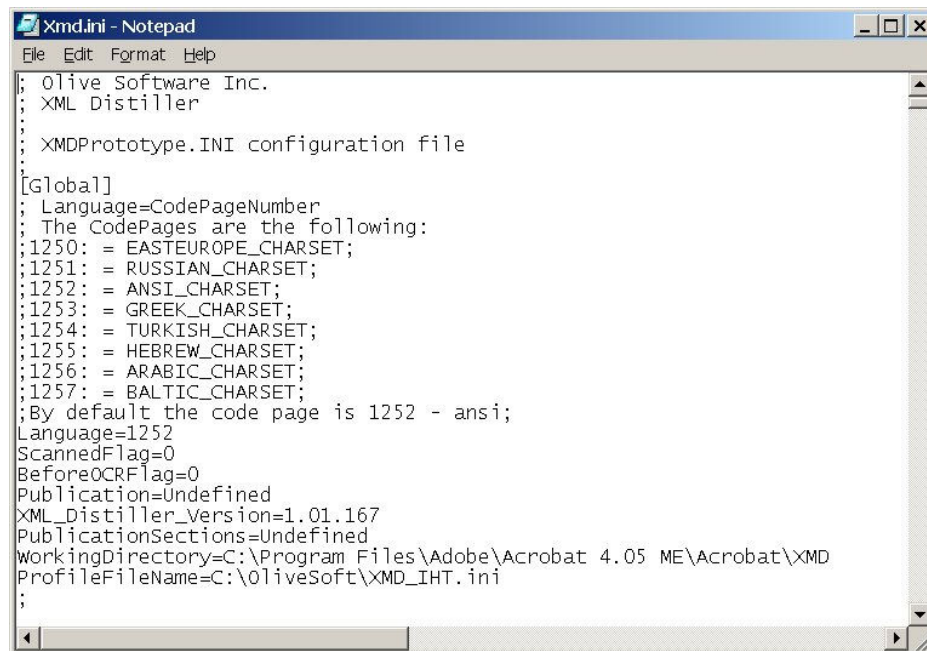
Users may modify default settings. Any setting specified in a loaded profile, however, overrides the setting in the Settings file described here.

► **To modify global and default settings:**

- In the **Olive** menu, choose the **Settings...** option.

The file (xmd.ini) is opened in Notepad for editing.

Changes in this file, especially to parameters not covered in this document, require the assistance of Olive Professional Services personnel.



```
Xmd.ini - Notepad
File Edit Format Help
; Olive Software Inc.
; XML Distiller
;
; XMDPrototype.INI configuration file
;
[Global]
; Language=CodePageNumber
; The CodePages are the following:
;1250: = EASTEUROPE_CHARSET;
;1251: = RUSSIAN_CHARSET;
;1252: = ANSI_CHARSET;
;1253: = GREEK_CHARSET;
;1254: = TURKISH_CHARSET;
;1255: = HEBREW_CHARSET;
;1256: = ARABIC_CHARSET;
;1257: = BALTIC_CHARSET;
;By default the code page is 1252 - ansi;
Language=1252
ScannedFlag=0
BeforeOCRFlag=0
Publication=Undefined
XML_Distiller_Version=1.01.167
PublicationSections=Undefined
WorkingDirectory=C:\Program Files\Adobe\Acrobat 4.05 ME\Acrobat\XMD
ProfileFileName=C:\OliveSoft\XMD_IHT.ini
;
```

Figure 99 – Olive Global Settings (xmd.ini file)

5 Setting and Programming Segmentation Rules

The identification of the elements in the document, their hierarchy, reading flow, relations between texts and pictures and between pictures and their captions, are all part of the basic segmentation capabilities of Olive's recognition engine. However the engine also uses basic information about the document's layout (mostly, how fonts are used throughout the document and some other various geometric conditions) to improve the accuracy of the segmentation and to customize the segmentation to specific requirements. This information is provided in the form of a *Segmentation Profile*, which is an ".ini" file containing several parameters that are specified for each publication. Profiles include information about the formatting styles of the elements, their position, and various patterns. The segmentation engine uses this information to make its segmentation decisions.

The global profile (xmd.ini) provided by Olive contains default settings for all parameters not specified in a loaded profile. See *Settings and Default Values* on page 182.

Enterprise documents are usually consistent in terms of their formatting style throughout the document, and they therefore use a single Profile. However, newspapers usually have multiple inserts, with independent formatting styles. In this case, a separate profile should be used for each insert.

ActivePaper/ViewPoint includes generic Profiles for the various types of documents: See *Loading a Profile* on pages 35 (periodicals) and 136 (enterprise documents). These generic Profiles are a starting point for the creation of the specific Profile for your publication. The creation of the first Profile is not covered in this guide as it is handled by Olive Software's Professional Services.

The first Profile that is created may not provide the complete coverage of all the layout occurrences of a publication. Therefore, over time, you may discover consistent errors in automatic segmentation. You can either improve the Profile by following the instructions below or report them to Olive's Professional Services personnel, and our quality assurance team will fine-tune your profile, improving the recognition results.



If you save a PDF file using the **Save As** command, or extract pages from the file, the most-recently-assigned profile is automatically reloaded.



When the PDF file is reopened, the most recently assigned profile is automatically reloaded.

5.1 Editing/Improving Profiles

The Profiles can be edited/improved via the **Profile Editor** screens, and by direct editing of the Profile INI file.

Enterprise document profiles include parameters for the recognition of the Header/Footer elements and the table of contents (TOC): for example, the number of levels for the TOC.

The recognition of these elements cannot be improved in XML Distiller. To improve their recognition accuracy, contact Olive's Professional Service department.

5.1.1 Improving Profiles with Profile Editor

The Profile Editor allows you to create a new profile or modify an existing one.

► **To open Profile Editor:**

- On the **Olive** menu, point to Profile, and then choose **Editor**.

The *Profile Editor* window opens. The left-hand side of the window contains a tree that shows the various entity and element types, while the right-hand side contains four tabs (**Page Properties** tab, **Advanced Settings** tab, **Fonts List** tab and **Font Monitor** tab) that are described below. The parameters on the right-hand side apply to the selected elements in the tree.

Page Properties

Page Properties defines the borders of the page for the segmentation (i.e. elements outside this border will not be segmented) and the borders of the page to be displayed in the front-end.

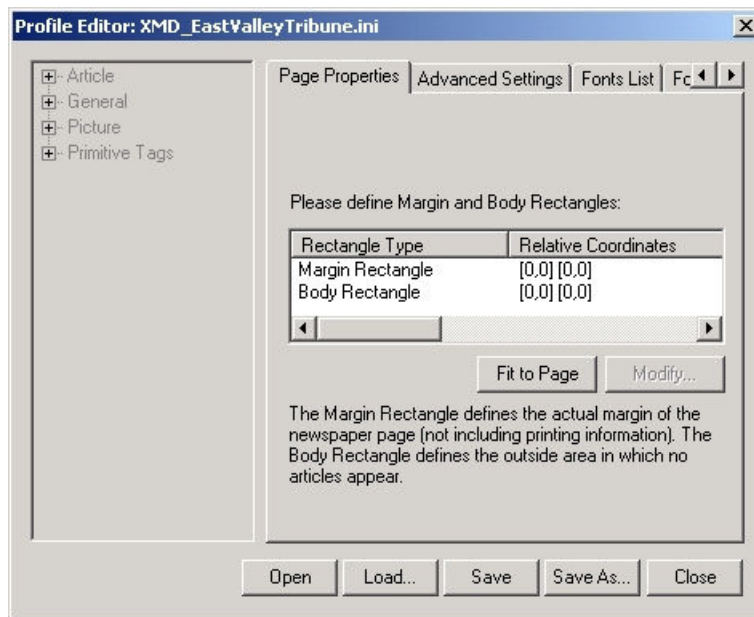


Figure 100 – Profile Editor: Page Properties Tab

The *Margin Rectangle* defines an ideal border for the page, which may be smaller than the actual border of the Acrobat page. You may find that the outer edge of the PDF pages contains crop marks or other unsightly markings. Defining a Margin Rectangle that excludes these marks prevents them from appearing to the end user. When the document is output to the XML Warehouse (Repository) at the end of the process, pages will be cropped around the Margin Rectangle.

The *Body Rectangle* is the area that contains the actual body of the newspaper to be segmented. White margins, footer lines, etc. can be excluded from this rectangle, as they can only confuse segmentation and do not contain actual articles, pictures or ads.

The Body Rectangle is generally included in the Margin Rectangle. Margins and footers are necessary and should be included in the Margin Rectangle (they have aesthetic and functional significance and *should* appear to the end user), however they should be excluded from the smaller Body Rectangle as they are not relevant to the segmentation process.

Automatic Resizing

XML Distiller can automatically resize the page to fit the page of the document without the bleed or printing marks.

► To resize the segmentation frame automatically:

1. Select **Margin Rectangle** or **Body Rectangle** in the **Rectangle Type** field.
2. Click the  button.

The frame are resized automatically to fit the page of the document.

Manual Margin/Body Definition

You can also specify exactly where the segmentation should be performed on the page (Body) and the size of the page that will be displayed to the end user (Margin).

► **To resize the segmentation frame manually:**

1. Select **Margin Rectangle** or **Body Rectangle** in the **Rectangle Type** field.
2. Click the  button.

The **Modify Margin Rectangle** or **Modify Body Rectangle** window opens.

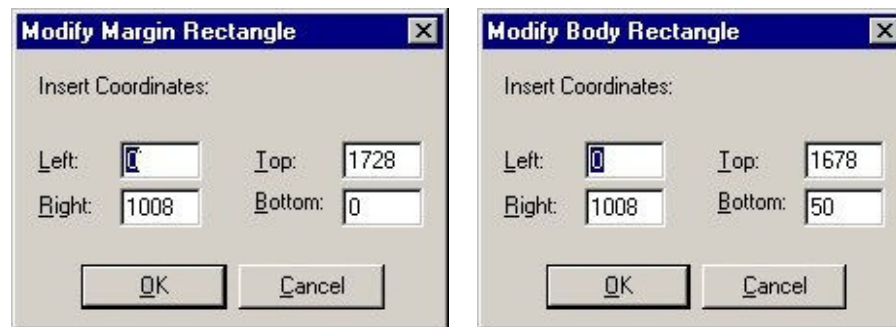


Figure 101 – Modify Margin/Body Rectangle

3. Enter the required coordinates in the fields of **Insert Coordinates**.
4. Click **OK** to apply the modified coordinates to the selected frame, or **Cancel** to cancel the coordinate changes.

You can also modify the frame margin interactively on the issue page. Click on the margin frame and drag it to a new position.

Advanced Settings

Element Pattern specifies recurring patterns of elements for their identification and for the extraction of relevant information. For example, the *Continuation To* element may always look like this: "Continued on page <page number>." Or it may look like this: "See <title> on page <page number>."

Element Positions specifies the position of a certain element whenever these elements constantly appear in the same place on the page. For example, in a newspaper, the *Section* element may always appear in the center of the page, below the masthead. In this case a rectangle can be defined, using coordinates, which will always contain this element. This makes segmentation of the element foolproof, since the engine knows exactly where to look for it.

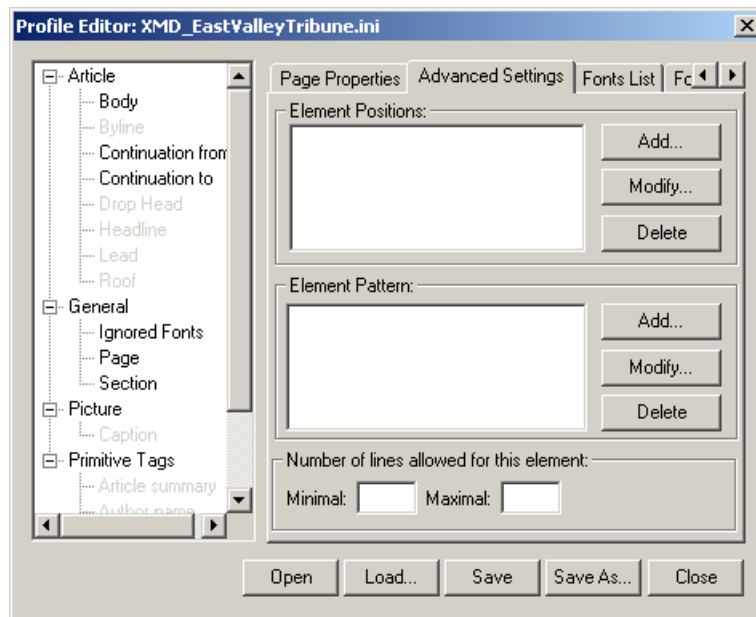


Figure 102 – Profile Editor: Advanced Settings Tab

You can add an element pattern/position by using the buttons in the Advanced Settings tab, or add them to the page itself.

Element Pattern should be used to specify the logical page number in Newspapers. The “logical page number” is the page number that appears on the page itself, which can be different from the sequential number in the PDF. For example, page “B-4” on the page can be page 15 in the PDF. Use the Element Pattern to specify the pattern used to identify the page number object on the page and to extract the page number itself from the page.

Assigning Element Position

► To add an element position through the page itself:

1. From the **element type tree**, select the **element type** (e.g., General > Section) to be assigned with a new position.
2. In the page itself, click and drag to create a selection frame for the required position. Move the **Profile Editor** dialog if it is obscuring a relevant page area.

When you finish drawing the frame, it appears in red.

3. Click the right mouse button on the red frame, then **Set As Element Position**.

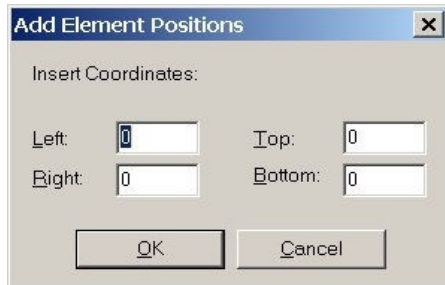
The coordinates of the position, assigned to the selected element type, appear in the **Element Position** field of the Advanced Settings tab.

When the position is assigned, the selection frame in the issue appears in green.

► **To add an element's position by using the add/modify buttons:**

1. Click the **Add** button to the right of the **Element Position** field to add a new position, Or select an existing position in the **Element Position** field list and then click the **Modify** button.

Either the **Add Element Positions**, or **Modify Element Positions** window opens accordingly:



2. Enter the required coordinates and confirm them by clicking **OK**, or cancel them by clicking **Cancel**.

► **To delete an element position:**

1. In the **Element Position** field, select the position to be deleted.
2. Click the  button to the right of the **Element Position** field:

The selected position is deleted.

Assigning Element Pattern

► **To add an element pattern through the selection on the page:**

1. From the **element type tree**, select the **element type** to be assigned with a new pattern.
2. In the page itself, click and drag to create a selection frame for the required pattern.

When you finish drawing the frame, it appears in red.

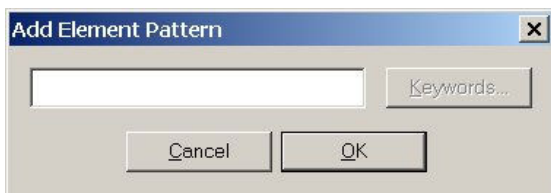
3. Click the right mouse button on the red frame, then **Set As Element Pattern**.

The pattern selected appears in the **Element Pattern** field of the Advanced Settings tab.

► **To add/modify an element pattern by using the add/modify buttons:**

1. Select the required element type (e.g., *Continuation from* under the *Article* item) in the tree.
2. Click the **Add** button to the right of the **Element Pattern** field to add a new pattern, Or select an existing pattern in the **Element Pattern** field list, and then click the **Modify** button – to modify the selected pattern.

Either the **Add Element Pattern** or the **Modify Element Pattern** window opens accordingly:



3. Type your words for a new pattern (for example, *To be continued on*) and/or use the keywords provided on the Keywords drop-down list.
4. Click **OK** to save the new pattern and to close the **Element Pattern** window, or **Cancel** – to close the window without saving the new pattern.

The new pattern appears in the list of the **Element Pattern** field.

► **To delete an element pattern:**

1. In the **Element Pattern** field select the pattern to be deleted.
2. Click the  button to the right of the **Element Position** field.

The selected pattern is deleted.

► **To define the number of lines allowed for an element type:**

1. Select an element type.
2. Fill in the **Minimal** and **Maximal** fields in the **Number of lines allowed** for each primitive of the element.

Fonts List

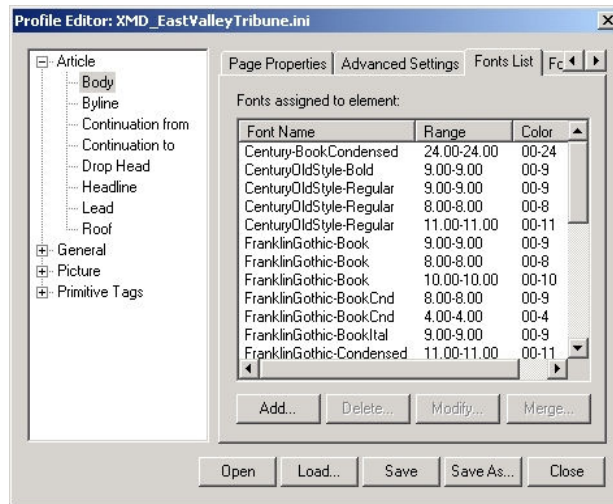


Figure 103 – Profile Editor: Fonts List Tab

Fonts are a crucial part of the profile, as they give the segmentation engine powerful clues on the structural significance of each piece of text. If you define Helvetica-Bold 9pt as the font generally used for bylines, the engine will recognize bylines effortlessly, except in some special cases where reasoning and “guessing” are required.

This tab allows you to add or remove fonts from the profile, and assign these to different elements. The font list on the right shows the fonts currently assigned to the element selected in the tree on the left. For example, the screenshot above shows the fonts assigned to the Body element.

Adding and deleting a font assignment from the current element can be performed by using the Add and Delete buttons of the tab (see the instructions below), or by selecting an existing font on the page.

► **To add a font by selecting it on the page:**

1. In the **element type tree**, click an **element type** (e.g., Article > Body), to which a font assignment/deletion will be performed.
2. On the page, drag-and-drop the selection frame of the required font.
When selected, the frame appears in red.
3. Click the right mouse button on the selected frame.

Depending on the selected element, the following context menu appears:



4. Choose **Assign to Body** to add the selected font to the selected element type.

The element type will be verified in accordance with the selection made in the element tree.

When a font is assigned, the font selection frame appears in blue.

► **To delete a font by selecting it in the page:**

1. Follow steps 1 – 4 in the above procedure, but in the context menu choose **Delete New Assigned Fonts Only** to cancel the font assignment performed.
2. To delete the assignment of all the fonts of the selection frame, choose **Delete All Occurred Fonts**.

► **To add a font assignment to an element type by using the buttons:**

1. Select an **element type** in the **element type** tree.
2. Click the **Add** button.

The following window opens:

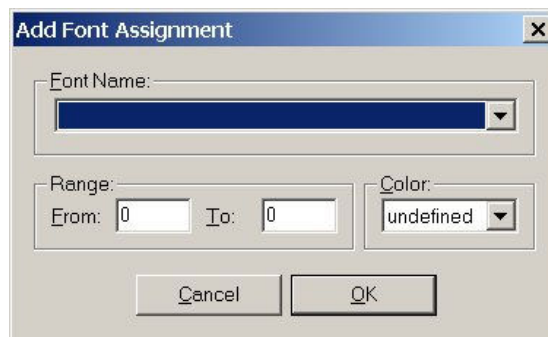


Figure 104 – Add Font Assignment

3. Select the font from the **Font Name** drop-down list, fill in its size range values, and select its color from the **Color** drop-down list, if required.
4. Click **OK**.

The selected element type is assigned a newly defined font.

► To delete a font assignment to an element type by using the buttons:

1. Select a font in the **Fonts assigned to element** field.
2. Click the **Delete** button.

The following window opens:

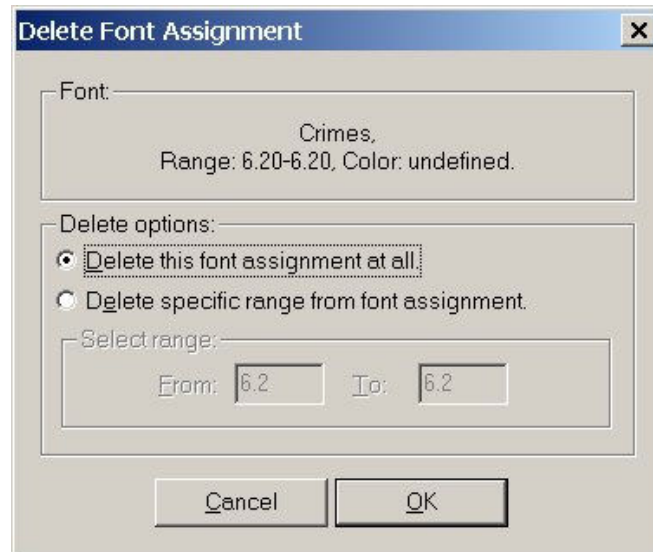


Figure 105 – Delete Font Assignment

3. Select one of the Delete options:

- **Delete this font assignment at all** (default)

or

- **Delete specific range from font assignment.**

If you check the latter option, the fields of the **Select range** section become available, so that you can define the new range of sizes for the selected font.

4. Click **OK** to apply the checked **Delete** option, or **Cancel** - to leave the selected font as it is.

► **To modify the assigned font:**

1. Select a font in the **Fonts assigned to element** field.
2. Click the **Modify** button.

The following window opens:

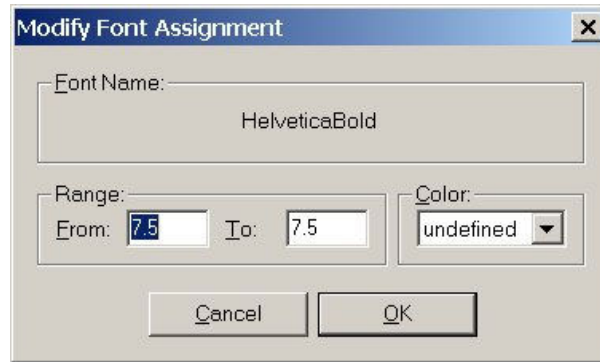


Figure 106 – Delete Font Assignment

3. Modify the parameters of the font displayed in the **Font Name** section.
4. Click **OK** to apply the changes, or **Cancel** to leave the selected font as it is.

► **To merge fonts with the same font name, but different sizes:**

1. In the **Fonts assigned to element** field, select the fonts to be merged by pressing and holding the Shift key while clicking the required fonts.
2. Click the **Merge...** button.

The selected fonts merge into one. The **Fonts assigned to element** field now displays only one row for this font name, the size range of which now covers the size ranges of all fonts that have been merged.

► **To exclude fonts from the segmentation process (Ignore fonts):**

1. In the element type tree, click **Ignore Fonts** under the **General** item.
2. In the issue page, draw a rectangle (by drag-and-drop) to select the fonts to be ignored.
3. Select **Assign to Ignore Fonts** from the context menu.
4. The selected fonts appear in the field of the **Fonts List** tab.

The selected fonts will be ignored during the segmentation process.



The fonts selected as ignored can be modified (by using the **Modify** button, described above).

Font Monitor

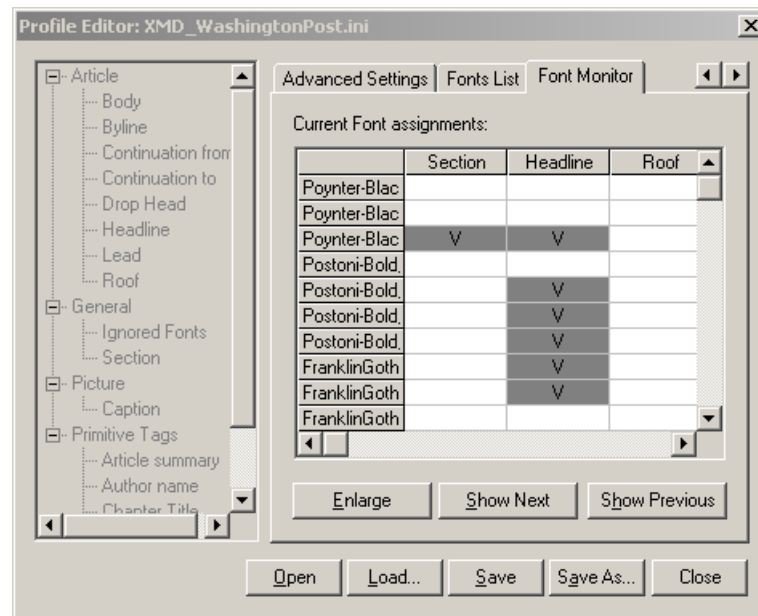


Figure 107 – Profile Editor: Font Monitor

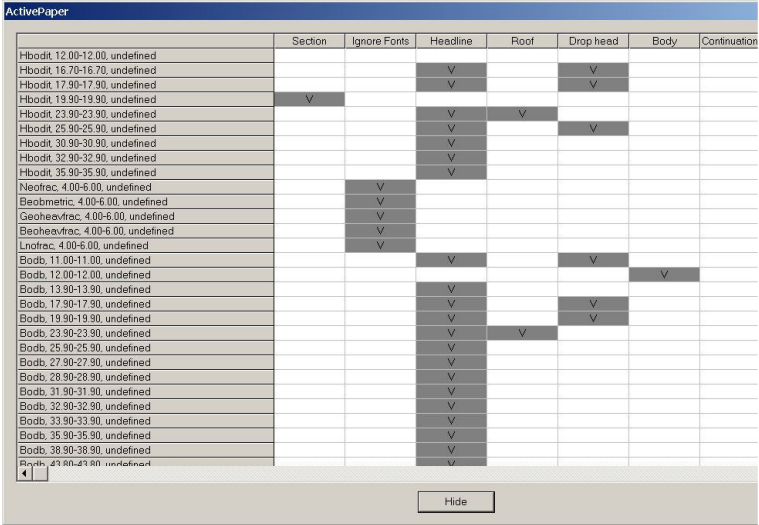
This tab is used for visualizing all the fonts defined for each element type in the profile. This visualization is important because it allows you to spot ambiguities.

Ambiguities can occur when the same font is assigned to more than one element type. For example, say Helvetica-Bold 9-16pt is assigned both to Body and to Picture Caption. When the segmentation engine finds text that is Helvetica-Bold and is between 9 and 16pt, it has to make an educated guess as to whether it is Body text or a Picture caption. In many cases the engine uses clues from the layout of the page to make a correct decision. However, in some cases it will make mistakes, causing errors in automatic segmentation.

The Font Monitor tab is a good place to troubleshoot profiles, because it allows you to identify font ambiguities that may have caused problems in automatic segmentation. For example, if you see that automatic segmentation very often identifies picture captions as body text, you might check to see if an ambiguity like the one described above exists in your profile.

If it does, you may be able to solve the problem by returning to the Font List tab and changing the font assignments.

The *Font Monitor* tab contains the following control buttons:

Button	Description
	Displays the enlarged list of fonts with the indication of element types in which each of these fonts is found:
	
	The enlarged display is closed with the Back to Editor button.
	Used for viewing next font occurrences of the font assignment.
	Used for viewing previous font occurrences of the font assignment.

While in the Advanced Settings or in the Fonts List tab, you can search the issue for occurrences of the element type selected in the element type tree.

► **To perform an element type search in the Fonts List tab (applies also to Enlarged view):**

1. Click an **element tab** or a **fonts** row of the **Font List** to see the matching occurrences in the currently open page.

The scanning bar appears. When the scanning is over, all the matching font occurrences are selected on the page.

If a font cannot be found, the following message appears:



If assignments have already been made in previous profile editing sessions, the occurrences cannot be shown and the following message appears:



You can also perform an element type search in the *Advanced Settings* tab.

► **To perform an element type search in the Advanced Settings tab:**

1. Click the right mouse button on the element type. The **Select** command pops up.
2. Click **Select** to view the font that matches the selected element type.

5.1.2 Improving Profiles in the INI File

This command opens the profile INI file in Notepad, allowing you to modify it. This file contains all the profile's settings and parameters. These include all the settings that can be modified by the Profile Editor (font assignments, body and margin rectangles, etc.), and other, more advanced settings that can only be modified directly in the INI.

The Profile Editor is in fact an easy-to-use front end that writes settings to this INI file. Any changes made to the profile with the Editor are reflected in the file, and the opposite is also true: Font assignments, etc. can be modified in the INI file and are automatically updated in the Profile Editor, once you reload the profile.

In general, the profile INI is used to modify a multitude of advanced settings that do not appear in the Profile Editor, but have an impact on how automatic segmentation is performed.

► **To open the profile:**

1. From the **Olive** menu choose **Profile**, and then **Open**.

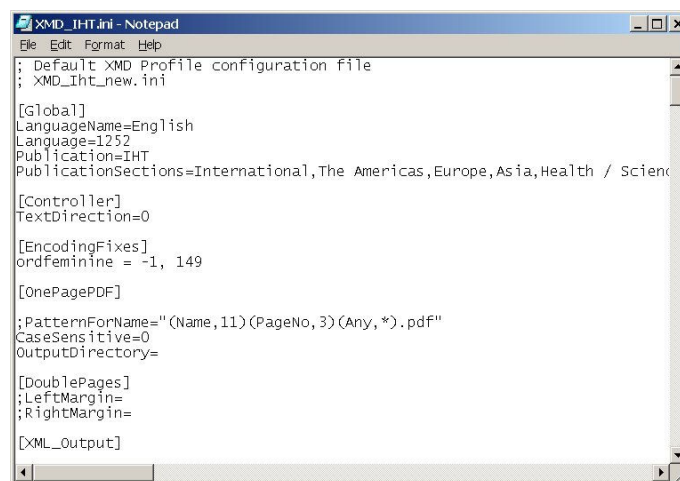


Figure 108 – Profile Editing

If no profile was loaded, the Open command calls up the general *Open* window for file selection. If there is a loaded profile, the command will open this loaded profile.

2. Modify the profile and save changes.



To apply the changes, reload the profile. (See the instructions above.)

5.1.3 Improving Profiles with Learning Mode

Learning Mode allows you to easily assign new fonts to elements in the profile. When Learning Mode is on, it picks up any segmentation that does not conform to the font assignments in the profile. For example, if the only font assigned to Body is Times Roman 12pt, and you have created a Body primitive containing text that is Helvetica-Bold 9pt, Learning Mode will automatically pick this up, and you will be able to update the profile with this new font assignment.

This can be useful in a number of instances:

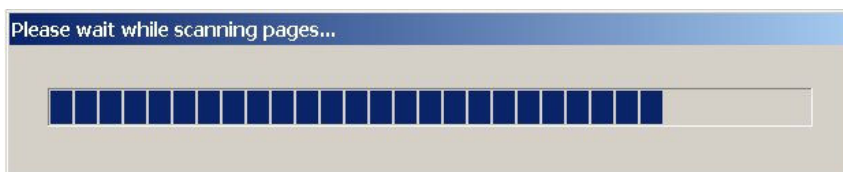
- When creating or fine-tuning a new profile, you can turn Learning Mode on and start doing manual segmentation. Fonts in the primitives you draw become font assignments in the profile. Alternatively, when running automatic segmentation to test your new profile, use Learning Mode to spot missing fonts and add them to the profile.
- Automatic segmentation often produces “garbage articles,” which are entities composed of unknown fonts, which the engine cannot identify. Turning on Learning Mode allows you to easily add these fonts to the profile with the correct element definitions.
- When the document’s design conventions change, you can use Learning Mode to quickly update the profile.



Use Learning Mode with discretion, keeping in mind that changes you make to the profile will not necessarily improve it. A good profile is a fine balance of font assignments that results in minimum ambiguities – adding font assignments may reduce errors in segmentation, but could cause new errors by disrupting this balance. Treat Learning Mode as an easy way to modify your profile, but always test the updated profile to see how well it works.

► To use Learning Mode:

1. From the **Olive** menu choose **Profile**, and then choose the **Learning Mode** option to activate Learning mode. A progress bar appears:



The Learning Mode Console and the Selected Segment Properties windows appear at the same time. The console becomes available as soon as the scanning of segmented pages is completed. (The controls of the Selected Segment Properties window only become available when a primitive has been selected.

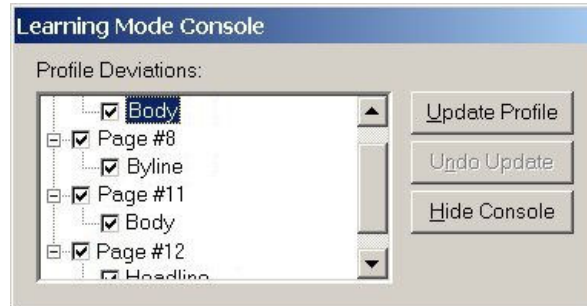


Figure 109 – Learning Mode Console

2. Inside the console, go through the displayed pages and elements to view and edit the deviations from the profile.



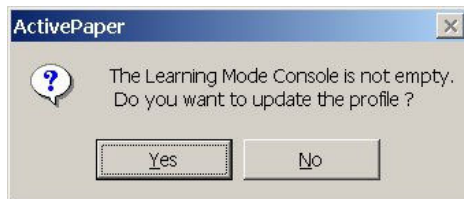
Editing the segmentation in Learning Mode can be performed through the Edit Segmentation Tool context menu, which pops up when you right-click a primitive.

3. Select the changes to be included in the profile by selecting the appropriate checkboxes in the **Learning Mode** console.
4. To include the checked changes in the updated profile, click **Update Profile** in the **Learning Mode Console** window.
5. Click **Hide Console** to remove the console window from the display.

You can also hide or view the console by clicking the  icon on the toolbar.

6. To exit **Learning Mode**, choose **Profile** and uncheck the **Learning Mode** option.

If you attempt to exit Learning Mode while the *Learning Mode Console* still contains unresolved profile deviations, the following dialog box appears:

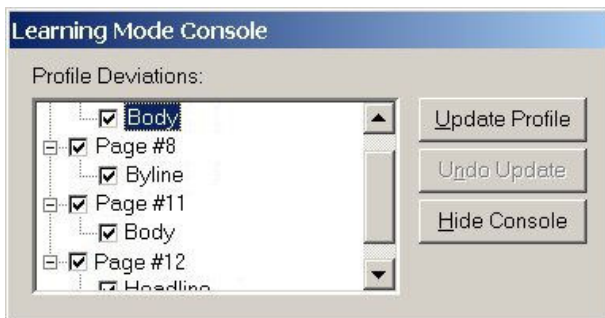


This dialog box contains no Cancel button, so returning to the last Learning Mode status requires several steps.

► **To retrieve the last Learning Mode status:**

1. Click **No** in the dialog box shown above.
2. Return to segmentation editing mode (through **Segmentation -> Edit** or by activating the  icon).
3. Reactivate Learning Mode as described at the beginning of this section.

The **Learning Mode Console** displays the pages that contain the deviations, and the deviating element types.



► **To retrieve the last Learning Mode status:**

1. Click the page number (e.g., Page #8) to view the page in the document.
2. Click the element type (e.g., Byline) to select it on the page opened in step 1. The relevant element is highlighted (black frame).
3. Go through the elements of the opened page one by one, to see which fonts may be considered for a new font assignment or which fonts may cause a problem.
4. Check/uncheck the elements in the console to be included in/left out of the profile.
5. Click the **Update Profile** button.



If the console displays several checks for the same element type, it is possible that the indication of a single font deviates from the profile. To verify if this is the case, check only the first one of these elements in the console, and click **Update Profile**. If a single font caused problems in all the occurrences of this element, then the operation described will provide a solution, and this element type will no longer appear in the **Learning Mode Console**.



After the file is updated, you can check the added fonts in the **Fonts List** tab of the *Profile Editor*.

5.2 Creating Scripts

XML Distiller includes a scripting language with which you can add metadata information to objects, provide segmentation instructions, and more. For example, you can create a script for the automatic insertion of the News Agency name, found in the picture's caption, as metadata to pictures. In this case, the script will include instructions for the identification of the caption within the page, its extraction and input into the designated metadata.

The scripts use *Layout Objects* as their input. These Layout Objects include *Text Blocks* (comprised of *text lines* and *quads*), *Horizontal/Vertical Lines*, *Graph Blocks* (including images), and *Table Blocks*. The input includes the article's text and fonts and also the text and fonts of neighbor elements on the page (used as an additional input for determining the identity of the object).

The scripts' output can be included in the Final Segmentation as *Tags*. These tags identify the objects. In our example, a "*News Agency*" tag can be created indicating the exact location of the news agency credit. A *Flag* in the script indicated the action that is performed by the script. In this case the Flag can be to extract the value inside the "News Agency" tag and insert it into the Article metadata.

Another example that can illustrate the possible usage of Scripts is the identification of Footnotes. The script adds "Footnote" Tags to the footnotes. At a later stage the system can be instructed to disregard the footnotes to improve the reading order.

Detailed reference and instructions for creating and using scripts can be found in the *Olive Recognition Language (ORL) Reference Guide*. The scripts are activated as part of the segmentation process. To include the scripts in the segmentation process, check the **Scripts** check box in the **Segmentation Properties** dialog (see page 146), or activate them in a separate step using the **Olive** menu **Layout Objects** sub-menu.

(A) Distiller Processing Steps

The following table summarizes the steps available for job management. Each Job Type is combination of one or more of these steps (see below and in *Steps and Job Types* on page 159).

Step	Description	Parameter Sources
XMDAnalyzerStep	Build Segmentation (Entities, Links, PageLabels, Sections). If [Global]: BeforeOCRFlag=0 then PDF Elements are processed; Otherwise, results of Image Recognition are used.	All parameters are written in Profile.
(XMDAnalyzerAfterOCRStep)	Obsolete. The same as XMDAnalyzerStep. Is used if Job needs to execute XMDAnalyzer twice: before Image Recognition and after.	
XMDSaveSegToPageDictStep	Save all text primitives to PDF page dictionary to be used as segmentation for OCR. "Title" primitives are saved with additional flag and can be processed by OCR with different parameters.	
XMDGetTitlesFromPDFStep	Sets the Titles of untitled Articles using text captured from the PDF. Text is found in "Headline" primitive blocks.	
XMDLinkAllPagesStep	Connect all articles by continuation link.	

XMDSetAllPublishStep	Set "To Publish" flag to all Entities. Used for correction results of auto segmentation if too many articles have been set "Not To Publish". This can occur as a result of errors in Info Recognition.	
XMDExportEntitiesStep	- not used	
XMDImportEntitiesStep	- not used	
XMDPreparePagesStep	- not used	
XMDBatchStep	Launch a batch passing two parameters: DSC-name and the status of the last executed step. The value of Status is one of the following: JOB_UNKNOWN_STATUS = -1, JOB_SUCCEEDED = 0, JOB_TERMINATED, was cancelled. JOB_FAILED_SYS_EXCEPTION, failed because of a system exception. JOB_STOPPED, stopped for special purposes (needed for XML Distiller Cluster). Batch is run as an asynchronous process.	Full name of batch is taken from DSC parameter: [BatchStep]: BATCH_PATH.
XMDRemovePreparedPagesStep	- not used	
XMDTrimByBBoxStep	Change Media Box of each page according to parameters in the Profile. TrimBBoxLeftMargin, TrimBBoxRightMargin, TrimBBoxTopMargin, TrimBBoxBottomMargin.	

Setting and Programming Segmentation Rules

Creating Scripts

DocPropertiesStep	Store Document Properties to PDF. Load document meta data from manifest if the manifest file is set. [ISSUE_FILES]: MANIFEST_PATHNAME (set via property dialog boxes). Set Date (default - current date) [ISSUE_PROPERTIES]: ISSUE_DATE (set via property dialog boxes if "IssueDate" is checked) Store PDF short name to Document Properties (as "Release" property). Store version of XML Distiller to Document Properties. Note that "Release" and "Publication" text boxes from the step-by-step Olive menu items are not used in the Job Manager. "Publication" is used by the Job Generator.	It is set in XMD.ini: [Global]: "XML_Distiller_Version").
CombinePagesStep	Merge several PDFs.	Naming convention and other parameters are taken. from DSC:[CombinePages] (set via property dialog boxes).
XMDImportPageIdsStep	Import PageID for all pages from text file to Page Property.	Name of text file is taken from DSC: [ImportPageIds]: FILE_NAME (set via the "ImportPageIds" tab).

XMDImportManifestStep	Import "AdFrame" predefined areas from "Manifest" - text file exported from Excel.	Name of Manifest file is taken from DSC: [ImportManifest]: FILE_NAME (set via the "ImportManifest" tab). Name of XML Setting file containing structure of Manifest records is taken from Profile: [Manifest]: SettingsPath.
XMDOCRonAreasStep	Run OCR on "AdFrame" Predefined Areas.	All parameters are taken from Profile: [OCROnAreas].
XMDRedistillStep	Performs re-distillation of PDF.	Parameters are taken from Profile.
XMDImportMetadataToRepository	Insert Metadata from DSC to TOC.xml.	All parameters and MD are taken from DSC: [ImportMetaDataSetToWarehouse (Repository)].
XMDIndexRepository	Index Issue in Warehouse (Repository).	All parameters are taken from DSC: [IndexWarehouse (Repository)].
XMDAutoPublishFinalStep	Used for OCP only. Clean Working folder and send notification to client.	Get all parameters from DSC: [AutoPublisher].
XMDOfficeDocumentConverting Step	Convert MS Word document to PDF.	All parameters are taken from DSC: [OfficeDocumentConverter].
XMDCheckInputStep	Check that PDF pages are not corrupted.	No parameters.
XMDExportDocCustomPropsToDscStep	Obsolete (used in DTIC). Writes to INI-file Metadata of Document kept in old obsolete format. Is used in some old projects	
XMDMergeDoublePagesStep	Merge Double Pages.	All parameters are taken from Profile.

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Creating Scripts

XMDImportSegmentationFromManifestStep	Create segmentation getting all information from Manifest XML file.	Full path is taken from DSC: [ISSUE_FILES]: MANIFEST_PATHNAME (set via the "Issue" tab).
XMDOutputJobStep	Output to Warehouse (Repository). If parameter is empty or not present in DSC, it is taken from the Profile.	Most of parameters are taken from Profile. Some of them can be taken from DSC (set via the "XMDOutput" tab).
COLvImageAnalyzerStep	Perform Image Layout Recognition.	Parameters are taken from DSC.
COLvOCRStep	Run OCR on Images extracted from Pages. Most of parameters can be set via property dialog boxes.	All parameters are taken from DSC: [OCR_EXECUTION] and [OCR_PARAMETERS].
COLvCorrectResolutionStep	Changes resolution of images in a scanned PDF.	
COLvImportIssueStep	Create PDF from Image files according to naming conventions.	All parameters are taken from DSC: [ISSUE_FILES] and [IMPORT_ISSUE].
COLvAnalyzeIssuePagesStep	Check page sizes of PDF created by COLvImportIssueStep.	All parameters are taken from DSC: [CHECK_PAGE_SIZE].
COLvApplyImportCorrectionStep	Apply page size correction (done manually after import issue and check page sizes) to PDF images.	
COLvCheckIssueErrorPagesFixed Step	Check pages after fixing.	
COLvRepairStep	For XML Distiller Cluster only. Prepare environment for Page Size correction. Can be used in XML Distiller Cluster only.	
COLvExportImageStep	Export Images from PDF pages. (See menus as well.)	All parameters are taken from DSC: [EXPORT_IMAGES].

COlvPdfCleanerStep	Remove data from PDF according flags. See list of flags in the Appendix.	All parameters are taken from DSC and can be set via property dialog boxes.
BasicSegmentationStep	Build Layout Objects.	All parameters are taken from Profile.
XMDHeaderFooterRecognitionStep	Recognize Page Footer and Header. Change some of OlvTextBlock objects.	All parameters are taken from Profile.
TOCRognitionStep	Recognize TOC of Document and its Items. Look for relevant text. Change some of OlvTextBlock objects and keep information for Final Segmentation.	All parameters are taken from Profile.
XMDScriptStep	Run one or more scripts. Script can analyze Layout Objects, create OlvTag objects and keep information for Final Segmentation.	The only parameter ScriptPath is taken from DSC: [SCRIPTING_PROPERTIES].
FinalSegmentationStep	Create segmentation using Layout Objects.	All parameters are taken from Profile.

(B) Job Types

For each of the four major categories of job types in this appendix, a summary of the job types is provided, as well as summary tables indicating which steps are used in each job type, and in what order.

Job Types for Layout-Object-Based Segmentation

Job Type	Comment
LOM_WithOCR	Used for Digital or Scanned Documents. In any case set in the Profile [Global]: BeforeOCRFlag=0. Performs Layout Object Based Segmentation including Header/Footer recognition, TOC recognition and execution of script(s). If input files are Image(s) then the first step imports them to PDF. If input file is PDF then Job should start with COlvPdfCleanerStep. If configuration is not OCP then the last step should be XMDOutputJobStep.
LOM_NoHeadFooter	The same as LOM_WithOCR but without Header/Footer recognition.
LOM_NoToc	The same as LOM_WithOCR but without TOC recognition.
LOM_NoHeadFooter_NoToc	The same as LOM_WithOCR but without Header/Footer and TOC recognition.
LOM_OFFICE	Used for MSOffice Word Documents. Performs Layout Object Based Segmentation including Header/Footer recognition, TOC recognition and execution of script(s). If configuration is not OCP then the last step should be XMDOutputJobStep.
LOM_OFFICE_NoHeadFooter	The same as LOM_OFFICE but without Header/Footer recognition.
LOM_OFFICE_NoToc	The same as LOM_OFFICE but without TOC recognition.
LOM_OFFICE_NoHeadFooter_NoToc	The same as LOM_OFFICE but without Header/Footer and TOC recognition.

Steps

Step	LOM_WithOCR	LOM_NoHead Footer	LOM_ NoToc	LOM_NoHead Footer_NoToc
COlvImportIssueStep	1☑	1☑	1☑	1☑
COlvImageAnalyzerStep	2☑	2☑	2☑	2☑
COlvPdfCleanerStep	3☑	3☑	3☑	3☑
COlvOCRStep	4☑	4☑	4☑	4☑
BasicSegmentationStep	5☑	5☑	5☑	5☑
XMDHeaderFooterRecognitionStep	6☑		6☑	
TOCRecognitionStep	7☑	6☑		
XMDScriptStep	8☑	7☑	7☑	6☑
FinalSegmentationStep	9☑	8☑	8☑	7☑
DocPropertiesStep	10☑	9☑	9☑	8☑
XMDImportMetadataToRepository	12☑	11☑	11☑	10☑
XMDIndexRepository	13☑	12☑	12☑	11☑
XMDOutputJobStep	11☑	10☑	10☑	9☑

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Step	LOM_OFFICE	LOM_OFFICE_ NoHeadFooter	LOM_ OFFICE_No Toc	LOM_OFFICE_ NoHeadFooter _NoToc
DocPropertiesStep	7☑	6☑	6☑	5☑
XMDImportMetadataToRepository	9☑	8☑	8☑	7☑
XMDIndexRepository	10☑	9☑	9☑	8☑
XMDOfficeDocumentConvertingStep	1☑	1☑	1☑	1☑
XMDOutputJobStep	8☑	7☑	7☑	6☑
BasicSegmentationStep	2☑	2☑	2☑	2☑
XMDHeaderFooterRecognitionStep	3☑		3☑	
TOCRecognitionStep	4☑	3☑		
XMDScriptStep	5☑	4☑	4☑	3☑
FinalSegmentationStep	6☑	5☑	5☑	4☑

LOM_WithOCR=COLvImportIssueStep,COLvImageAnalyzerStep,COLvPdfCleanerStep,COLvOCRStep,BasicSegmentationStep,XMDHeaderFooterRecognitionStep,TOCRecognitionStep,XMDScriptStep,FinalSegmentationStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

LOM_NoHeadFooter=COLvImportIssueStep,COLvImageAnalyzerStep,COLvPdfCleanerStep,COLvOCRStep,BasicSegmentationStep,TOCRecognitionStep,XMDScriptStep,FinalSegmentationStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

LOM_NoToc=COLvImportIssueStep,COLvImageAnalyzerStep,COLvPdfCleanerStep,COLvOCRStep,BasicSegmentationStep,XMDHeaderFooterRecognitionStep,XMDScriptStep,FinalSegmentationStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

LOM_NoHeadFooter_NoToc=COLvImportIssueStep,COLvImageAnalyzerStep,COLvPdfCleanerStep,COLvOCRStep,BasicSegmentationStep,XMDScriptStep,FinalSegmentationStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

LOM_OFFICE=XMDOfficeDocumentConvertingStep,BasicSegmentationStep,XMDHeaderFooterRecognitionStep,TOCRecognitionStep,XMDScriptStep,FinalSegmentationStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

LOM_OFFICE_NoHeadFooter=XMDOfficeDocumentConvertingStep,BasicSegmentationStep,TOCRecognitionStep,XMDScriptStep,FinalSegmentationStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

LOM_OFFICE_NoToc=XMDOfficeDocumentConvertingStep,BasicSegmentationStep,XMDHeaderFooterRecognitionStep,XMDScriptStep,FinalSegmentationStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

LOM_OFFICE_NoHeadFooter_NoToc=XMDOfficeDocumentConvertingStep,BasicSegmentationStep,XMDScriptStep,FinalSegmentationStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

Job Types for Scanned Documents ("Body per Page" Segmentation)

In the profile, the parameter(s) in section [Parameters4OneBodyOnPage] should be set.

■

Job Type	Comment
AP_MICROFILM_NO_ANALYZER2	If input files are Image(s) then the first step imports them to PDF. If input file is PDF then Job should start with COlvPdfCleanerStep. If configuration is not OCP then the last step should be XMDOutputJobStep.
OFFICEDOCUMENTCONVERT	Used for MSOffice Word Documents. If configuration is not OCP then the last step should be XMDOutputJobStep.
AP_PDF_DIGITAL_OR_SCANNED	Used if a document is to be combined from several PDFs (usually a separate PDF per page). If configuration is not OCP then the last step should be XMDOutputJobStep.

Setting and Programming Segmentation Rules

Creating Scripts

Steps

Step	AP_MICRO FILM_NO_ ANALYZER 2	OFFICE DOCUMENT CONVERT	AP_PDF_DIGITAL_OR_ SCANNED
XMDAnalyzerStep	5☑	3☑	4☑
XMDSaveSegToPageDictStep	6☑		
DocPropertiesStep	8☑	4☑	5☑
CombinePagesStep			1☑
XMDImportMetadataToRepository	10☑	6☑	7☑
XMDIndexRepository	11☑	7☑	8☑
XMDOfficeDocumentConvertingStep		1☑	
XMDOutputJobStep	9☑	5☑	6☑
COLvImageAnalyzerStep	4☑		
COLvOCRStep	7☑		3☑
COLvImportIssueStep	1☑		
COLvAnalyzeIssuePagesStep	2☑		
COLvPdfCleanerStep	3☑	2☑	2☑

AP_MICROFILM_NO_ANALYZER2=COLvImportIssueStep,COLvAnalyzeIssuePagesStep,COLvPdfCleanerStep,COLvImageAnalyzerStep,XMDAnalyzerStep,XMDSaveSegToPageDictStep,COLvOCRStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

OFFICE DOCUMENT CONVERT=XMDOfficeDocumentConvertingStep,COLvPdfCleanerStep,XMDAnalyzerStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

AP_PDF_DIGITAL_OR_SCANNED=CombinePagesStep,COLvPdfCleanerStep,COLvOCRStep,XMDAnalyzerStep,DocPropertiesStep,XMDOutputJobStep,XMDImportMetadataToWarehouse (Repository),XMDIndexWarehouse (Repository)

General Microfilm Job Types (for scanned PDF or TIFF files)

Job Type	Comment
MICROFILM_NO_ANALYZER2	Used for TIFF input. Build segmentation using Image Recognition results.
MICROFILM_COMBINE	Used for PDF input. Build segmentation using Image Recognition results. If Input is "combined" PDF then should be started from COLvPdfCleanerStep.
MICROFILM_XMDEXportImages	Export Images from PDF. Used for scanned PDFs.

Steps

Step	MICROFILM_NO_ANALYZER2	MICROFILM_COMBINE	MICROFILM_XMDEXportImages
COLvImportIssueStep	1☑		
CombinePagesStep		1☑	
COLvExportImageStep			1☑
COLvAnalyzeIssuePagesStep	2☑		
COLvPdfCleanerStep	3☑	2☑	
COLvImageAnalyzerStep	4☑	3☑	
XMDAnalyzerStep	5☑	4☑	
XMDSaveSegToPageDictStep	6☑	5☑	
COLvOCRStep	7☑	6☑	
DocPropertiesStep	8☑	7☑	
XMDOutputJobStep	9☑	8☑	

MICROFILM_NO_ANALYZER2=COLvImportIssueStep,COLvAnalyzeIssuePagesStep,COLvPdfCleanerStep,COLvImageAnalyzerStep,XMDAnalyzerStep,XMDSaveSegToPageDictStep,COLvOCRStep,DocPropertiesStep,XMDOutputJobStep

Setting and Programming Segmentation Rules

Creating Scripts

MICROFILM_COMBINE=CombinePagesStep,COlvPdfCleanerStep,COlvImageAnalyzerStep,XMDAnalyzerStep,XMDSaveSegToPageDictStep,COlvOCRStep,DocPropertiesStep,XMDOutputJobStep

MICROFILM_XMDEXportImages=COlvExportImageStep

Digital Input Job Types: Style-Based Segmentation

Job Type	Comments
DIGITAL	The basic job type for digital input.
DIGITALOCR	The same as DIGITAL but additionally OCR is run on all pages. Is used for digital documents containing some texts as images or vector graphics. Note that XMDOCROnAreasStep runs OCR removing all digital text before it.
DIGITALMANIFEST	Used if there is Manifest for Ads.
DIGITALMANIFEST4SEGMENTATION	Used if there is Manifest containing full segmentation.
MICROFILMMANIFEST4SEGMENTATION	The same as DIGITALMANIFEST4SEGMENTATION but gets scanned material and runs OCR.

Steps

Step	DIGITAL	DIGITAL OCR	DIGITAL MANIFEST	DIGITAL MANIFEST 4SEGMENTATION	MICRO FILM MANIFEST 4SEGMENTATION
XMDAnalyzerStep	3☑	4☑	4☑		
XMDSaveSegToPageDictStep					5☑
DocPropertiesStep	4☑	5☑	5☑	5☑	7☑
CombinePagesStep	1☑	1☑		1☑	1☑
XMDImportPageIdsStep			2☑		
XMDImportManifestStep			3☑		
XMDOCROnAreasStep		2☑			

XMDMergeDoublePagesStep				4☑	4☑
XMDImportSegmentationFromManifestStep				3☑	3☑
XMDOutputJobStep	5☑	6☑	6☑	6☑	7☑
COLvImageAnalyzerStep					
COLvOCRStep					6☑
COLvPdfCleanerStep	2☑	3☑	1☑	2☑	2☑

DIGITAL=CombinePagesStep, COLvPdfCleanerStep, XMDAnalyzerStep, DocPropertiesStep, XMDOutputJobStep

DIGITALOCR=CombinePagesStep, XMDOCROnAreasStep, COLvPdfCleanerStep, XMDAnalyzerStep, DocPropertiesStep, XMDOutputJobStep

DIGITALMANIFEST=COLvPdfCleanerStep, XMDImportPageIdsStep, XMDImportManifestStep, XMDAnalyzerStep, DocPropertiesStep, XMDOutputJobStep

DIGITALMANIFEST4SEGMENTATION=CombinePagesStep, COLvPdfCleanerStep, XMDImportSegmentationFromManifestStep, XMDMergeDoublePagesStep, DocPropertiesStep, XMDOutputJobStep

MICROFILMMANIFEST4SEGMENTATION=CombinePagesStep, COLvPdfCleanerStep, XMDImportSegmentationFromManifestStep, XMDMergeDoublePagesStep, XMDSaveSegToPageDictStep, COLvOCRStep, DocPropertiesStep, XMDOutputJobStep

(C) DSC Template Descriptions

Each of the following descriptions of job descriptor templates includes the associated job type and profile, and a process summary. DSC files are found in the Olive\Batch\Templates folder.

JOB_EP_GENERAL_DOCUMENT_OFFICE.dsc

Descriptor for common office documents

JOB_EP_TECHNICAL_DOCUMENT_OFFICE.dsc

Descriptor for technical office documents with tables and graphs

JOB_EP_GENERAL_DOCUMENT_PDF.dsc

Descriptor for common PDF documents

JOB_EP_TECHNICAL_DOCUMENT_PDF.dsc

Descriptor for technical PDF documents with tables and graphs

JOB_EP_GENERAL_DOCUMENT_TIFF.dsc

Descriptor for common TIFF documents

JOB_EP_TECHNICAL_DOCUMENT_TIFF.dsc

Descriptor for technical TIFF documents with tables and graphs

JOB_AP_OFFICEDOCUMENTCONVERT.dsc

Job Type - OFFICEDOCUMENTCONVERT

Profile - AP_OfficeDocument.ini

Process -

- Conversion of MS Document to PDF format
- "Body per Page" segmentation and union of all pages as continuation for one article
- Output to Warehouse (Repository)
- Import MetaData written in DSC
- Index Warehouse (Repository).

JOB_AP_PDF_DIGITAL_OR_SCANNED_MultiPage.dsc

Job Type - AP_PDF_DIGITAL_OR_SCANNED

Profile - AP_PDFDocument.ini

Process -

- "Body per Page" segmentation and union of all pages as continuation for one article
- OCR on scanned pages
- Output to Warehouse (Repository)
- Import MetaData written in DSC
- Index Warehouse (Repository).

JOB_AP_MICROFILM_OnePage.dsc

Job Type - AP_MICROFILM

Profile - AP_PDFDocument.ini

Process -

- Import TIFF files to PDF with pages (Letter size)
- "Body per Page" segmentation and union of all pages as continuation for one article
- OCR
- Output to Warehouse (Repository)
- Import MetaData written in DSC
- Index Warehouse (Repository).

JOB_AP_BOOKS.dsc

Job Type - LOM_WithOCR

Profile - AP_PDF_Books.ini

Script - AP_FindHeadlines.js

Process -

- Creation of Layout Objects
- Header/Footer Recognition

Creating Scripts

- TOC Recognition
- Title recognition on the first page using script
- Final Segmentation
- Output to Warehouse (Repository)
- Import MetaData written in DSC
- Index Warehouse (Repository).
-

Note that the input should be digital.

For scanned or mixed documents, the **START_STEP** should be `OCR`.

JOB_AP_SCIENTIFIC.dsc

Job Type - LOM_NoHeadFooter_NoToc

Profile - AP_PDFScientific.ini

Script - AP_Scientific.js

Process -

- Creation of Layout Objects
- Recognition using script:
 - Find the TextBlock with the Document's Headline.
 - Identify main chapters, find Abstract and References.
 - Extract abstract contents as metadata for Articles.
- Final Segmentation
- Output to Warehouse (Repository)
- Import MetaData written in DSC
- Index Warehouse (Repository).

Note that the input should be digital.

For scanned or mixed documents, the **START_STEP** should be `OCR`.

(D) Parameters for Clean PDF

This section specifies PDF cleaner setting options for the section [PDF_CLEANER] in the Settings file:

```
[PDF_CLEANER]
CLEAN_METADATA=
CLEAN_CONTENT=
```

CLEAN_METADATA Flags

Flag name	Flag value	Flag description
segmentation	0x00000001	Segmentation
image_layout	0x00000002	Image recognition results
OCR_layout	0x00000004	List of text zones for OCR
OCR_statistics	0x00000008	OCR quality statistics
prepared_pages	0x00000010	Prepared pages
import_log	0x00000020	Import issue log
predefined_areas	0x00000040	Predefined areas
XMD_annotations	0x00000080	XMDAnnotations
page_labels	0x00000100	Page Labels
sections	0x00000200	Sections
page_ids	0x00000400	Page Ids
gatefold_info	0x00000800	Gatefold Info
page_image_descriptors	0x00001000	Page image descriptors
entity_titles	0x00002000	Titles of entities
layout_objects	0x00004000	Layout objects
page_metadata	0x00008000	Metadata of pages
document_metadata	0x00010000	Metadata of Document

CLEAN_CONTENT Flags

Flag name	Flag value	Flag description
texts	0x00000001	Delete all text elements
pathes	0x00000002	Delete all path elements
alt_images	0x00000004	Delete alternate images
OCR_text	0x00000008	Delete text generated by OCR
invisible_text	0x00000010	Delete all invisible text

(E) Ad Frames from Manifest

Metadata Reader v2.0

Depending on the software that you are using, creating an advertisement manifest can be a natural part of the newspaper printing process. However, when generating a manifest make sure that it contains the following the elements described in this section.

When preparing for manifest processing, make sure that you complete the following:

- Configure Manifest
- Set PageId via `log_<name>.txt` File
- Create Manifest Table – `Manifest_<name>.txt`
- Update Profile/Settings
- Metadata.

Once you have completed the above, you will be ready to follow the procedure under *Creating Ad Frames from Manifest* on page 53.

Configure Manifest

The **manifest** file is interpreted according to the **manifest configuration** file:

ManifestSettings_<name>.xml

This file describes the following:

- Shifts method for Ad frames coordinates from manifest file and how to calculate them. Ads created according to coordinates written in the manifest are shifted according to values of `<ShiftX>` and `<ShiftY>`.
- For Settings `MODE=n`:
 - For Settings `MODE="1"`, parameters are taken from `SHIFT_TYPE="FixShift"` and all Ads are shifted the same.
 - For Settings `MODE="2"`, parameters are taken from `SHIFT_TYPE="PerPublication"` with value equal to Publication Name for Issue.
 - For Settings `MODE="3"`, parameters are taken from `SHIFT_TYPE="PageSizes"` with a value equal to PageSize of the correspondent page.

Keep in mind that:

- All values are always in inches.
- `ShiftY<0` causes shifting up.
- `ShiftX<0` causes shifting right.

Example

The following is an example of a **Shifts** section:

```
<Shifts MODE="2">
  <Shift SHIFT_TYPE="FixShift">
    <ShiftX>0.4</ShiftX>
    <ShiftY>-1</ShiftY>
  </Shift >
  <Shift SHIFT_TYPE="PerPublication">
    <Publication PUBLICATION_NAME="WPost">
      <ShiftX>0.75</ShiftX>
      <ShiftY>-1</ShiftY>
    </Publication>
    <Publication PUBLICATION_NAME="ETS">
      <ShiftX>0.38</ShiftX>
      <ShiftY>0</ShiftY>
    </Publication>
  </Shift>
  <Shift SHIFT_TYPE="PageSizes" TOLERANCE="0.2">
    <PageSize WIDTH="12.5" HEIGHT="21.54">
      <ShiftX>0.2</ShiftX>
      <ShiftY>0.7</ShiftY>
    </PageSize>
  </Shift>
</Shifts>
```

- The value separating (delimiting) each record in the table.
- The value separating (delimiting) each field in a record.
- The coordinate system, that identifies how the frame is measured on the page. The possible values are:
 - DOWN_RIGHT
 - UP_RIGHT
 - DOWN_LEFT
 - UP_LEFT

- The values indicating the coordinates that you are specifying in the manifest. These values conform to one of the following:

X_Y_WIDTH_HEIGHT Four values. Two values that specify the X and Y coordinates of a corner, and two values that describe the width and height of the ad frame.

RECTANGLE_POINTS Four values. Two values that specify the X and Y coordinates of a corner, and two values that specify the X and Y coordinates of the opposite corner.

- The value describing the measurement unit type. You can choose from the following:
 - INCH
 - CENTIMETER
 - MILLIMETER

- The **Code page** of the input manifest file. Refer to the MS Platform SDK for possible values (**Code-Page Identifiers** http://msdn.microsoft.com/library/en-us/intl/unicode_81rn.asp).
- Each of the column headings in the table in the **Manifest** file.



If any of the above is omitted from the **manifest configuration** file, manifest processing will fail.

The user should set correct values of **<CoordinateSystem>** and **<Units>** in header and pay special attention to the **mandatory fields description**.

Example

The following is an example of the **ManifestDesc** section:


```

<ManifestDesc>
  <RecordDelimiter>\n</RecordDelimiter>
  <!-- LAST_FIELD_IS_OPEN values:
    T - record delimiter is also delimiter for last field;
    F - last field has delimiter followed by record delimiter;
  -->
  <FieldDelimiter LAST_FIELD_IS_OPEN="T">\t</FieldDelimiter>
  <ZoneDelimiter>,</ZoneDelimiter>
  <Comment>;;</Comment>
  <!-- possible values: DOWN_RIGHT, UP_RIGHT, DOWN_LEFT, UP_LEFT
-->
  <CoordinateSystem>DOWN_RIGHT</CoordinateSystem>
  <!-- possible values: X_Y_WIDTH_HEIGHT ,RECTANGLE_POINTS -->
  <Coordinates>X_Y_WIDTH_HEIGHT</Coordinates>
  <!-- possible values: INCH, CENTIMETER, MILLIMETER -->
  <Units>INCH</Units>
  <!-- Refer to Code-Page Identifiers table in MS Paltform SDK for
possible values -->
  <CodePage>1252</CodePage>
  <Fields>
    <!--
      Types:

      OLV_COORDINATE_1 -_u99 ?oordinate 1
      OLV_COORDINATE_2 -_u99 ?oordinate 2
      OLV_COORDINATE_3 -_u99 ?oordinate 3
      OLV_COORDINATE_4 -_u99 ?oordinate 4
      ZONE               -_u108 ?ist of strings with delimiter
      PAGE_ID            -_u115 ?tring, unique page label

      Formats:

      INT      - integer number
      NUMBER  - floating point number
      STRING   - generic string

      Mandatory flags:

      T - true
      F - false

    -->
    <Field NAME="PDF_NAME" TYPE="PAGE_ID" FORMAT="STRING"
MANDATORY="T"/>
    <Field NAME="PUBLICATION" FORMAT="STRING" MANDATORY="F"/>
    <Field NAME="AD_HEIGHT" TYPE="OLV_COORDINATE_4" FORMAT="NUMBER"
MANDATORY="T"/>
    <Field NAME="AD_WIDTH" TYPE="OLV_COORDINATE_3" FORMAT="NUMBER"
MANDATORY="T"/>
    <Field NAME="AD_X" TYPE="OLV_COORDINATE_1" FORMAT="NUMBER"
MANDATORY="T"/>
    <Field NAME="AD_Y" TYPE="OLV_COORDINATE_2" FORMAT="NUMBER"
MANDATORY="T"/>
  </Fields>
</ManifestDesc>

```

Set PageId via log_<name>.txt File

When processing ad frames, if the original PDF files contain more than one page per PDF file, you need to create a text file for each PDF file that assigns a page ID for each page in the PDF.

See page 52 for more information.

Create Manifest Table – Manifest_<name>.txt

The *manifest file* is a text file that contains a table with cells (or fields) separated by tabs, often referred to as CSV format. Any standard spreadsheet application should be able to export and read CSV files. We recommend that you use Microsoft Excel.

The first line of the **manifest** file must contain a header consisting of a series of field names separated by tabs. Each of these field names represents a column heading, and must be defined in the **<fields>** section of the manifest configuration file:

ManifestSettings_<name>.xml.

For example, the configuration file ManifestSettings_<name>.xml on page 226 defines the **table fields** as:

- **PDF_NAME** – PAGE_ID
- **AD_X** – OLV_COORDINATE_1
- **AD_Y** – OLV_COORDINATE_2
- **AD_WIDTH** – OLV_COORDINATE_3
- **AD_HEIGHT** – OLV_COORDINATE_4

Example

The CSV that this field would describe would look like this:

PDF_NAME	AD_HEIGHT	AD_WIDTH	ADX	ADY
001	5.8	6.05	3.25	14.23
001	4.4	7.8	3.23	6.0



The fields in the CSV do not need to be in a specific order. However the column names in the header must be referred to correctly.

Please note that some word-processing programs might display the above table like this:

PDF_NAME	AD_HEIGHT	AD_WIDTH	ADX	ADY
001	5.8	6.05	3.25	14.23
001	4.4	7.8	3.23	6.0

The following information must be included in the text file:

- **PageId** – A unique value associated with a page in the publication.

Creating Scripts

- X and Y values for at least one of the four points of the Ad frame.
- Height and Width of the Ad frame,
or
A second set of X and Y values.



If you choose a second set of X and Y values, these X and Y values must be those of the opposite corner of that of the previous X and Y values. The second X value must not equal the previous X value and the second Y value must not equal the previous Y value.

Update Profile/Settings

To generate Ads from a **Manifest** file, the user needs to create (in the **Profile** or the **Settings** file) a section labeled [**Manifest**], and include two parameters:

- full path to **<ManifestSettings>.xml**
- `useOldVersionManifest`

Example

```
[Manifest]
SettingsPath=<path>\ManifestSettings.xml
UseOldVersionManifest=0 ; use MD Reader v2.0
```

Add parameter to the Profile's [**Controller**] section

Ads that are imported from Manifest can be located one inside another and in such way they can have common text. By default each text is written only once in the Entity XML, but in case of Manifest Ads we want it to be written in all Ads containing it. This is accomplished by adding, in the Profile's [Controller] section, the parameter **TextCanBelongDifPrims=1**.



Parameters defined in the Profile override identical parameters in the Settings file.

Metadata

Starting from Version 2.5.613 of ActivePaper, metadata from manifests is inserted during Ad Frames creation, and included in XML output.

Example of ManifestSettings.xml

The following is an example of a **ManifestSettings.xml** file:

```

<?xml version='1.0' encoding='utf-8'?>
<Settings>
  <!-- Possible modes:
    1 - fixed
    2 - by publication
    3 - by page size
  -->
  <Shifts MODE="2">
    <Shift SHIFT_TYPE="FixShift">
      <ShiftX>0.4</ShiftX>
      <ShiftY>-1</ShiftY>
    </Shift>
    <Shift SHIFT_TYPE="PerPublication">
      <Publication PUBLICATION_NAME="WPBroad">
        <ShiftX>0.75</ShiftX>
        <ShiftY>-1</ShiftY>
      </Publication>
      <Publication PUBLICATION_NAME="WPTabloid">
        <ShiftX>0.38</ShiftX>
        <ShiftY>0</ShiftY>
      </Publication>
      <Publication PUBLICATION_NAME="Classified">
        <ShiftX>0.47</ShiftX>
        <ShiftY>-1</ShiftY>
      </Publication>
    </Shift>
    <Shift SHIFT_TYPE="PageSizes" TOLERANCE="0.2">
      <PageSize WIDTH="12.5" HEIGHT="21.54">
        <ShiftX>0.2</ShiftX>
        <ShiftY>0.7</ShiftY>
      </PageSize>
    </Shift>
  </Shifts>
  <ManifestDesc>
    <RecordDelimiter>\n</RecordDelimiter>
    <!-- LAST_FIELD_IS_OPEN values:
      T - record delimiter is also delimiter forelast field;
      F - last field has_u100 ?elimiter followed by record
delimiter;
    -->
    <FieldDelimiter LAST_FIELD_IS_OPEN="T">\t</FieldDelimiter>
    <ZoneDelimiter>,</ZoneDelimiter>
    <Comment>;;</Comment>
    <!-- possible values: DOWN_RIGHT, UP_RIGHT, DOWN_LEFT,
UP_LEFT -->
    <CoordinateSystem>DOWN_RIGHT</CoordinateSystem>
    <!-- possible values: X_Y_WIDTH_HEIGHT ,RECTANGLE_POINTS -->
    <Coordinates>X_Y_WIDTH_HEIGHT</Coordinates>
    <!-- possible values: INCH, CENTIMETER, MILLIMETER -->
    <Units>INCH</Units>
    <!-- Refer to Code-Page Identifiers table in MS PaltformeSDK for
possible values -->
    <CodePage>1252</CodePage>
    <Fields>
      <!--

```

```

Types:

    OLV_COORDINATE_1 - coordinate 1
    OLV_COORDINATE_2 - coordinate 2
    OLV_COORDINATE_3 - coordinate 3
    OLV_COORDINATE_4 - coordinate 4
    ZONE              - list of strings with delimiter
    PAGE_ID           - string, unique page label

Formats:

    INT      - integer number
    NUMBER   - floating point number
    STRING   - generic string

Mandatory flags:

    T - true
    F - false

-->
<Field NAME="PDF_NAME" TYPE="PAGE_ID" FORMAT="STRING"
MANDATORY="T"/>
<Field NAME="PUBLICATION" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="PARENT_ACCT_NUM" FORMAT="STRING"
MANDATORY="F"/>
<Field NAME="PARENT_NAME" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="AGENCY_PARENT" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="AGENCY" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="AGENCY_ACCT_NUM" FORMAT="STRING"
MANDATORY="F"/>
<Field NAME="ADVERTISER" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="ADVERTISER_ACCT_NUM" FORMAT="STRING"
MANDATORY="F"/>
<Field NAME="AD_NUMBER" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="INSERTION_NUMBER" FORMAT="STRING"
MANDATORY="F"/>
<Field NAME="PUB_DATE" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="PRODUCT" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="SECTION_NAME" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="PAGE_NUMBER" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="ZONE_CODE" TYPE="ZONE" MANDATORY="F"/>
<Field NAME="ZONE" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="CATCHLINE" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="AD_SIZE" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="AD_SIZE_UNITS" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="AD_SQIN" FORMAT="NUMBER" MANDATORY="F"/>
<Field NAME="COLOR" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="PAGETYPE" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="PRODUCTION_METHOD" FORMAT="STRING"
MANDATORY="F"/>
<Field NAME="AD_HEIGHT" TYPE="OLV_COORDINATE_4"
FORMAT="NUMBER" MANDATORY="T"/>
<Field NAME="AD_WIDTH" TYPE="OLV_COORDINATE_3"
FORMAT="NUMBER" MANDATORY="T"/>

```

```
<Field NAME="AD_X" TYPE="OLV_COORDINATE_1" FORMAT="NUMBER"
MANDATORY="T"/>
<Field NAME="AD_Y" TYPE="OLV_COORDINATE_2" FORMAT="NUMBER"
MANDATORY="T"/>
<Field NAME="DAT_SERVICE" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="DAT_FILE_NAME" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="SALES_ID" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="SALES_NAME" FORMAT="STRING" MANDATORY="F"/>
<Field NAME="SOURCE" FORMAT="STRING" MANDATORY="F"/>
</Fields>
</ManifestDesc>
</Settings>
```